Surface Response to Idealised Stratospheric Intrusions: A High-Order Finite Element Approach to Quasigeostrophic Potential Vorticity Inversion and Evolution

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Abstract

Cut-off lows are responsible for a substantial share of the extreme rainfall and flash flooding over South Africa, and their surface impact depends on how far the stratospheric intrusion at their core reaches towards the ground. We investigate how the structure of an idealised intrusion, and the background atmosphere in which it is embedded, determine its surface response, building upon the quasigeostrophic potential vorticity (QGPV) framework used by Barnes et al. A fifth-order finite element solver for QGPV inversion is built in Firedrake and verified with the method of manufactured solutions. A matrix-free conjugate gradient solver with p-multigrid pre-conditioning scales linearly with the number of degrees of freedom, so that a full atmospheric state can be constructed, inverted and analysed in under three seconds, making extensive parameter sweeps feasible. The sweeps show that the surface response is governed mainly by the height of the tropopause and the depth and horizontal size of the intrusion, while the jet stream, the treatment of the tropopause and the stratospheric stability have little effect at the surface. Once states outside the validity of quasigeostrophic theory are excluded, the minimum height of the dynamical tropopause is only weakly to moderately correlated with the surface response. A derivative-free optimiser finds intrusions whose surface pressure anomaly and wind are about ten times those of the control state, but all such states have Rossby numbers of at least 1.5, so these results are qualitative. Finally, an upwind discontinuous Galerkin scheme with a bound-preserving limiter and third-order strong stability preserving time stepping is developed and verified. Evolving the control state shows the intrusion drifting poleward under the potential vorticity gradient of the jet, so the closed domain must be revised before the longevity of intrusions can be assessed.

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1 Introduction

The intersection of applied mathematics and atmospheric science provides a rigorous framework for modelling complex meteorological phenomena. This work introduces foundational atmospheric concepts – such as atmospheric stratification, potential vorticity, stratospheric intrusions, and cut-off lows – from an applied mathematics perspective, with a specific focus on their dynamics and significant weather impacts in South Africa.

The theoretical core of this project is governed by the quasigeostrophic potential vorticity (QGPV) equation [HMR85]. We specify the mathematical formulation of the QGPV equation, detailing its inherent limitations, physical assumptions, and the requisite background states. To approximate solutions to this equation, we design and implement an efficient numerical solver. This solver is designed to scale effectively in parallel computational environments and remain computationally tractable for time-stepping, with its mathematical correctness rigorously verified.

A primary objective of this research is to construct an idealised model of stratospheric intrusions caused by cut-off lows, building directly upon the foundational work of Fehlmann [Feh97], Sprenger [Spr07], and Barnes et al. [Bar+22]. We aim to replicate their time-independent experiments to validate our computational approach, subsequently comparing results and conducting additional numerical experiments to expand on their findings. Finally, this project investigates the computational feasibility of time-stepping the QGPV equation. By implementing time-dependent simulations, we seek to explore the time-evolution and dynamic lifecycle of these idealised atmospheric systems.

2 Atmospheric Science Concepts

For readers who are not familiar with atmospheric science, but who are familiar with Applied Mathematics, a brief overview of terminology and atmospheric properties is given.

2.1 Layers of the Atmosphere

The atmosphere is divided into a number of distinct layers, a characteristic referred to as **stratification**. We will focus on the two layers closest to the Earth's surface: the troposphere and the stratosphere. The boundary between these two layers is known as the *tropopause* [1] [Val17].

- The **troposphere** extends from the Earth's surface ($z = 0$) up to the tropopause. This layer is mostly transparent to incoming solar (shortwave) radiation, meaning it is primarily heated from below by terrestrial (longwave) radiation emitted by the Earth's surface. This bottom-up heating results in a negative vertical temperature gradient ($\frac{\partial T}{\partial z} < 0$) [Val17]. As a practical example, as you climb higher up a mountain, the temperature tends to decrease with increasing altitude. The warmer air close to the surface is less dense than the colder air at higher altitudes, which leads to convective instability [DLA03]. Consequently, this layer experiences high turbulence [Spr07] and substantial dissipation of energy; almost all typical weather phenomena occur here.
- Above the troposphere lies the **stratosphere**. This layer contains a higher concentration of ozone, which absorbs high-energy ultraviolet (UV) solar radiation, heating the layer from within. Solar radiation first heats the ozone at higher altitudes, losing energy as it penetrates deeper into the lower stratosphere. This results in a positive temperature gradient ($\frac{\partial T}{\partial z} > 0$) [Val17]. Because warmer, less dense air sits atop cooler, denser air, the stratosphere is highly stable with respect to vertical convection [Bar+22].

[1] Italicised text is used to show terms that will be defined in more detail later on in the text.

- The **tropopause** serves as the boundary between the negative temperature gradient of the troposphere and the positive temperature gradient of the stratosphere. At this boundary, the temperature gradient experiences a discontinuity [Pan85]. The altitude of the tropopause primarily varies with latitude (φ), occurring at an altitude of about 16 km in equatorial regions and dropping to approximately 8 km in polar regions [Val17].

2.2 Terminology

The meaning of certain relevant terms that will be used later on is given. We assume that the reader is familiar with the quantities temperature (T), pressure (p), density (ρ) and acceleration due to gravity (g).

- An **adiabatic** process involves a system exchanging no heat with its surroundings. In the case of the movement of an air parcel, the process is adiabatic if temperature only changes due to changes in pressure [Val17].

- The **specific gas constant for dry air** (R) is $287.05\,\mathrm{J\,kg^{-1}\,K^{-1}}$. It is used to relate p, ρ and T using the bulk ideal gas law $p = \rho RT$ [Val17].

- The **specific heat capacity of dry air** (c_p) is the amount of heat necessary to heat up 1 kg of dry air by 1 K. We will take it to be $1005\,\mathrm{J\,kg^{-1}\,K^{-1}}$

- **Potential temperature** (θ) is the temperature an air parcel would attain if it were brought adiabatically to some reference pressure p_0. p_0 is usually taken to be 1000 hPa [Spr07].

$$\theta = T\left(\frac{p_0}{p}\right)^{R/c_p} \tag{1}$$

  In a purely adiabatic process potential temperature is conserved.

$$\frac{D\theta}{Dt} = \frac{\partial\theta}{\partial t} + \boldsymbol{u}\cdot\nabla\theta = 0 \tag{2}$$

- The **Brunt–Väisälä frequency** (N) is a measure of the stability of an air parcel to vertical displacements, such as those caused by convection [Spr07]. The stratosphere tends to have larger values of N^2 than the troposphere [Bar+22], representing greater stability and less turbulent flow.

$$N^2 = \frac{g}{\theta}\frac{\partial\theta}{\partial z} \tag{3}$$

- **Relative vorticity** ($\boldsymbol{\omega}$) is defined as the curl of the wind velocity vector, i.e. $\boldsymbol{\omega} = \nabla\times\boldsymbol{u}$. It physically represents the local spin of a fluid element relative to the Earth's surface [Val17].

- The **Coriolis parameter** (f) is a local measure of the effect that the Earth's rotation has on the movement of fluids (such as the atmosphere) on the Earth's surface [Val17]. It is calculated as $f = 2|\Omega|\sin(\varphi)$, where φ is latitude, and $\boldsymbol{\Omega}$ is the angular momentum of the Earth [Spr07]. By convention f is negative in the Southern Hemisphere [Bar+22].

- The **zonal** and **meridional** directions refer to the west-to-east and south-to-north directions respectively. This text will use the convention that the x-axis points east and the y-axis points north.

- **Hydrostatic equilibrium** is an assumption often made to simplify large-scale atmospheric models. It presumes that the downward gravitational force on a fluid parcel is precisely balanced by the upward force due to the pressure gradient. It can be written as [Val17]:

$$\frac{\partial p}{\partial z} = -\rho g \tag{4}$$

- **Geostrophic** describes a flow in which the force induced by horizontal pressure gradients is exactly balanced by the Coriolis force [Wor92]. This causes the resulting geostrophic wind to blow parallel to contours of constant pressure. This balance holds approximately for large-scale weather system in the mid latitudes, as long as the the *Rossby number* is assumed to be small [Val17].
- **Geostrophic vorticity** (ζ_g) is the vertical component of the vorticity of geostrophic wind, measuring the rotation about the vertical axis [HH13b; Wor92]. It is quantitatively defined as follows, where ψ_g is the geostrophic stream function [Val17].

$$\zeta_g = \hat{\boldsymbol{k}} \cdot \boldsymbol{\omega_g} = \frac{\partial v_g}{\partial x} - \frac{\partial u_g}{\partial y} = \frac{\partial^2 \psi_g}{\partial x^2} + \frac{\partial^2 \psi_g}{\partial y^2} \tag{5}$$

- The **Rossby number** (Ro) is the ratio of the typical flow speed U to f times the typical length scale L. A typical value in the mid-latitude atmosphere is Ro ~ 0.1. The natural scale for ζ_g is U/L, so the Rossby number can be approximated as [HH13b; Val17]:

$$\mathrm{Ro} = \frac{U}{fL} \approx \left| \frac{\zeta_g}{f} \right| \tag{6}$$

  Quasigeostrophic theory is only formally valid when Ro $\ll 1$, but remains qualitatively useful despite this restrictive condition [HH13b].

2.3 Potential Vorticity

In one of his most significant contributions, Ertel showed that under adiabatic and frictionless conditions, there exists a conserved quantity in the idealised atmosphere. We will call this quantity Ertel's potential vorticity (EPV) and denote it as Q [Feh97; Sch+04]:

$$\frac{DQ}{Dt} = \frac{D}{Dt}\left(\frac{(\boldsymbol{\omega} + 2\boldsymbol{\Omega}) \cdot \nabla \theta}{\rho} \right) = 0 \tag{7}$$

where the material derivative $\frac{D}{Dt} = \frac{\partial}{\partial t} + \mathbf{u} \cdot \nabla$ is used and Q has dimensions of $M^{-1}L^2T^{-1}\Theta$ [Bar+22].

2.4 Rossby Waves

Rossby waves are the large scale meanders (wave lengths $\sim 3000\,\mathrm{km}$) of the mid-latitude prevailing westerly winds, manifesting in the upper troposphere as a series of alternating troughs and ridges [Stu15] along the jet stream and the tropopause [HMR85]. They owe their existence to a background gradient of PV, most simply that provided by the variation of f with latitude; The potential vorticity of an air parcel displaced towards the pole or equator is conserved, so according to equation (7), its relative vorticity ($\boldsymbol{\omega}$) may change to compensate for the change in planetary vorticity ($2\boldsymbol{\Omega}$) [Val17]. The induced circulation pushes parcels back towards their original latitude, but a quarter wavelength out of phase with the displacement, so the pattern propagates westward relative to the mean flow. In terms of potential vorticity, a Rossby wave is a wavy displacement of PV contours on a surface of constant θ (isentropic surface). The sharp PV gradient at the tropopause tends to act as a waveguide along which these waves travel [HMR85; Stu15].

At small amplitude these undulations are reversible: the PV contours are displaced, but return to their original positions once the wave has passed. As the amplitude grows, however, the contours no longer simply undulate but overturn, and the wave breaks. **Rossby wave breaking** is therefore a strongly nonlinear and irreversible deformation of the PV contours, analogous to the breaking of ocean waves on a beach [HMR85; Val17; NW10]. During equatorward breaking, tongues of high-PV stratospheric air are drawn equatorward and downward, appearing as a tropopause fold in the tropopause in a vertical cross-section [SWB07; HMR85; Stu15]. These filaments can detach entirely from the main stratospheric reservoir of high-PV air, forming pools of high-PV air surrounded by low-PV air in the troposphere [SWB07].

2.5 Cut-off Lows

When such a filament detaches, the isolated high-PV anomaly induces a closed cyclonic[2] circulation beneath it that is cut off equatorward from the prevailing westerlies: a cold-cored **cut-off low** (COL) [Wor92]. Rossby wave breaking is the dominant formation mechanism: 89% of Southern Hemisphere COLs are linked to RWB events, with most of the remainder linked to PV intrusions in which high-PV air is advected equatorward without the contours overturning [NW10]. With a more generous spatial and temporal search window, 99.3% of COLs are found to be co-located with at least one RWB event [BNL21].

COLs differ widely in vertical extent, and fall largely into two groups. Shallow COLs are confined to the upper troposphere and fade out around the 500 hPa level[3] or higher, whereas deep COLs extend all the way to the surface, and make up 58% of all COLs. Very few COLs reach 700 to 925 hPa without reaching the surface, warranting the pseudo-binary distinction between deep and shallow COLs. Shallow COLs are most common in the subtropics, while deep COLs dominate poleward of 30°S in the extratropics. Over the southern African sector only 49% of COLs reach the surface. Shallow COLs are most frequent in summer [BNL21], when the tropopause is higher [SR06], whereas deep COLs peak in autumn [BNL21].

The distinction matters because only deep COLs are accompanied by a surface cyclonic circulation, adding strong winds, large waves and storm surge to the heavy rainfall that COLs of all depths can produce [BNL21; NW10]. In South Africa, COLs contribute 16% of the total annual rainfall on the Cape south coast, and deep moist convection within them produces short bursts of extreme rainfall responsible for 20% of all flash-flooding events [BNL21].

On its western flank, a deep COL draws cold polar air equatorward [BNL21; NW10; SR07]. This dense air raises the surface pressure and contributes to a ridging high pressure zone behind the low [BNL21]. The tight pressure gradient between the two drives moist air onshore, nearly perpendicular to the coast [Nda+18; Bar+21], against the steep coastal topography of South Africa, where it is forced to rise [SR07]. Combined with the COL's own cyclonic forcing, this uplift can produce extreme rainfall [Bar+21; SR07]. Some COLs are quasi-stationary [NW10], and the topography itself can hold the associated surface low in place [SR07], so that these conditions persist over the same region for several days [BNL21].

The high-PV air at the heart of a COL is a **stratospheric intrusion**: highly stable stratospheric air drawn down sloping isentropic surfaces into the troposphere [HMR85; Stu15; Bar+22]. Stratospheric air has high PV mainly because of its strong static stability [HMR85; GTB10], and in the absence of diabatic and frictional effects it conserves this PV as it descends [Stu15]. Within the intrusion the PV anomaly is realised partly as increased static stability and partly as increased cyclonic vorticity, the balance depending on its shape [HMR85]. Its influence, however, extends beyond the anomaly itself: the circulation it induces penetrates vertically, and beneath it the isentropes bow apart, so that the weakly stratified air below trades static stability for cyclonic rotation [HMR85]. The circulation induced by an intrusion that reaches low enough can extend all the way to the surface [Bar+22], where it advects the latitudinal temperature gradient, creating a warm anomaly ahead of the intrusion that induces a cyclonic circulation of its own [HMR85].

Potential vorticity ties these processes together through the **invertibility principle**: given a suitable balance condition, the distribution of PV on isentropic surfaces together with θ on the lower boundary determines all other dynamical fields (wind, pressure, temperature, static stability) [HMR85]. From this point of view, a COL can be understood as the interaction of two anomalies [BNL21]. The stratospheric intrusion is a cyclonic (in the Southern Hemisphere, negative) upper-level PV anomaly [Bar+22], and a warm surface potential-temperature (θ) anomaly is equivalent to a cyclonic PV anomaly concentrated at the ground [HMR85]. Each induces cyclonic circulation of its own, and when the warm surface anomaly lies just downstream (east) of the upper anomaly, their circulations reinforce one another and the two anomalies amplify together [HMR85; BNL21]. The depth of the stratospheric intrusion is therefore important in determining the depth of the COL, but it is not the whole story: the surface θ anomaly

[2] Clockwise in the Southern Hemisphere and anticlockwise in the Northern Hemisphere; the sense in which geostrophic wind flow circulates around a low pressure zone [Wor92].

[3] Corresponds to an altitude of ≈ 5.5 km [Stu15]

must also be in phase with the upper anomaly. An out-of-phase surface anomaly can inhibit the COL from extending to the surface: only 12% of shallow COLs have an in-phase surface anomaly when they form, compared with 66% of deep COLs [BNL21]. In 53% of deep COLs with a closed surface low (57% over southern Africa), the surface cyclone develops at or after the formation of the COL rather than pre-existing, and the two generally develop almost simultaneously [BNL21]. Once formed, a COL over a warm ocean can generate a surface θ anomaly of its own through surface heat fluxes, coupling the surface to the upper-level anomaly and allowing the system to intensify [SR07; BNL21].

3 Objectives

As discussed above, COLs and Rossby wave breaking are responsible for a substantial share of the extreme weather experienced over and around South Africa. Both are associated with stratospheric intrusions: tongues of high-PV air that extend deep into the troposphere and induce a circulation that can reach the surface.

The primary aim of this project is to quantify how the structure of a stratospheric intrusion, and the background atmosphere in which it is embedded, determine its surface response, and hence its capacity to produce severe weather.

The approach rests on the PV invertibility principle. Inverting the full Ertel PV (equation (7)) is a difficult numerical problem because of its strong nonlinearity. We therefore use the *quasigeostrophic* approximation, in which the inversion reduces to a linear elliptic problem for the geostrophic streamfunction (equation (37)). This is considerably more tractable, though still non-trivial, while retaining the essential dynamics of an intrusion.

The specific objectives of this project are to:

1. recreate the idealised background atmospheric conditions of the control state used by Barnes et Al. [Bar+22]
2. develop and validate a numerical solver for the QGPV inversion problem (equation (37)) for idealised intrusions embedded in a prescribed background state;
3. optimise this solver, both for use on consumer-grade hardware and a high-performance computing (HPC) cluster, so that large parameter sweeps become computationally feasible;
4. assess the limitations of the quasigeostrophic PV framework to determine which results are physically meaningful
5. use systematic parameter sweeps to isolate the effect of individual intrusion and background-state parameters on the surface fields, and to characterise the relationship between derived quantities, such as the minimum height of the dynamical tropopause, and the strength of the surface response;
6. implement an optimisation scheme that automatically searches parameter space for the intrusions and background states that produce the most severe surface response; and
7. integrate these configurations forward in time using the prognostic equation (45) to assess their longevity and evolution.

This text therefore focuses on solutions to equations (37) and (45) and their physical implications. It builds on the PV inversion framework of Fehlmann [Feh97] and Sprenger [Spr07], and on the South African COL work of Barnes et al. [Bar+22].

4 The Quasigeostrophic Potential Vorticity Equation

4.1 Derivation

A summarised derivation from equation (7) to equation (37) is given, noting assumptions and terms dropped along the way. Throughout, synoptic scaling is assumed, with Burger number $\mathrm{Bu} \equiv \left(\overline{N}H/f_0L\right)^2 =$

$\mathcal{O}(1)$ [Val17], and orders of magnitude are stated relative to the leading-order (background) contribution to Q.

The full absolute vorticity in equation (7) is given by [HH13a]:

$$\boldsymbol{\omega} + 2\boldsymbol{\Omega} = \left(\frac{\partial w}{\partial y} - \frac{\partial v}{\partial z}, \quad 2|\Omega| \cos\varphi + \frac{\partial u}{\partial z} - \frac{\partial w}{\partial x}, \quad f + \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) \tag{8}$$

At synoptic scales ($U \sim 10\,\mathrm{m\,s^{-1}}$, $W \sim 10^{-2}\,\mathrm{m\,s^{-1}}$, $L \sim 10^6\,\mathrm{m}$, $H \sim 10^4\,\mathrm{m}$) [HH13a], $|\partial_y w / \partial_z v| \sim |\partial_x w / \partial_z u| \sim (W/U)(H/L) \sim 10^{-5}$, so the horizontal gradients of w may be neglected in the vorticity. Furthermore, $\frac{2|\Omega|\cos(\varphi)\,\partial_y \theta}{f\,\partial_z \theta} \sim \cot\varphi \, \frac{H}{L} \, \frac{\mathrm{Ro}}{\mathrm{Bu}} \sim 10^{-2}\mathrm{Ro}$, allowing the cos term to be discarded as well, consistent with the traditional approximation [HH13a; Val17]. The resulting approximation is:

$$\boldsymbol{\omega} + 2\boldsymbol{\Omega} \approx \left(-\frac{\partial v}{\partial z}, \quad \frac{\partial u}{\partial z}, \quad f + \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) \tag{9}$$

Making a change of variables using the hydrostatic assumption $\partial p / \partial z = -\rho g$ (equation (4)) and substituting into equation (7) yields EPV in pressure coordinates [HMR85], where $\nabla_p = (\partial_x, \partial_y, \partial_p)$ and $\boldsymbol{u} = (u, v, 0)$ is the horizontal velocity. Once hydrostatic balance is assumed this change of variables is exact: the extra terms arising from $(\partial_x)_z \to (\partial_x)_p$ [Val17] cancel identically.

$$\begin{aligned} Q &= -g \left[(f + \zeta_p) \frac{\partial \theta}{\partial p} - \frac{\partial v}{\partial p} \left(\frac{\partial \theta}{\partial x} \right)_p + \frac{\partial u}{\partial p} \left(\frac{\partial \theta}{\partial y} \right)_p \right] \\ &= -g (f \hat{\boldsymbol{k}} + \nabla_p \times \boldsymbol{u}) \cdot \nabla_p \theta \end{aligned} \tag{10}$$

$$\zeta_p \equiv \left(\frac{\partial v}{\partial x} \right)_p - \left(\frac{\partial u}{\partial y} \right)_p \tag{11}$$

It is assumed that θ can be split into two components: a horizontally uniform 'reference potential temperature profile' $\overline{\theta}(p)$, and the potential temperature anomaly θ^* [HMR85; HH13b].

$$\theta(x, y, p) = \overline{\theta}(p) + \theta^*(x, y, p) \tag{12}$$

The anomaly is assumed to be weak in the sense that its vertical gradient is small compared with the reference stratification, $\partial_p \theta^* / (d\overline{\theta}/dp) = \mathcal{O}(\mathrm{Ro})$, which holds for geostrophically balanced anomalies when $\mathrm{Bu} = \mathcal{O}(1)$ [Val17]. The density and geopotential are split in the same way, $\rho = \overline{\rho} + \rho^*$ and $\Phi = \overline{\Phi} + \Phi^*$, and the reference buoyancy frequency $\overline{N}$ is defined from $\overline{\theta}$ by equation (3). Substituting into equation (10) yields the following without further approximation (cf. the analogous height-coordinate expansion of [Val17]):

$$\begin{aligned} Q &= -g \Bigg[\underbrace{f \frac{d\overline{\theta}}{dp}}_{\mathcal{O}(1)} + \underbrace{\zeta_p \frac{d\overline{\theta}}{dp} + f \frac{\partial \theta^*}{\partial p}}_{\mathcal{O}(\mathrm{Ro})} + \underbrace{\zeta_p \frac{\partial \theta^*}{\partial p} - \frac{\partial v}{\partial p} \frac{\partial \theta^*}{\partial x} + \frac{\partial u}{\partial p} \frac{\partial \theta^*}{\partial y}}_{\mathcal{O}(\mathrm{Ro}^2)} \Bigg] \\ &= -g \frac{d\overline{\theta}}{dp} \left[f + \zeta_p + \frac{f}{\partial_p \overline{\theta}} \frac{\partial \theta^*}{\partial p} \right] + \mathcal{O}\left(\mathrm{Ro}^2\right) \end{aligned} \tag{13}$$

The term $\zeta_p \partial_p \theta^*$ is the product of two $\mathcal{O}(\mathrm{Ro})$ quantities. The last two terms can be shown to be $\mathcal{O}\left(\mathrm{Ro}^2\right)$ using thermal wind balance [Val17]. To do so we introduce the function $\mathcal{R}(p)$, where $\Pi(p) = c_p \left(p/p_0\right)^{R/c_p}$ is the Exner function [HMR85]:

$$\mathcal{R}(p) = \frac{d\Pi}{dp} = \frac{R\Pi(p)}{c_p p} = \frac{RT}{p\theta} = \frac{1}{\rho\theta} \tag{14}$$

In pressure coordinates the full geopotential is defined as $\Phi(x, y, p) = g \cdot z(x, y, p)$. Combining $d\Phi = g\,dz$, equation (4) and equation (14) gives hydrostatic equilibrium in terms of the geopotential [HH13b; HMR85]:

$$\frac{\partial \Phi}{\partial p} = -\frac{1}{\rho} = -\mathcal{R}\theta \tag{15}$$

Since $\mathcal{R}$ depends only on p, this holds separately for the reference state and for the anomalies, $\partial_p \Phi^* = -\mathcal{R}\theta^*$ [HMR85]. Combining with geostrophic balance, $(u_g, v_g) = f_0^{-1}\left(-\partial_y \Phi^*, \partial_x \Phi^*\right)$ [HH13b], gives the thermal wind relations $f_0 \partial_p v_g = -\mathcal{R}\,\partial_x \theta^*$ and $f_0 \partial_p u_g = \mathcal{R}\,\partial_y \theta^*$ [HH13b], so that

$$-\frac{\partial v}{\partial p}\frac{\partial \theta^*}{\partial x} + \frac{\partial u}{\partial p}\frac{\partial \theta^*}{\partial y} \approx \frac{f_0}{\mathcal{R}}\left|\frac{\partial \boldsymbol{u}_g}{\partial p}\right|^2 \tag{16}$$

Relative to the leading term $f_0\, d\overline{\theta}/dp$, and using equation (35) below, this has magnitude [Val17]

$$\frac{|\partial_p \boldsymbol{u}_g|^2}{|\mathcal{R}\, d\overline{\theta}/dp|} = \frac{|\partial_z \boldsymbol{u}_g|^2}{\overline{N}^2} = \mathrm{Ri}^{-1} \sim \frac{\mathrm{Ro}^2}{\mathrm{Bu}} \tag{17}$$

Invoking the β-plane approximation for the Coriolis parameter $f \approx f_0 + \beta y$ [HH13b] allows a further $\mathcal{O}\left(\mathrm{Ro}^2\right)$ term to be separated out using the assumption that $\beta L/f_0 \in \mathcal{O}\left(\mathrm{Ro}\right)$ [Val17].

$$Q = -g\frac{d\overline{\theta}}{dp}\left[f + \zeta_p + \frac{f_0}{\partial_p \overline{\theta}}\frac{\partial \theta^*}{\partial p}\right] - \underbrace{g\beta y \frac{\partial \theta^*}{\partial p}}_{\mathcal{O}(\mathrm{Ro}^2)} + \mathcal{O}\left(\mathrm{Ro}^2\right) \tag{18}$$

giving the further approximation

$$Q \approx A(p)B, \qquad A(p) \equiv -g\frac{d\overline{\theta}}{dp} > 0, \qquad B \equiv f + \zeta_p + f_0 \frac{\partial \theta^*}{\partial p}\left(\frac{d\overline{\theta}}{dp}\right)^{-1} \tag{19}$$

where $A > 0$ for a statically stable reference state [HH13a].

The invertibility principle concerns the distribution of PV on surfaces of constant θ (isentropic surfaces) [HMR85], so we make the change of variables of the height coordinate from p to θ, without approximation, using the exact formula [Val17]

$$\left(\frac{\partial Q}{\partial x}\right)_\theta = \left(\frac{\partial Q}{\partial x}\right)_p + \frac{\partial Q}{\partial p}\left(\frac{\partial p}{\partial x}\right)_\theta \tag{20}$$

First we calculate $(\partial_x p)_\theta$ on the isentrope $(\partial_x \theta)_\theta = 0$.

$$0 = \left(\frac{\partial \theta^*}{\partial x}\right)_p + \left(\frac{d\overline{\theta}}{dp} + \frac{\partial \theta^*}{\partial p}\right)\left(\frac{\partial p}{\partial x}\right)_\theta \tag{21}$$

Let Δp_H be the depth (in pressure) of the anomaly and Δp_R the depth over which the reference stratification changes. Since $\partial_p \theta^* / \partial_p \overline{\theta} = \mathcal{O}(\mathrm{Ro})$, we have $\theta^*/\partial_p \overline{\theta} \sim \mathrm{Ro}\,\Delta p_H$, so the slope of the isentropes is $\mathcal{O}(\mathrm{Ro}\,\Delta p_H/L)$ [Val17]:

$$\left(\frac{\partial p}{\partial x}\right)_\theta = -\frac{(\partial_x \theta^*)_p}{\partial_p \overline{\theta} + \partial_p \theta^*} = -\underbrace{\frac{1}{\partial_p \overline{\theta}}\left(\frac{\partial \theta^*}{\partial x}\right)_p}_{\mathcal{O}(\mathrm{Ro}\,\Delta p_H/L)}\left[1 + \mathcal{O}(\mathrm{Ro})\right] \tag{22}$$

The $\mathcal{O}(\mathrm{Ro})$ relative correction in square brackets may be dropped. We write

$$b \equiv \frac{B - f_0}{f_0} = \frac{\beta y}{f_0} + \frac{\zeta_p}{f_0} + \frac{\partial_p \theta^*}{\partial_p \overline{\theta}} = \mathcal{O}(\mathrm{Ro}) \tag{23}$$

And hence $B \equiv f_0(1+b) = f_0(1 + \mathcal{O}(\mathrm{Ro}))$. Apply the product rule:

$$\frac{\partial Q}{\partial p} = A'(p)B + A(p)\frac{\partial B}{\partial p} = \underbrace{f_0 A'}_{\boxed{1}} + \underbrace{f_0 A' b}_{\boxed{2}} + \underbrace{A\frac{\partial B}{\partial p}}_{\boxed{3}} \tag{24}$$

We have $\partial_p B \sim \mathrm{Ro}\, f_0/\Delta p_H$ and $A' \sim A/\Delta p_R$. Each term enters equation (20) multiplied by $(\partial_x p)_\theta \sim \mathrm{Ro}\,\Delta p_H/L$, and must be compared with the leading term $(\partial_x Q)_p = A\,\partial_x B = A f_0\,\partial_x b \sim A f_0 \mathrm{Ro}/L$:

$$\begin{aligned}
\boxed{1}\frac{(\partial_x p)_\theta}{A\,\partial_x B} &\sim \frac{f_0 A}{\Delta p_R}\,\frac{\mathrm{Ro}\,\Delta p_H}{L}\,\frac{L}{A f_0 \mathrm{Ro}} = \frac{\Delta p_H}{\Delta p_R} \\
\boxed{2}\frac{(\partial_x p)_\theta}{A\,\partial_x B} &\sim \mathrm{Ro}\,\frac{\Delta p_H}{\Delta p_R} \\
\boxed{3}\frac{(\partial_x p)_\theta}{A\,\partial_x B} &\sim A\frac{\mathrm{Ro}\, f_0}{\Delta p_H}\,\frac{\mathrm{Ro}\,\Delta p_H}{L}\,\frac{L}{A f_0 \mathrm{Ro}} = \mathrm{Ro}
\end{aligned} \tag{25}$$

Terms $\boxed{2}$ and $\boxed{3}$ are therefore $\mathcal{O}(\mathrm{Ro})$ relative to the leading term, i.e. $\mathcal{O}\left(\mathrm{Ro}^2\right)$ overall, for anomalies of any depth, and are dropped. Term $\boxed{1}$ is retained: it is comparable with the leading term when the anomaly depth is comparable with the scale of the reference stratification, and becomes negligible only for shallow anomalies ($\Delta p_H \ll \Delta p_R$), in which case the result below reduces to its Boussinesq form [HMR85; HH13b]. To the order required,

$$\frac{\partial Q}{\partial p} \approx f_0 A' = -g f_0 \frac{d^2\overline{\theta}}{dp^2} \tag{26}$$

Combining with equation (22) yields:

$$\begin{aligned}
\frac{\partial Q}{\partial p}\left(\frac{\partial p}{\partial x}\right)_\theta &\approx \left(-g f_0 \frac{d^2\overline{\theta}}{dp^2}\right)\left[-\frac{1}{\partial_p \overline{\theta}}\left(\frac{\partial \theta^*}{\partial x}\right)_p\right] \\
&= \frac{g f_0 \partial_p^2 \overline{\theta}}{\partial_p \overline{\theta}}\frac{\partial \theta^*}{\partial x} = A(p)\left[-\frac{f_0 \partial_p^2 \overline{\theta}}{\left(\partial_p \overline{\theta}\right)^2}\frac{\partial \theta^*}{\partial x}\right]
\end{aligned} \tag{27}$$

Because A only depends on p,

$$\left(\frac{\partial Q}{\partial x}\right)_p = A(p)\left(\frac{\partial B}{\partial x}\right)_p \tag{28}$$

Substituting into equation (20) gives:

$$\left(\frac{\partial Q}{\partial x}\right)_\theta \approx A(p)\left(\frac{\partial}{\partial x}\right)_p \left[f + \zeta_p + \underbrace{f_0\frac{\partial_p \theta^*}{\partial_p \overline{\theta}} - \frac{f_0 \theta^* \partial_p^2 \overline{\theta}}{\left(\partial_p \overline{\theta}\right)^2}}_{f_0 \frac{\partial}{\partial p}\left(\frac{\theta^*}{\partial_p \overline{\theta}}\right)}\right] \tag{29}$$

Applying the same process in the y direction (where $\partial_y f = \beta$ is retained), we define the absolute quasigeostrophic potential vorticity $\tilde{q}$ in pressure coordinates as follows, relating to Q by equation (31) [HMR85].

$$\tilde{q} \equiv f + \zeta_p + f_0 \frac{\partial}{\partial p}\left[\theta^*\left(\frac{d\overline{\theta}}{dp}\right)^{-1}\right] \tag{30}$$

$$\nabla_\theta Q \approx -g\frac{d\overline{\theta}}{dp}\nabla_h \tilde{q} \tag{31}$$

Here $\nabla_\theta \equiv ((\partial_x)_\theta, (\partial_y)_\theta)$ and $\nabla_h \equiv ((\partial_x)_p, (\partial_y)_p)$ are the horizontal gradients along isentropic and isobaric surfaces respectively [HMR85]. The relation holds only for these horizontal components: the vertical derivative $\partial Q/\partial p$ is dominated by the background PV gradient $f_0 A'(p)$, which has no counterpart in $\partial \tilde{q}/\partial p$.

We define the stream function as $\psi = \Phi^*/f_0$, a scaling of the geopotential anomaly Φ^*, so that the geostrophic wind is $(u_g, v_g) = (-\partial_y \psi, \partial_x \psi)$ [HMR85; Val17]. Applying equation (15) to the anomaly fields and rearranging we get [HMR85]

$$\theta^* = -\frac{f_0}{\mathcal{R}}\frac{\partial \psi}{\partial p} \tag{32}$$

Approximating the relative vorticity by its geostrophic value, $\zeta_p \approx \zeta_g = \nabla_h^2 \psi$ (equation (5)), which introduces a relative error of $\mathcal{O}(\mathrm{Ro})$ [HH13b, p. 181], and substituting equation (32) into equation (30) gives us QGPV in terms of the stream function [HMR85; HH13b]:

$$\tilde{q} = f + \nabla_h^2 \psi + f_0^2 \frac{\partial}{\partial p}\left(\frac{-1}{\mathcal{R}\partial_p \overline{\theta}}\frac{\partial \psi}{\partial p}\right) \tag{33}$$

We apply hydrostatic equilibrium (equation (4)) once more using the reference density $\overline{\rho}(z)$[4]:

$$\frac{d\overline{\theta}}{dp} = \frac{d\overline{\theta}}{dz}\frac{dz}{dp} = -\frac{1}{\overline{\rho} g}\frac{d\overline{\theta}}{dz} \tag{34}$$

Combine with equation (3) and equation (14) [HMR85]:

$$-\mathcal{R}\frac{d\overline{\theta}}{dp} = \frac{1}{g^2\overline{\rho}^2}\frac{g}{\overline{\theta}}\frac{d\overline{\theta}}{dz} = \frac{\overline{N}^2}{g^2\overline{\rho}^2} \tag{35}$$

And finally substitute into the last term of equation (33):

$$\begin{aligned} f_0^2 \frac{\partial}{\partial p}\left(\frac{-1}{\mathcal{R}\partial_p \overline{\theta}}\frac{\partial \psi}{\partial p}\right) &= f_0^2 \frac{\partial}{\partial p}\left(\frac{g^2\overline{\rho}^2}{\overline{N}^2}\frac{\partial \psi}{\partial p}\right) \\ &= f_0^2 \left(\frac{-1}{g\overline{\rho}}\frac{\partial}{\partial z}\right)\left[\frac{g^2\overline{\rho}^2}{\overline{N}^2}\left(\frac{-1}{g\overline{\rho}}\frac{\partial \psi}{\partial z}\right)\right] \\ &= \frac{f_0^2}{\overline{\rho}}\frac{\partial}{\partial z}\left(\frac{\overline{\rho}}{\overline{N}^2}\frac{\partial \psi}{\partial z}\right) \end{aligned} \tag{36}$$

Now we can at last put equation (33) and equation (36) together and define the relative quasi-geostrophic potential vorticity $q \equiv \tilde{q} - f$ [Spr07] to yield the main equation of study by Barnes et al. [Bar+22] (cf. [Val17]):

$$\boxed{q(x,y,z,t) = \frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} + \frac{f_0^2}{\overline{\rho}(z)}\frac{\partial}{\partial z}\left(\frac{\overline{\rho}(z)}{\overline{N}(z)^2}\frac{\partial \psi}{\partial z}\right)} \tag{37}$$

We impose Neumann horizontal boundary conditions [Bar+22; Spr07], which follow from the geostrophic relations $u = -\psi_y$ and $v = \psi_x$ [Bar+22], and which prescribe the boundary-parallel (tangential) wind on each lateral wall:

$$\boxed{\begin{aligned} \psi_x(0,y,z) &= \psi_x(L_x,y,z) = \overline{v}(y,z) \\ \psi_y(x,0,z) &= \psi_y(x,L_y,z) = -\overline{u}(x,z) \end{aligned}} \tag{38}$$

Changing equation (32) to height coordinates, and using the background state of equation (14) ($\mathcal{R} = 1/\left(\overline{\rho}\overline{\theta}\right)$) yields

$$\frac{f_0}{g\overline{\rho}}\frac{\partial \psi}{\partial z} = \mathcal{R}\theta^* \quad \Longrightarrow \quad \frac{\partial \psi}{\partial z} = \frac{g\theta^*}{f_0\overline{\theta}} \tag{39}$$

Which we use for our vertical boundary conditions [Bar+22]:

$$\boxed{\begin{aligned} \psi_z(x,y,0) &= \frac{g\theta^*}{f\overline{\theta}(0)} \\ \psi_z(x,y,H) &= \frac{g\theta^*}{f\overline{\theta}\,(H)} \end{aligned}} \tag{40}$$

[4] This is a simplification that makes the derivation more compact, but not as correct as that of Hoskins et al. In the full derivation a new density ρ_s and potential temperature θ_0 are introduced, where $\rho_s/\overline{\rho} = \overline{\theta}/\theta_0$. In addition z becomes a pseudo-height coordinate defined by $z = g^{-1}\theta_0\,[\Pi(p_0) - \Pi(p)]$, as opposed to the true geometric height [HMR85].

To derive the prognostic equation for q, we change equation (7) into isentropic coordinates using the same approach as equation (20):

$$\frac{DQ}{Dt} = \left(\frac{\partial Q}{\partial t}\right)_\theta + \frac{Dx}{Dt}\left(\frac{\partial Q}{\partial x}\right)_\theta + \frac{Dy}{Dt}\left(\frac{\partial Q}{\partial y}\right)_\theta + \frac{D\theta}{Dt}\frac{\partial Q}{\partial t} = 0 \tag{41}$$

It follows from the definition of D/Dt that $Dx/Dt = u$, and $Dy/Dt = v$. Reapplying our assumption of adiabatic flow (equation (2)) yields $D\theta/Dt = 0$, simplifying the equation to

$$\left(\frac{\partial Q}{\partial t}\right)_\theta + \boldsymbol{u} \cdot \nabla_\theta Q = 0 \tag{42}$$

Reusing the logic from equation (20) through to equation (29) with $x \to t$ gives

$$\left(\frac{\partial Q}{\partial t}\right)_\theta \approx A(p)\left(\frac{\partial \tilde{q}}{\partial t}\right)_p \tag{43}$$

Substituting into equation (42) along with equation (31), and dividing by $A(p)$ yields

$$\frac{\partial \tilde{q}}{\partial t} + \boldsymbol{u} \cdot \nabla_h \tilde{q} = \frac{D\tilde{q}}{Dt} \approx 0 \tag{44}$$

For $\beta = 0$ this reduces to the final nonlinear transport form of the prognostic equation for q:

$$\boxed{\frac{\partial q}{\partial t} - \frac{\partial \psi}{\partial y}\frac{\partial q}{\partial x} + \frac{\partial \psi}{\partial x}\frac{\partial q}{\partial y} = 0} \tag{45}$$

4.2 Background States and Coefficients

We aim to recreate the control state of the atmosphere used by Barnes et al. to use as the benchmark for the diagnostic solver (section 5). In order to capture larger jet streams in section TODO, we use an increased vertical domain of $H = 25\,\text{km}$, as opposed to the $H = 20\,\text{km}$ used by Barnes et al. [Bar+22]. The same f as Barnes et al. is used with $\varphi = -42°$ and $\beta = 0$. A horizontal domain size of $L_x = L_y = 7500\,\text{km}$ is set.

We will specify $\overline{N}(z)$ using the same values as Barnes et al. [Bar+22], from which we can calculate $\overline{\rho}(z)$ using the same methodology as Fehlmann [Feh97].

Barnes et al. specify a stability profile such that $\overline{N}_{\text{strat}} = 0.03\,\text{s}^{-1}$ in the stratosphere and $\overline{N}_{\text{trop}} = 0.01\,\text{s}^{-1}$ in the troposphere [Bar+22]. This discontinuity at the tropopause is smoothed out by the following family of kink functions [Feh97]:

$$\kappa_\delta(x) = \begin{cases} 0 & x \le 0 \\ \frac{1}{2}(2x)^\delta & 0 < x \le \frac{1}{2} \\ 1 - \frac{1}{2}[2(1-x)]^\delta & \frac{1}{2} < x \le 1 \\ 1 & x > 1 \end{cases} \tag{46}$$

where $\delta > 1$ is a parameter which determines the 'sharpness' of the kink. We can then rescale and translate κ_δ to specify our reference Brunt–Väisälä frequency profile, where z_{trop} is the altitude of the centre of the tropopause and w_{trop} is the total vertical width of the smoothing zone. Initially we will take $\delta = 2$, $z_{\text{trop}} = 12\,500\,\text{m}$ [Bar+22] and $w_{\text{trop}} = 1000\,\text{m}$ [Feh97].

$$\overline{N}(z) = \left(\overline{N}_{\text{strat}} - \overline{N}_{\text{trop}}\right)\kappa_\delta\left(\frac{z - z_{\text{trop}}}{w_{\text{trop}}} + \frac{1}{2}\right) + \overline{N}_{\text{trop}} \tag{47}$$

We can then determine the reference potential temperature profile by solving for $\overline{\theta}$ in the definition of the Brunt–Väisälä frequency from equation (3),

$$\overline{\theta}(z) = \overline{\theta}(0)\exp\left(\frac{1}{g}\int_0^z \overline{N}(\xi)^2\,d\xi\right) \tag{48}$$

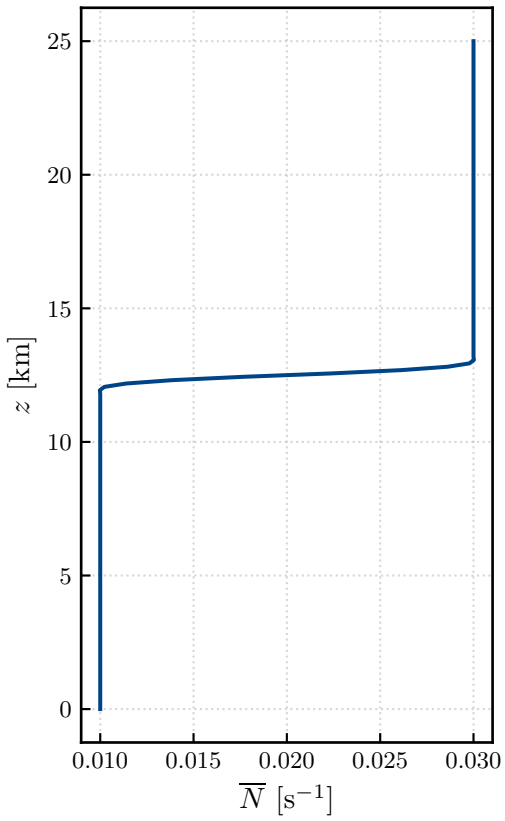


Figure 1: Reference Brunt–Väisälä frequency – equation (47)

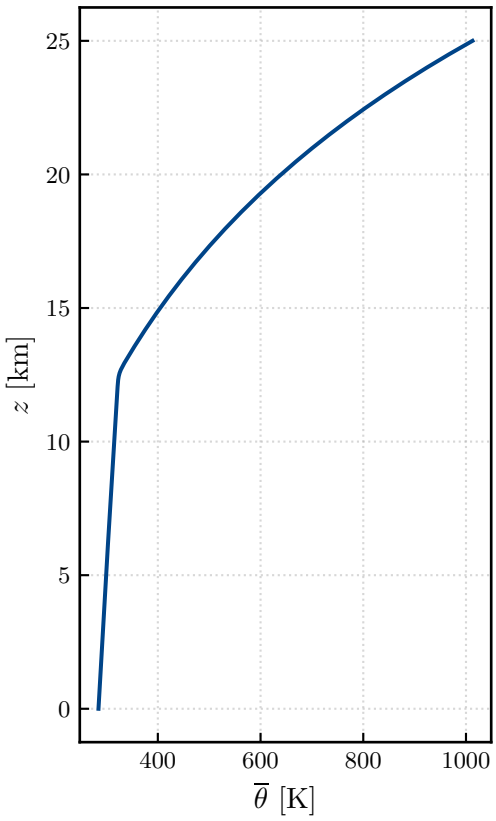


Figure 2: Reference potential temperature profile – equation (48)

The unrealistically large $\overline{N}_{\text{strat}} = 0.03\,\text{s}^{-1}$ used by Barnes et al. causes the potential temperature to take on unrealistic values above the tropopause. At 42°S a typical value would be $\overline{\theta}(20\,\text{km}) \approx 480\,\text{K}$ [Bir06], however figure 2 shows a $\overline{\theta} > 600\,\text{K}$ at the same altitude. This could be fixed by using a more realistic $\overline{N}_{\text{strat}} = 0.022\,\text{s}^{-1}$ [GTB10], but it will be left as-is to maintain parity with Barnes et al. for the control state.

We combine the ideal gas law $p = \rho RT$ and the the definition of θ (equation (1)) to express the reference density profile, where $\kappa = R/c_p$:

$$\overline{\rho}(z) = \frac{\overline{p}}{R\overline{\theta}} \left(\frac{p_{\text{surf}}}{\overline{p}} \right)^{\kappa} \tag{49}$$

To determine the pressure profile $\overline{p}(z)$ we use the hydrostatic assumption (equation (4)), substitute in equation (49), and solve the ODE to yield [Feh97].

$$\overline{p}(z) = \overline{p}(0) \left(1 - \frac{\kappa g}{R} \left(\frac{p_{\text{surf}}}{\overline{p}(0)} \right)^{\kappa} \int_0^z \frac{d\xi}{\overline{\theta}(\xi)} \right)^{1/\kappa} \tag{50}$$

The approximation $\overline{p}(0) = p_{\text{surf}} = 1000\,\text{hPa}$ will be made for simplicity. It can be numerically calculated that the relative error between ψ when $\overline{p}(0) = 1000\,\text{hPa}$ and $\overline{p}(0) = 800\,\text{hPa}$ is ~ 0.01, which justifies this approximation, and the decision to treat $\overline{p}(0)$ as constant across varying parameter sets.

4.3 Boundary Conditions

To specify the horizontal boundary conditions, we need to choose values of u and v. We will define a jet stream aiming to recreate the one shown in [Bar+22]. The exact function used is not given, but we can specify a function that also has a maximum magnitude of $u_{\text{jet}} = 35\,\text{m}\,\text{s}^{-1}$, is centred about the middle of the tropopause vertically, is centred about the middle of the domain in the y-direction ($y_{\text{jet}} = L_y/2$),

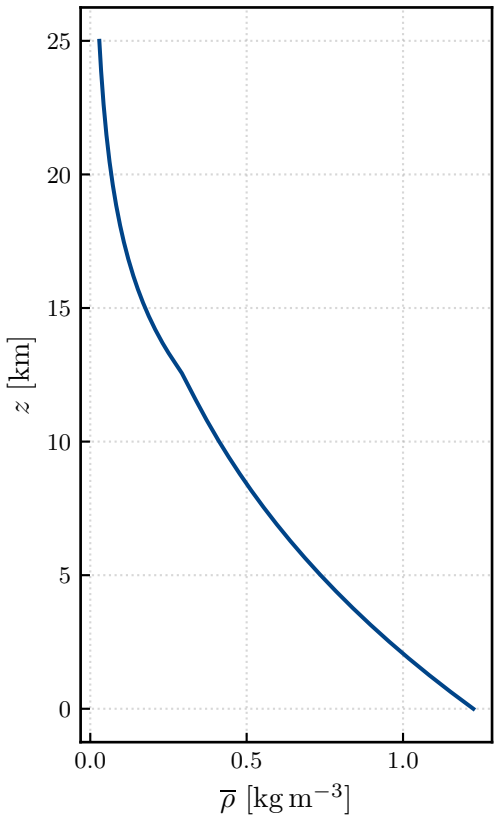


Figure 3: Reference density profile – equation (49)

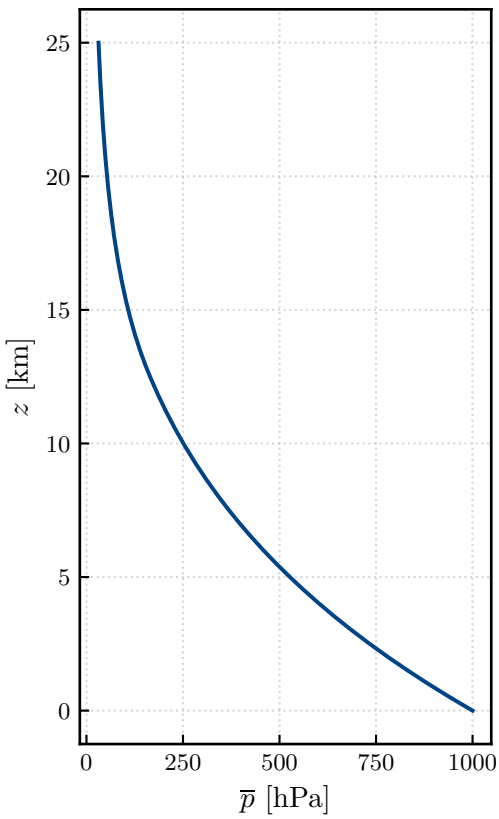


Figure 4: Reference pressure profile – equation (50)

is uniform in the y-direction, has a Gaussian shape, and has a similar overall size to the one shown in [Bar+22] ($L_{\text{jet}} \approx 500\,\text{km}$ and $z_{\text{jet}} \approx 4\,\text{km}$). We will let

$$\overline{u}(y,z) = U_{\text{jet}} \exp\left[-\left(\frac{y-y_{\text{jet}}}{L_{\text{jet}}}\right)^2 - \left(\frac{z-z_{\text{trop}}}{z_{\text{jet}}}\right)^2\right] \tag{51}$$

$$\overline{v} = 0 \tag{52}$$

With regards to the vertical boundary conditions (equation (40)), the potential temperature anomaly θ^* will be calculated in section 5.5 to ensure satisfaction of the so-called *Neumann solvability condition*.

4.4 Source Term

To calculate the source term q we must first specify the background state $\overline{q}$ and the QGPV associated with the anomaly, q^*. We integrate equations (51) and (52) using $\overline{u} = -\overline{\psi}_y$ and $\overline{v} = \overline{\psi}_x$ to calculate the background stream function $\overline{\psi}$:

$$\overline{\psi}(y,z) = -\frac{\sqrt{\pi}}{2} U_{\text{jet}}\, L_{\text{jet}}\, \text{erf}\left(\frac{y-y_{\text{jet}}}{L_{\text{jet}}}\right) \exp\left[-\left(\frac{z-z_{\text{trop}}}{z_{\text{jet}}}\right)^2\right] \tag{53}$$

The background QGPV $\overline{q}$ can then be calculated using equation (37), with $\overline{\zeta_g} = \nabla_h^2 \overline{\psi}$ (equation (5)):

$$\overline{q} = \overline{\zeta_g} + \frac{f_0^2}{\overline{\rho}} \frac{\partial}{\partial z}\left(\frac{\overline{\rho}(z)}{\overline{N}^2} \frac{\partial \overline{\psi}}{\partial z}\right) \tag{54}$$

Barnes et al. specify the anomaly in EPV, which has different units to QGPV, so the following approximation is used to calculate $\overline{Q}$ [Feh97]:

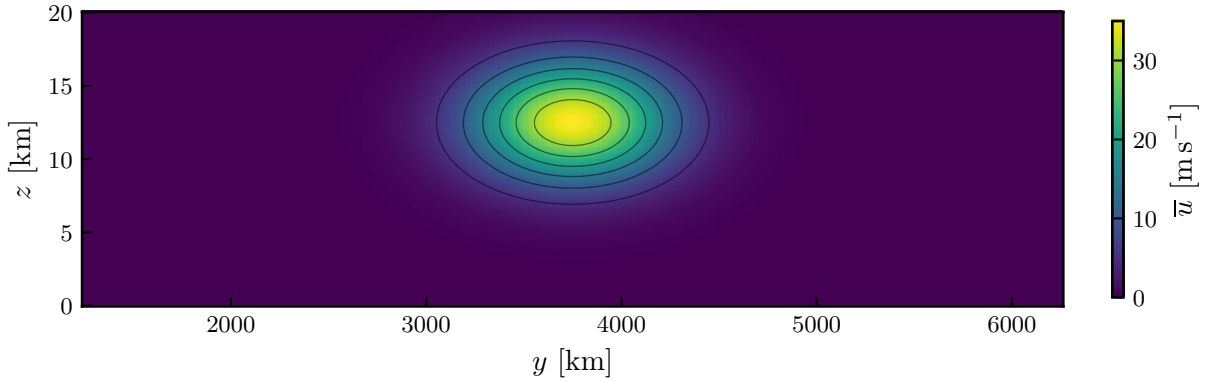


Figure 5: Jet stream specified in equation (51). A contour interval of $5\,\mathrm{m\,s^{-1}}$ is used.

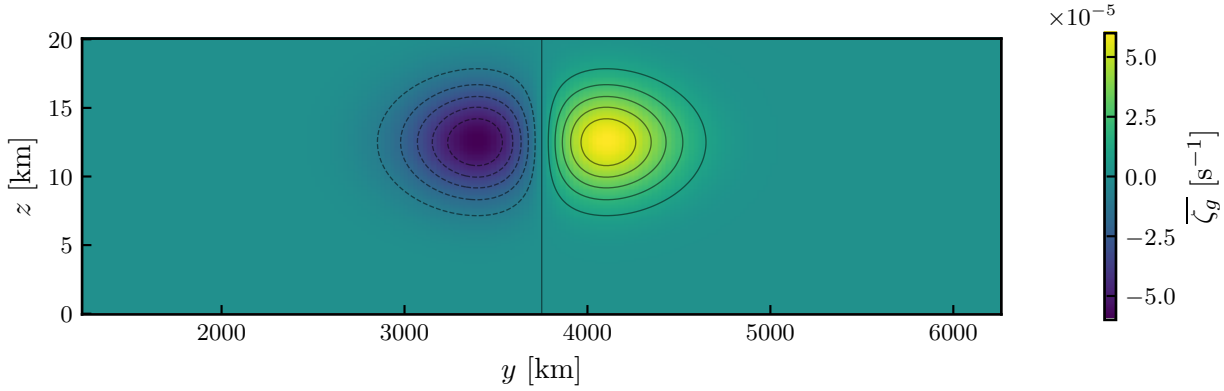


Figure 6: Background geostrophic vorticity calculated from equation (53) using equation (5)

$$q \approx \frac{\overline{\rho} g}{\overline{N}^2 \overline{\theta}} Q - f \tag{55}$$

$$\overline{Q} \approx \frac{(\overline{q} + f)\,\overline{N}^2 \overline{\theta}}{\overline{\rho} g} \tag{56}$$

We specify the same PV anomaly as the one used by Barnes et Al. [Bar+22] bar the apparent typographical errors. We take $Q^*_{\mathrm{mag}} = -4 \times 10^{-6}\,\mathrm{K\,m^2\,s^{-1}\,kg^{-1}}$, and $Q^*_{\mathrm{clip}} = -1.5 \times 10^{-6}\,\mathrm{K\,m^2\,s^{-1}\,kg^{-1}}$. The control anomaly will be placed at the centre of the domain ($x_{\mathrm{ano}} = L_x/2$ and $y_{\mathrm{ano}} = L_y/2$). The anomaly sizes $x_{\mathrm{size}} = y_{\mathrm{size}} = 200\,\mathrm{km}$ and $z_{\mathrm{size}} = 2000\,\mathrm{m}$ will be used [Bar+22].

$$Q^*(x,y,z) = \mathrm{softmax}_\tau \left\{ Q^*_{\mathrm{clip}},\ Q^*_{\mathrm{mag}} \exp\left[-\left(\frac{x - x_{\mathrm{ano}}}{x_{\mathrm{size}}}\right)^2 - \left(\frac{y - y_{\mathrm{ano}}}{y_{\mathrm{size}}}\right)^2 - \left(\frac{z - z_{\mathrm{trop}}}{z_{\mathrm{size}}}\right)^2 \right] \right\} \tag{57}$$

A $\mathrm{softmax}_\tau\{\}$ function is used in place of the $\max\{\}$ function, so as to improve numerical conditioning. $\mathrm{softmax}_\tau$ is C^∞ – infinitely smooth for $\tau > 0$. The smoothing value chosen was $\tau = 0.075|Q^*_{\mathrm{mag}}|$.

$$\mathrm{softmax}_\tau\{a,b\} \equiv \max\{a,b\} + \tau \ln\left[1 + \exp\left(-\frac{|a-b|}{\tau}\right)\right] \tag{58}$$

EPV inversion comes with the condition that $fQ > 0$ everywhere in the domain; a necessity to ensure equation (7) remains elliptic and not ill-posed [HMR85]. Equation (37) is elliptic and linear, so the inversion will not fail for q that corresponds to $fQ \leq 0$. QGPV is however an approximation of EPV in this case [Feh97], so the $fQ > 0$ constraint must be checked manually. As is seen in figure 7, $Q = \overline{Q} + Q^*$ has regions where $fQ \leq 0$, implying an EPV inversion over this atmospheric state would

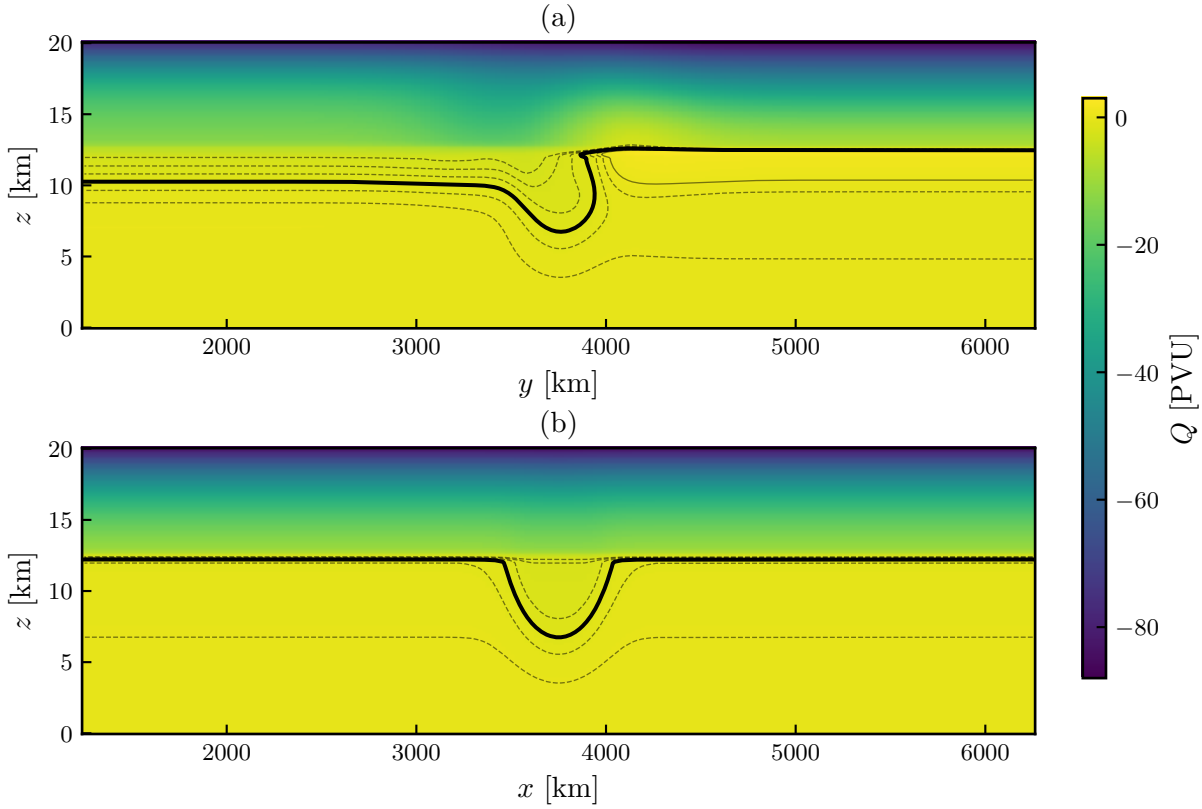


Figure 7: EPV distribution specified in equations (56) and (57). A contour interval of 0.5 PVU (Potential Vorticity Units) is used. $1\,\mathrm{PVU} = 10^{-6}\,\mathrm{K\,m^2\,s^{-1}\,kg^{-1}}$. The thick solid line represents the dynamical tropopause – the $-1.5\,\mathrm{PVU}$ contour. Thin solid lines represent contours where $Q > 0$, indicating a non-physical background state.

fail. No explanation is offered as to why Barnes et al. chose a background state that has this feature. It must however be acknowledged that Barnes et al. did not publish all information necessary to recreate their background state exactly[5], so whether this holds is unknown [Bar+22].

Nevertheless, we establish a metric of the volume of 'invalid EPV' compared to the total domain volume as a diagnostic:

$$\mathcal{V}_- \equiv \frac{1}{L_x L_y H} \int_V \vartheta(-fQ)\, dV, \qquad \vartheta(x) \equiv \begin{cases} 1 & x > 0 \\ 0 & x < 0 \end{cases} \tag{59}$$

For a truly correct background state, we need $\mathcal{V}_- = 0$, however the control state has $\mathcal{V}_- \sim 0.04$. To maintain parity with Barnes et al., we will use this background state for now.

Numerical experimentation showed that $\mathcal{V}_-$ is driven largely by $\partial\overline{u}/\partial z$, meaning that the most effective way to send $\mathcal{V}_- \to 0$ is by decreasing U_{jet}, or increasing z_{jet}.

Despite these issues, we arrive at the expression for the source term of equation (37) by using approximation (55) again:

$$q \approx \frac{\overline{\rho} g}{\overline{N}^2 \overline{\theta}} \left(\overline{Q} + Q^*\right) - f \tag{60}$$

5 Numerical QGPV Inversion

By adopting a smoothing value of $\delta = 2$ for equation (46), the integral in equation (48) is of the following general form [Wol26]:

[5] Quantities used but not explicitly given by Barnes et al. include but are not limited to w_{trop}, δ and L_{jet} [Bar+22].

$$\int e^{z^4}\,dz = -\frac{z\Gamma\left(\frac{1}{4}, -z^4\right)}{4\sqrt[4]{-z^4}} + C \tag{61}$$

Because this integral lacks a closed-form solution, the coefficients (equation (49)) and the source term (equation (60)) cannot be readily expressed analytically. Consequently, numerical integration is required, for the vertical integrals in equations (48) and (50), as well as solving for ψ in equation (37). The solver set-up for equation (37) is given in the following subsections.

5.1 Choice of Solver Framework

While spectral methods offer exponential convergence and efficient memory usage, they trigger non-physical oscillations, known as the Gibbs phenomenon, when applied to localised functions (such as equation (60)) or kink functions (such as equation (47)) [Tre00]. Finite difference schemes are straightforward to implement, but require discretising the Neumann boundary conditions (equations (38) and (40)) [Feh97]. Furthermore, although finite volume methods excel at tracking fluxes in hyperbolic conservation laws (such as equation (45)), scaling to higher-order accuracy requires implementing more complex schemes [Dur10]. To overcome these limitations, we employ a finite element method (FEM). FEM handles Neumann boundary conditions in the weak form as a boundary integral [Seg11], effortlessly scales to higher-order accuracy and supports formulations (such as Discontinuous Galerkin) that strictly enforce flux conservation [Ham+23].

This work will use the Python library Firedrake as the computational framework to build up the numerical solver [Rat+16]. It allows a high level specification of PDEs, automatic highly optimised C code generation and automatic Jacobian computation. These features significantly reduce development time, ensuring the project remains within the scope of an honours thesis. Furthermore, Firedrake provides access to some of the most efficient matrix solvers and preconditioners, such as conjugate gradient and polynomial multigrid, by simply changing a few lines in the solver parameters [Ham+23; Bal+25; Bal+97; Dal+11].

Unfortunately Firedrake does not offer a simple way to evaluate purely vertical integrals (such as equations (48) and (50)) efficiently, so a non-trivial alternative method is explained in full in section A.2.

5.2 Building the Mesh

To optimize mesh generation, a two-dimensional base mesh is extruded into a three-dimensional domain [McR+16; Ber+16]. This extrusion methodology was found to significantly reduce the computational cost of mesh creation compared to direct three-dimensional mesh initialisation. The corresponding implementation is detailed in section A.1.

By restricting the base mesh to quadrilateral elements [HH16], the vertical extrusion produces a mesh composed exclusively of hexahedral cells. This hexahedral topology facilitates the construction of tensor-product function spaces across the elements [McR+16]. Consequently, during form assembly and the application of matrix-free operators [KM18], the framework inherently leverages sum factorisation [Ham+23; HKH17]. This technique decomposes multi-dimensional integrals into a sequence of one-dimensional operations, thereby reducing the per-cell computational complexity from $\mathcal{O}(p^6)$ to $\mathcal{O}(p^4)$ for matrix-free evaluations [HKH17; KM18], and from $\mathcal{O}(p^9)$ to $\mathcal{O}(p^7)$ for explicit sparse matrix assembly [HKH17], where p denotes the polynomial degree of the function space.

It should be noted that the asymptotic complexity advantages conferred by sum factorisation during matrix-free matrix-vector multiplications do not fully translate to assembled matrix methods. For explicitly assembled matrices, these performance improvements are strictly confined to the assembly phase [HKH17; KM18].

5.3 Defining the Function Space

Firedrake offers a large number of supported finite elements, each with its own advantages and disadvantages [Ham+23]. Since the unknown stream function is a scalar field and the extruded mesh consists of hexahedral cells, the vector- and tensor-valued elements, as well as those defined only on simplices, are not applicable. Of the remaining options, the simplest are the continuous and discontinuous Galerkin elements, which on hexahedra are constructed directly as tensor products of one-dimensional interval elements [HKH17]; this structure is also what permits sum factorisation. The continuous Galerkin space is the natural choice for the second-order elliptic inversion problem. The discontinuous Galerkin space offers greater flexibility, including the representation of discontinuous fields, at the cost of additional interface terms and a penalty parameter that must be chosen sufficiently large to ensure stability [LB13]. The specific Firedrake variants used for meshes constructed with tensor products are the `'Q'` and `'DQ'` function spaces [Ham+23].

Equation (37) is essentially an advection-diffusion equation with a source term. If the equation is advection-dominated we need to take care that the selected function space does not result in non-physical oscillations. The `'Q'` function space simply handles diffusion, but can result in non-physical oscillations if the flow is advection-dominated [Dur10]. The `'DQ'` function space allows for stable upwinding and explicit conservation of flux between cells, but introduces a larger number of degrees of freedom for the same mesh and requires a more complex formulation such as the Symmetric Interior Penalty Galerkin (SIPG) method to handle diffusion [Ham+23].

Since there is no advection in the x and y directions in equation (37), we expand the vertical derivative term to determine which process dominates:

$$\frac{f_0^2}{\overline{\rho}}\frac{\partial}{\partial z}\left(\frac{\overline{\rho}}{\overline{N}^2}\psi_z\right) = \psi_z \underbrace{\frac{f_0^2}{\overline{\rho}}\frac{\partial}{\partial z}\frac{\overline{\rho}}{\overline{N}^2}}_{C_1(z)} + \psi_{zz}\underbrace{\frac{f_0^2}{\overline{N}^2}}_{C_2(z)} \tag{62}$$

We compute the ratio of advection to diffusion:

$$\frac{C_1(z)}{C_2(z)} = \frac{\overline{N}^2}{\overline{\rho}}\frac{\partial}{\partial z}\frac{\overline{\rho}}{\overline{N}^2} = \frac{\partial}{\partial z}\ln\left(\frac{\overline{\rho}}{\overline{N}^2}\right) \tag{63}$$

Numerically calculating the absolute maximum of equation (63) for $z \in [0, H]$ reveals that $|C_1/C_2| < 0.005$, meaning that in the vertical direction equation (37) is heavily dominated by diffusion.

It is likely that increasing p will stabilise these oscillations on a 3D mesh (as it does on a 1D and 2D mesh [Cai+14]). Nevertheless, we calculate a **heuristic** stability parameter based on the concept of a mesh Péclet number [DH03]. Here h is the local mesh spacing.

$$\mathrm{Pe} = \frac{hC_1}{2C_2} \tag{64}$$

For linear ($p = 1$) 1D continuous Galerkin formulations, the constraint $\mathrm{Pe} \leq 1$ prevents spurious oscillations. We will adopt this constraint as 'a best guess' of the stability bound of the `'Q'` function space, as no oscillation criterion is available for $p > 2$ in 3D [DH03].

When calculating Pe (64), h can not be taken to be constant. Even though we take Δz as constant, Firedrake by default places nodes at Gauss-Lobatto-Legendre (GLL) points. These points are not equally spaced to improve numerical conditioning [Ham+23] and mitigate the Runge phenomenon [Tre00]. The maximum spacing between GLL nodes for some p is given by table 1.

Consequently, the maximum spacing of nodes in a cell of size Δz can be expressed as $h_{\max} = \Delta x_{\max}(p)\Delta z/2$, and Pe (64) for $p = 4$ can be bounded by

$$\mathrm{Pe}_{p=4} \leq \frac{h_{\max}}{2}\left|\frac{C_1}{C_2}\right| < \frac{\Delta z}{1222} \tag{65}$$

We have the strictly heuristic stability constraint $\Delta z < 1222\,\mathrm{m}$, however the smallest feature we wish to capture is the smoothed tropopause which has a vertical thickness of $w_{\mathrm{trop}} = 1000\,\mathrm{m}$, so a safer limit

Table 1: Position of GLL nodes in the range $x_i \in [-1, 1)$ for various polynomial degrees [AS65, Table 25.6]. For even p, $|x|_{\min}$ corresponds exactly to the maximum spacing between GLL nodes on the interval $[-1, 1]$. For odd p it corresponds to half the spacing.

p	$\lvert x\rvert_{\min}(p)$	$\Delta x_{\max}(p)$
1	1.0000	2.0000
2	1.0000	1.0000
3	0.44721	0.89443
4	0.65465	0.65465
5	0.28523	0.57046
6	0.46885	0.46885
7	0.20930	0.41860
8	0.36312	0.36312
9	0.16528	0.33056

would be $\Delta z \leq 500\,\mathrm{m}$.

It is shown that a solution using the `'Q'` function space will likely not be polluted by spurious oscillations caused by high advection. Since the solutions to equation (37) appear to be reasonably smooth [Bar+22], we can take advantage of the high order convergence high degree polynomials provide, using fewer degrees of freedom than a standard 2nd order finite difference scheme, while achieving far greater accuracy [DH03] (see figure 10). We will initially take $p = 4$; a relatively arbitrary choice. The definition with Firedrake is written as

```
p = solver_params.polynomial_order # For now p=4
cg_space = FunctionSpace(mesh, 'Q', p) # Sometimes called 'V'
```

Note that the total number of degrees of freedom can be retrieved programatically with `V.dim()` [Ham+23] or analytically [HKH17]:

$$\mathrm{DOFs} = (N_x p + 1)(N_y p + 1)(N_z p + 1) \tag{66}$$

where if $N_x = N_y = N_z = 40$ and $p = 4$, the total number of degrees of freedom is $\approx 8 \times 10^6$. Barnes et al. solve the inversion with the finite-difference scheme of Sprenger [Spr07], which uses three-point centred differences on a uniform grid [Bar+22] and is therefore second-order accurate [LeV07]. Barnes et al. used a second order finite difference scheme and a total of 12×10^6 mesh points [Bar+22].

5.4 Weak Form

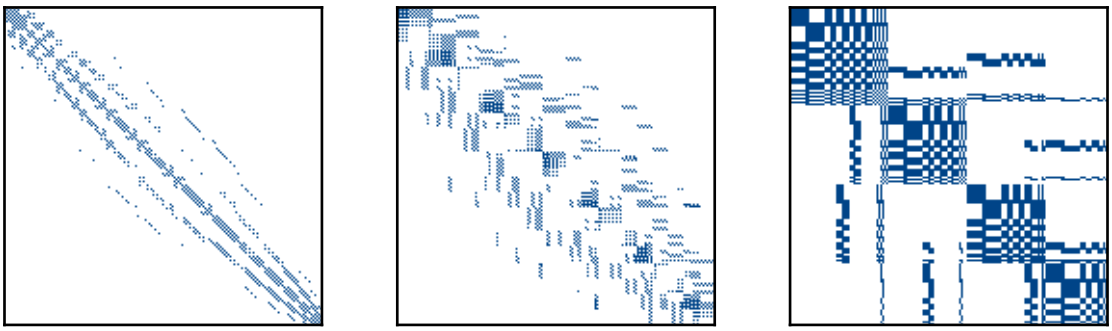
To set up equation (37) to be solvable with the finite element method, we multiply by a test function ϕ and integrate over the 3D domain V. Before integrating we also multiply the whole equation by $\overline{\rho}$, in order to allow for additional optimisations later on. We use the horizontal Laplace operator $\nabla_h^2 = \partial_x^2 + \partial_y^2$ to make this process more concise.

$$\int_V \overline{\rho}\phi q \, dV = \underbrace{\int_V \overline{\rho}\phi \nabla_h^2 \psi \, dV}_{I_h} + f_0^2 \underbrace{\int_V \phi \frac{\partial}{\partial z}\left(\frac{\overline{\rho}}{\overline{N}^2}\psi_z\right) dV}_{I_z} \tag{67}$$

Now we apply Green's first identity [Kel26], which can be written as equation (68) for a general number of spatial dimensions. Here A is the surface that encloses V.

$$\int_V \phi \nabla^2 \psi \, dV = \int_A \phi \nabla\psi \cdot \hat{\boldsymbol{n}} \, dA - \int_V \nabla\psi \cdot \nabla\phi \, dV \tag{68}$$

This yields the following when applied to I_h and I_z (67). The horizontal normal vector is denoted as $\hat{\boldsymbol{n}}_{\boldsymbol{xy}} = (n_x, n_y, 0)$.



$p = 1$, nnz/row ≈ 21 $p = 2$, nnz/row ≈ 49 $p = 4$, nnz/row ≈ 161

Figure 8: Sparsity pattern of the matrix assembled from the bilinear form α (equation (73)) for $p = 1, 2, 4$. The mesh is refined as p decreases ($N = 8, 4, 2$), so that every matrix has the same $9^3 = 729$ degrees of freedom and thus dimensions. Shaded entries are non-zero; nnz/row is the mean number of stored non-zero entries per row, which grows with p as each basis function overlaps with more of its neighbours.

$$I_h = \underbrace{\int_A \overline{\rho}\phi\nabla_h\psi\cdot\hat{\boldsymbol{n}}_{\boldsymbol{xy}}\,dA}_{I_{hb}} - \int_V \overline{\rho}\,\nabla_h\psi\cdot\nabla_h\phi\,dV \tag{69}$$

$$I_z = \underbrace{\int_A \frac{\overline{\rho}}{\overline{N}^2}\phi\psi_z n_z\,dA}_{I_{zb}} - \int_V \frac{\overline{\rho}}{\overline{N}^2}\phi_z\psi_z\,dV \tag{70}$$

We can then expand the boundary integrals I_{hb} and I_{zb} to implement the boundary conditions (equations (38) and (40)):

$$I_{hb} = \int_A \overline{\rho}\phi\left(\psi_x n_x + \psi_y n_y\right)\,dA = \int_A \overline{\rho}\phi\left(\overline{v}n_x - \overline{u}n_y\right)\,dA \tag{71}$$

$$I_{zb} = \int_A \phi\frac{\overline{\rho}}{\overline{N}^2}\frac{g\theta^*}{f_0\overline{\theta}}n_z\,dA \tag{72}$$

We then substitute everything into equation (67) and move terms around to get the form $a(\psi,\phi) = L(\phi)$:

$$\underbrace{\int_A \overline{\rho}\phi\left(\overline{v}n_x - \overline{u}n_y\right)\,dA + f_0^2\int_A \phi\frac{\overline{\rho}}{\overline{N}^2}\frac{g\theta^*}{f\overline{\theta}}n_z\,dA - \int_V \overline{\rho}\phi q\,dV}_{L(\phi)} = \underbrace{\int_V \overline{\rho}\,\nabla_h\psi\cdot\nabla_h\phi\,dV + \int_V \frac{f_0^2\overline{\rho}}{\overline{N}^2}\phi_z\psi_z\,dV}_{a(\psi,\phi)} \tag{73}$$

Now equation (37), with the boundary conditions embedded, is in a weak form that can easily be expressed in Firedrake's syntax[6].

Establishing certain properties of the matrix arising from α will aid in optimising the numerical solver later on. We notice that α is symmetric, as $\alpha(\psi,\phi) = \alpha(\phi,\psi)$. It can be shown that α is positive semi-definite by evaluating

$$\alpha(\phi,\phi) = \int_V (\phi_x^2 + \phi_y^2)\overline{\rho}\,dV + \int_V \frac{f_0^2\overline{\rho}}{\overline{N}^2}\phi_z^2\,dV \tag{74}$$

Since every term in equation (74) is ≥ 0, it follows that $\alpha(\phi,\phi) \geq 0$ for all real ϕ. These properties are transferred to the matrix arising from equation (73) [LB13].

[6] shown in section A.3

5.5 Solvability Condition

Equation (37) and its boundary conditions (equations (38) and (40)) only contain derivatives of ψ. This means that if ψ is a solution, $\psi + C$ will also be a solution. To ensure the weak form (73) is solvable, we calculate $\theta^* \in \mathbb{R}$ such that the constant solution is valid. This is called the Neumann solvability condition, and it requires that the test function $\phi = C$ satisfies equation (73) [LB13]. In other words, we solve for θ^* in the equation $\alpha(\psi, C) = L(C)$. As the bilinear form only contains derivatives of the test function, it follows that $\alpha(\psi, C) = 0$. Additionally, C can be pulled out of the integrals in $L(C)$ and divided away, leaving the following solution condition:

$$\oint_A \overline{\rho}\,(\overline{v}n_x - \overline{u}n_y)\,dA + \underbrace{f_0^2 \int_A \frac{\overline{\rho}}{\overline{N}^2}\frac{g\theta^*}{f\overline{\theta}} n_z\,dA}_{I_1} - \int_V \overline{\rho} q\,dV = 0 \tag{75}$$

The integrand of I_1 exclusively depends on z, meaning the integral can be evaluated directly.

$$I_1 = \theta^* L_x L_y f_0 g \left(\frac{\overline{\rho}(H)}{\overline{N}(H)^2 \overline{\theta}(H)} - \frac{\overline{\rho}(0)}{\overline{N}(0)^2 \overline{\theta}(0)} \right) \tag{76}$$

Rearranging equation (75) yields the formula for θ^*:

$$\theta^* = \frac{\displaystyle\int_V \overline{\rho} q\,dV - \int_A \overline{\rho}\,(\overline{v}n_x - \overline{u}n_y)\,dA}{L_x L_y f_0 g \left(\dfrac{\overline{\rho}(H)}{\overline{N}(H)^2 \overline{\theta}(H)} - \dfrac{\overline{\rho}(0)}{\overline{N}(0)^2 \overline{\theta}(0)} \right)} \tag{77}$$

Experimentally it was found that $\theta^* \sim -0.1\,\mathrm{K}$, representing a very small anomaly in potential temperature, smaller than the approximate $-2\,\mathrm{K}$ from [Spr07].

To check if equation (77) is correct, the flux of the linear form can be numerically calculated by comparing how close $L(1) - \alpha(\psi, 1)$ is to zero. As shown above, $\alpha(\psi, 1) = 0$, so only $L(1)$ needs to be evaluated. This is done by making the substitution $\phi = 1$ and evaluating `L` (defined in section A.3) [Aln+14]:

```
flux = assemble(replace(L, {phi: Constant(1)}))
```

Executing the above expression shows that the net flux across the whole domain is $\sim 10^{-3}$. Given the domain volume of $\sim 10^{18}\,\mathrm{m}^3$, a net flux of $\sim 10^{-3}$ is on the order of machine precision, confirming that equation (77) is satisfied up to floating-point truncation error.

5.6 Solver Settings

With the equation fully specified in Firedrake's syntax, we can proceed with the solver setup and parameter selection.

Since the coefficients of the bilinear form in equation (73) are time-independent and equation (37) is linear, the Jacobian remains constant. Explicitly instructing Firedrake to retain this constant Jacobian[7] resulted in significant speed improvements during repeated solves.

As discussed previously, this is a pure Neumann problem, making it necessary to pass a constant nullspace to Firedrake[8]. This prevents the solver from attempting to invert a singular matrix with a rank deficiency of one.

To finally solve the quasigeostrophic potential vorticity equation (equation (37)), an efficient set of solver parameters must be specified. A benchmarking search was conducted across over 40 permutations of theoretically reasonable solver parameters on the primary testing hardware (see section B.1). Pre-conditioners such as Fast Diagonalisation (the `'firedrake.FDMPC'` pre-conditioner [Ham+23]) and polynomial multigrid (p-multigrid) were evaluated, alongside a wide variety of smoothers and solvers for

[7] shown in section A.4
[8] shown in section A.4

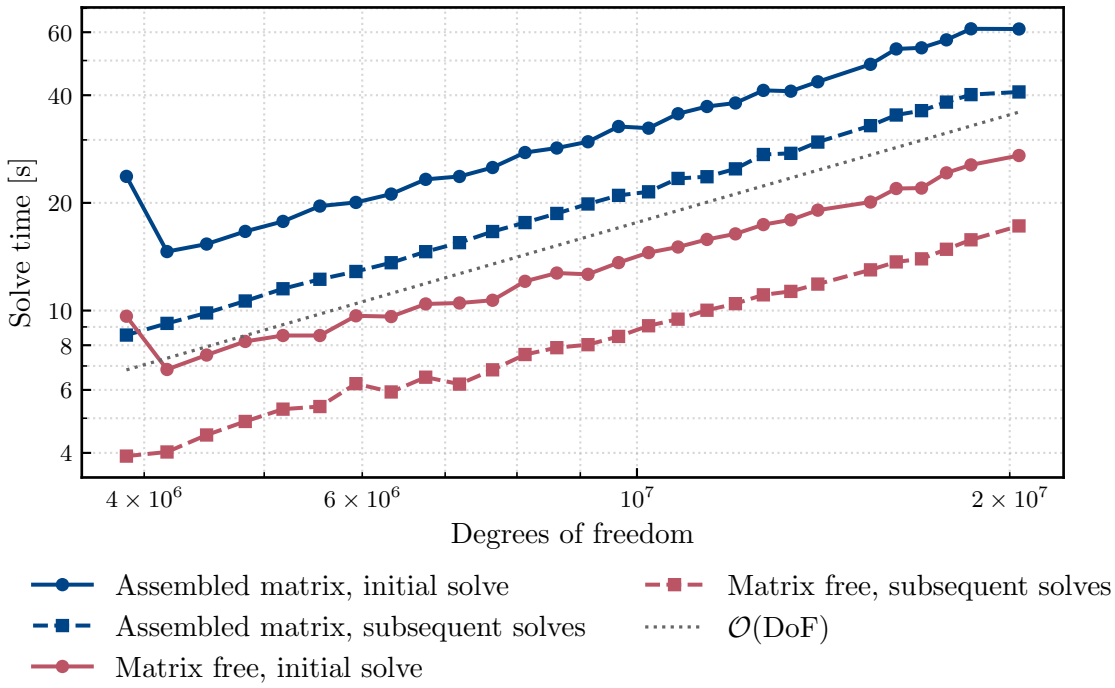


Figure 9: Log-log plot of solve time versus degrees of freedom with $p = 4$, comparing optimised matrix-free parameters (section 5.6) to optimised assembled matrix parameters (section A.4.1). Achieves $\mathcal{O}$(DoF) time complexity. Computed with the hardware in section B.2.

the coarse $p = 1$ mesh. The performance of matrix-free methods was also compared against assembled matrix methods. From this reasonably extensive search, the fastest combination of solver parameters was found to be:

- **Matrix Format:** Matrix-free. The sparse matrix (AIJ format) [Bal+25] is never built on the refined levels; instead, each time the solver needs the product of the matrix with a vector, that product is calculated directly from the weak form, cell by cell [Ham+23; KM18].
- **Krylov Subspace Solver:** The Conjugate Gradient (CG) method. This can be used because it was shown that the matrix arising from the bilinear weak form (equation (73)) is symmetric and positive semi-definite [Saa03].
- **Pre-conditioner:** A p-multigrid scheme using the `'firedrake.PMGPC'` pre-conditioner [Ham+23].
- **Multigrid Smoother:** A Chebyshev iteration, pre-conditioned with point-Jacobi, is applied across the refined multigrid levels ($p = 2, 3, 4$). A Chebyshev smoother is designed to quickly remove the rapidly oscillating (high-frequency) components of the error, which is exactly the job of a smoother. To do so, it only needs an estimate of the matrix's largest eigenvalue [Bal+25]. The Jacobi diagonal can be obtained cheaply from a matrix-free operator [Ham+23]. The same combination serves as the basic smoother of large-scale matrix-free multigrid solvers [Mün24].
- **Coarse Grid Solver:** Only the $p = 1$ coarse operator is assembled, using the `'firedrake.AssembledPC'` pre-conditioner inside the otherwise matrix-free multigrid hierarchy [Ham+23]. A sparse Cholesky factorisation (via MUMPS [Bal+25; Ame+01; Ame+06]) is conducted exactly once at the first solve, with null-pivot detection enabled to handle the rank deficiency of the matrix. Each subsequent solve using the constant Jacobian is a highly efficient $\mathcal{O}$(non-zero entries) operation [Dav06].

A detailed implementation of the solver set-up is given in section A.4.

Matrix-free methods are typically faster than assembled matrix methods for $p \geq 2$ [KSM23], because memory bandwidth and capacity generally become bottlenecks at these higher orders. Additionally,

on 3D hexahedral cells, assembled sparse matrix-vector multiplication requires $\mathcal{O}(p^6)$ operations per cell, whereas matrix-free methods using sum-factorisation reduce the operation count to $\mathcal{O}(p^4)$ per cell [HKH17; KM18].

These theoretical advantages were verified in practice: experiments involving repeated solutions of equation (73) showed that the matrix-free configuration described above was consistently faster than every assembled configuration tested, on both the primary testing hardware and the HPC cluster (see section B). Matrix-free methods were developed largely because solving massive systems of equations demands significant memory [Mün24]. An assembled matrix remains viable only while it is not itself the bottleneck, since every non-zero entry must be loaded from main memory (RAM) at each matrix-vector product [Mün24]. On modern computer architectures, fetching data from main memory causes the processor to stall, wasting valuable CPU cycles. The CPU must wait for data to propagate from RAM through the L3, L2, and L1 caches before finally reaching the registers [HP17]. Sparse matrix-vector multiplication performs roughly 1/6 floating-point operations per byte loaded, far below the ≈ 10 at which modern hardware is balanced [KM18]. Matrix-free methods aim to minimize these stalls by performing technically redundant or duplicate operations on the fly. Because modern CPUs can compute operations much faster than memory can supply data, this trade-off is often highly performant [Mün24].

It was initially expected that the spatially varying coefficients of equation (73) would erode this advantage. These coefficients lack an analytical expression (as discussed alongside equation (61)), and consequently must be stored in memory. For a mesh where $N_x = N_y = N_z = 35$ and $p = 4$, the system has 2 803 221 degrees of freedom (see equation (66)). Storing the source term (equation (60)) with double precision floating points requires $8\,\mathrm{B} \times 2\,803\,221 \approx 22\,\mathrm{MB}$ of memory. This exceeds the L3 cache capacity of the primary testing hardware (see section B.1), meaning the source term must be fetched entirely from RAM every time it is evaluated.

However, an assembled matrix must also be fetched from RAM at every matrix-vector product, and it is far larger. On the same mesh, each row of the $p = 4$ matrix holds on average about 212 non-zero entries, giving roughly 5.9×10^8 non-zeros in total[9]. Stored in AIJ format with a double-precision value and a four-byte column index per entry [Bal+25], this requires approximately 7 GB of memory, over 300 times that of the source term. The same disparity has been observed in Firedrake, where assembled three-dimensional operators consume well over an order of magnitude more memory per degree of freedom than their matrix-free equivalents [KM18]. Furthermore, even when coefficients are pre-computed and loaded from RAM at every quadrature point, making the matrix-free evaluation memory-bound, its throughput remains more than an order of magnitude higher than that of a sparse matrix-vector product at the same polynomial degree [KSM23].

In essence, matrix-free methods typically excel by minimizing slow memory accesses [Mün24]. Although our specific application must fetch its coefficient fields from RAM regardless, this traffic is small compared with streaming an assembled high-order sparse matrix. As a result, matrix-free methods outperform fully assembled sparse matrices in this context, and assembly is best reserved for the small $p = 1$ coarse operator, where it enables an exact direct solve.

Nevertheless, the following caveats should be noted:

- Assembling the matrix costs approximately ten matrix-free operator applications [KM18]. This cost was expected to be recovered over repeated solves with the constant Jacobian, but in practice each assembled solve remained slower than its matrix-free equivalent, so the cost was never recovered.

- The memory footprint of the matrix-free solver is substantially smaller than that of the assembled alternative. Only the $p = 1$ coarse matrix is stored (approximately 1.2×10^6 non-zeros, or 14 MB before Cholesky fill-in, for the mesh above), which permits far larger meshes on consumer-grade hardware (see section A.4).

[9] Evaluating `assemble(a).petscmat.getInfo()["nz_used"]` yields 594 823 321 stored entries, or about 212 per row. Here `a` is the bilinear form defined in section A.3. The number of stored entries per row is higher than in figure 8, because a mesh with $N = 2$ has only boundary cells which neighbour fewer cells than interior ones (comprising most of an $N = 35$ mesh).

- A transformation for equation (37) might exist that absorbs the spatially varying coefficients into ψ itself, or into a rescaled coordinates x', y', z'. Such a formulation would remove the remaining coefficient memory traffic, potentially widening the performance gap in favour of matrix-free methods further.

- On systems with abundant L1 and L2 cache, such as HPC clusters with dozens of cores per node, the performance gap between matrix-free and assembled methods was expected to narrow. Repeating the benchmark on such a cluster (see section B.2) confirmed that the matrix-free configuration remained the fastest.

- This advantage relies on sum-factorisation, which applies only to tensor-product cells [HKH17]. On simplicial meshes, where Firedrake did not offer sum-factorisation, assembled sparse matrix-vector products were found to be faster than matrix-free ones [KM18], so these conclusions should not be assumed to transfer to tetrahedral meshes.

5.7 Solver Verification

To verify the solver itself is correct, the method of manufactured solutions will be used [Roa02]. We choose some analytically expressible function $\psi_a(x, y, z)$ that has similar properties to the true solution to equation (37). The source term and boundary conditions that arise from ψ_a can then be directly calculated. These boundary conditions and source term are then provided to the numerical solver we defined above. If the solver's numerical solution matches ψ_a to some high degree of precision, the solver is likely correctly implemented.

The domain, mesh and all other parameters are left at their control values, but the coefficient profiles of equation (37) are simplified so that the source term stays short enough to write down:

- Constant reference stability profile: $\overline{N}(z) = N = 0.01\,\mathrm{s}^{-1}$ [Val17]

- An exponential density profile $\overline{\rho}(z) = \rho_0 \exp(-z/H_\rho)$, with $\rho_0 = 1.225\,\mathrm{kg\,m}^{-3}$ and $H_\rho = 8.5\,\mathrm{km}$ [Stu15]

Equation (37) then reduces to a simple advection-diffusion equation with constant coefficients:

$$q = \frac{\partial^2 \psi_a}{\partial x^2} + \frac{\partial^2 \psi_a}{\partial y^2} + \frac{f_0^2}{N^2}\left(\frac{\partial^2 \psi_a}{\partial z^2} - \frac{1}{H_\rho}\frac{\partial \psi_a}{\partial z}\right) \tag{78}$$

The manufactured solution is built from the three structures the true solution possesses – the jet stream, the response to the PV anomaly of equation (57), and the vertically uniform θ^* adjustment of section 5.5:

$$\psi_a(x, y, z) = \overline{\psi}(y, z) + \Psi_a \left[1 + \frac{(x - x_{\mathrm{ano}})^2 + (y - y_{\mathrm{ano}})^2}{L_a^2} + \left(\frac{z - z_a}{h_a}\right)^2\right]^{-1} + \Lambda z \tag{79}$$

Here $\overline{\psi}$ is the background streamfunction of equation (53) at the control jet parameters, and the anomaly term is centred on the control anomaly position $(x_{\mathrm{ano}}, y_{\mathrm{ano}}) = (L_x/2, L_y/2)$. The anomaly term is a Lorentzian rather than a Gaussian because inverting equation (37) spreads the Gaussian source of equation (57) into a profile with far broader tails. A least-squares fit against the numerical solution at the control parameters gives $\Psi_a = 8 \times 10^6\,\mathrm{m^2\,s^{-1}}$, $L_a = 500\,\mathrm{km}$, $h_a = 5\,\mathrm{km}$, $z_a = 8.75\,\mathrm{km}$ and $\Lambda = 45\,\mathrm{m\,s^{-1}}$, at which equation (79) is within 3 % of that solution in the relative L^2 norm. The solver is therefore verified against a problem carrying the same length scales, gradients and magnitudes as the one it is built to solve. Substituting equation (79) into equation (78) supplies the source term, and its first derivatives supply the boundary data: $\overline{v} = \partial\psi_a/\partial x$ and $\overline{u} = -\partial\psi_a/\partial y$ in equation (38), while $\partial\psi_a/\partial z$ is prescribed directly on the top and bottom boundaries in place of equation (40). Since ψ is only determined up to an additive constant, the mean of each field is removed before the relative L^2 error between the numerical solution and ψ_a is taken (section A.6).

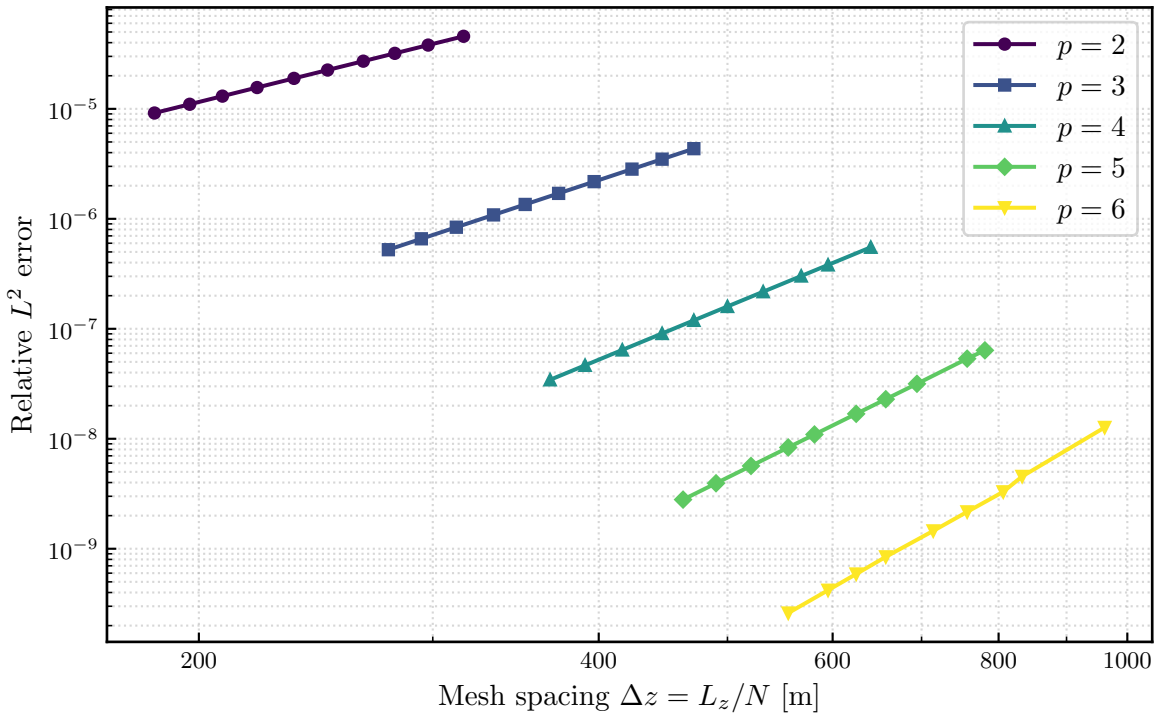


Figure 10: Log-log error convergence at varying p of the manufactured solution outlined in section 5.7. Each polynomial order is limited to DoF $\in [4 \times 10^6, 2 \times 10^7]$. Error is calculated using the method from section A.6 with $H = 20\,\text{km}$. Each line's slope is within 0.03 of $p+1$, so $\mathcal{O}\left(\Delta z^{p+1}\right)$ error convergence is achieved.

The relative error of the numerical solution for varying p against equation (79) is given in figure 10. Since the error of the $p = 4$ scheme is satisfactory and the solver settings were optimised for $p = 4$, we will keep the arbitrarily chosen $p = 4$. It should also be noted that according to figure 10, this 5th order scheme should provide orders of magnitude less error than the second order scheme used by Barnes et al, at a fraction of the degrees of freedom.

6 Post-Inversion Diagnostics

6.1 Derived Atmospheric Quantities

Using the numerically calculated stream function ψ that satisfies equation (37), we can finally calculate and plot[10] the post-inversion state of the control atmosphere using the invertibility principle.

- **Wind components** $\boldsymbol{u} = (u, v, 0)$: Calculated directly using properties of the stream function. The control state is plotted in figures 11 and 12.

$$u = -\frac{\partial \psi}{\partial y}, \qquad v = \frac{\partial \psi}{\partial x} \tag{80}$$

- **Geostrophic vorticity** ζ_g: Calculated using equation (5) and u and v from equation (80). The control state is plotted in figure 13.

- **Pressure anomaly** p^*: Since $\psi = \Phi^*/f_0$, and to leading order a geopotential anomaly corresponds hydrostatically to a pressure anomaly $p^* \approx \overline{\rho}\,\Phi^*$ [Val17],

$$p^*(x, y, z) = \overline{\rho}(z)\, f_0\, [\psi(x, y, z) - \psi_0(z)] \tag{81}$$

[10] Plots are generated by sampling points using Firedrake's `PointEvaluator` [Nix+24].

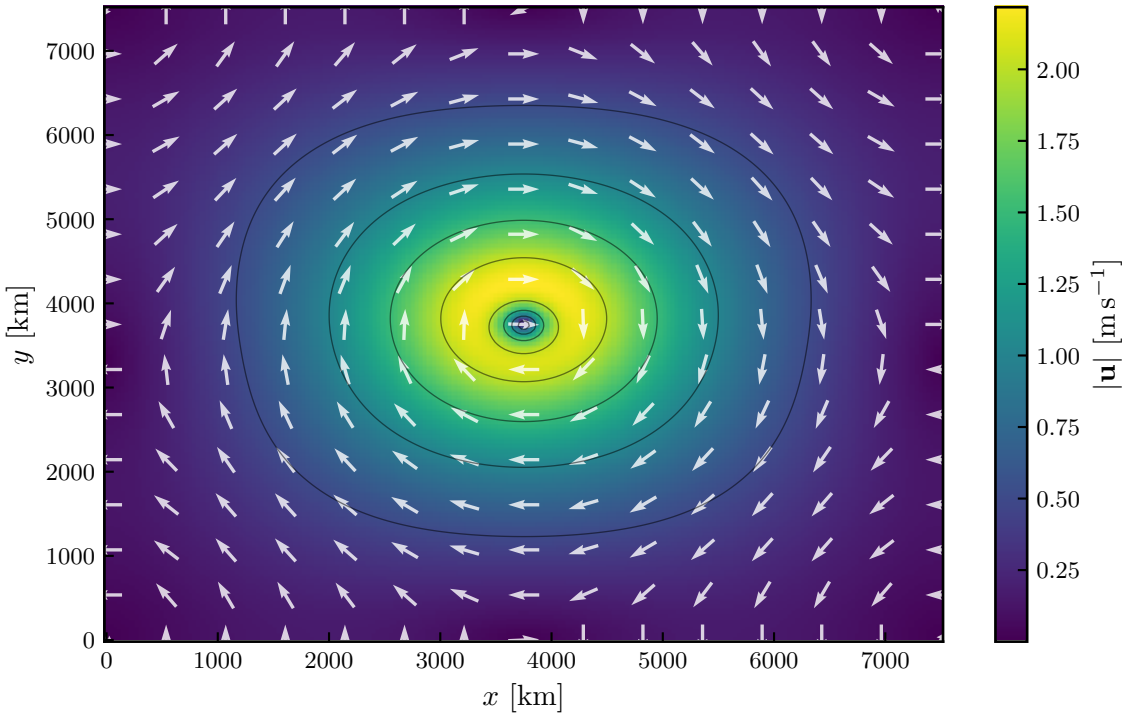


Figure 11: Surface wind of the control state, calculated using equation (80). A contour interval of $0.5\,\mathrm{m\,s^{-1}}$ is used, so that contour density can be more easily compared in later plots with stronger surface wind. This plot shows weak cyclonic rotation induced by the anomaly defined in equation (57).

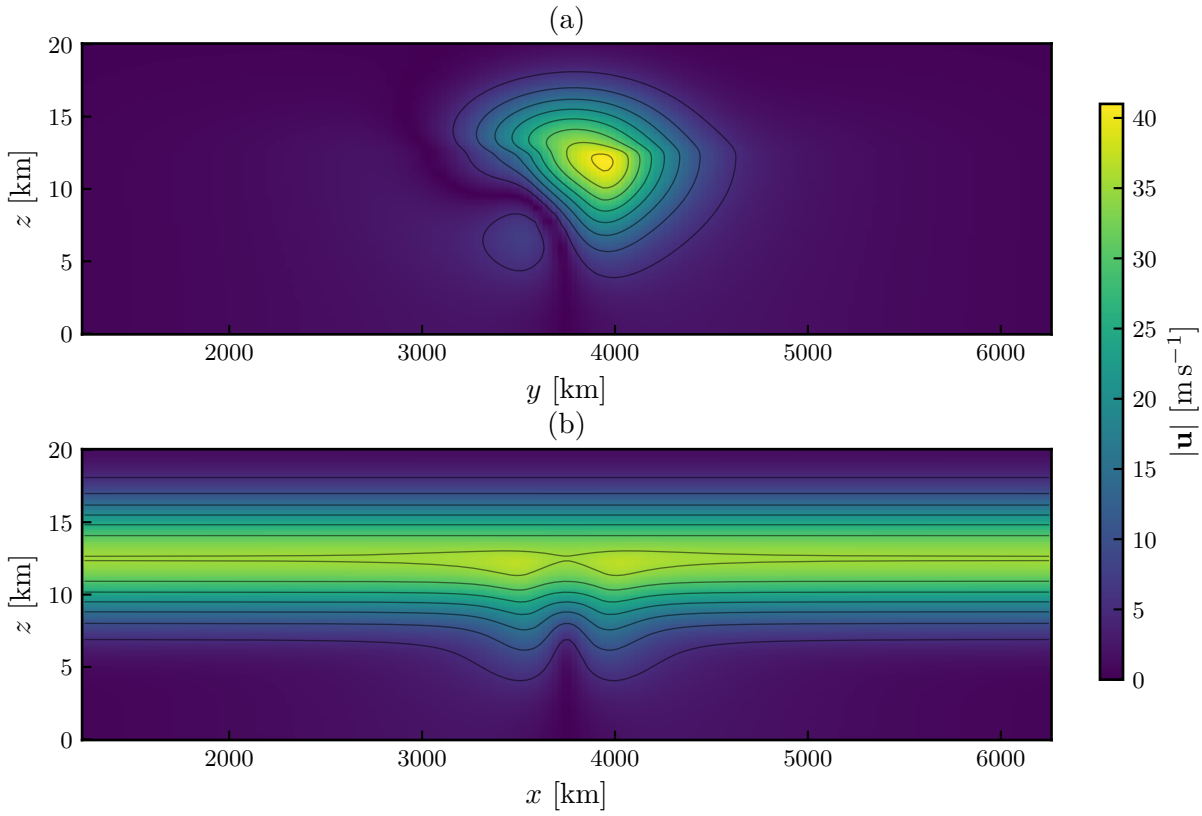


Figure 12: Vertical slices of wind speed of the control state, calculated using equation (80). A contour interval of $5\,\mathrm{m\,s^{-1}}$ is used. In this case the jet is the most significant source of wind, with the anomaly distorting it at the domain centre.

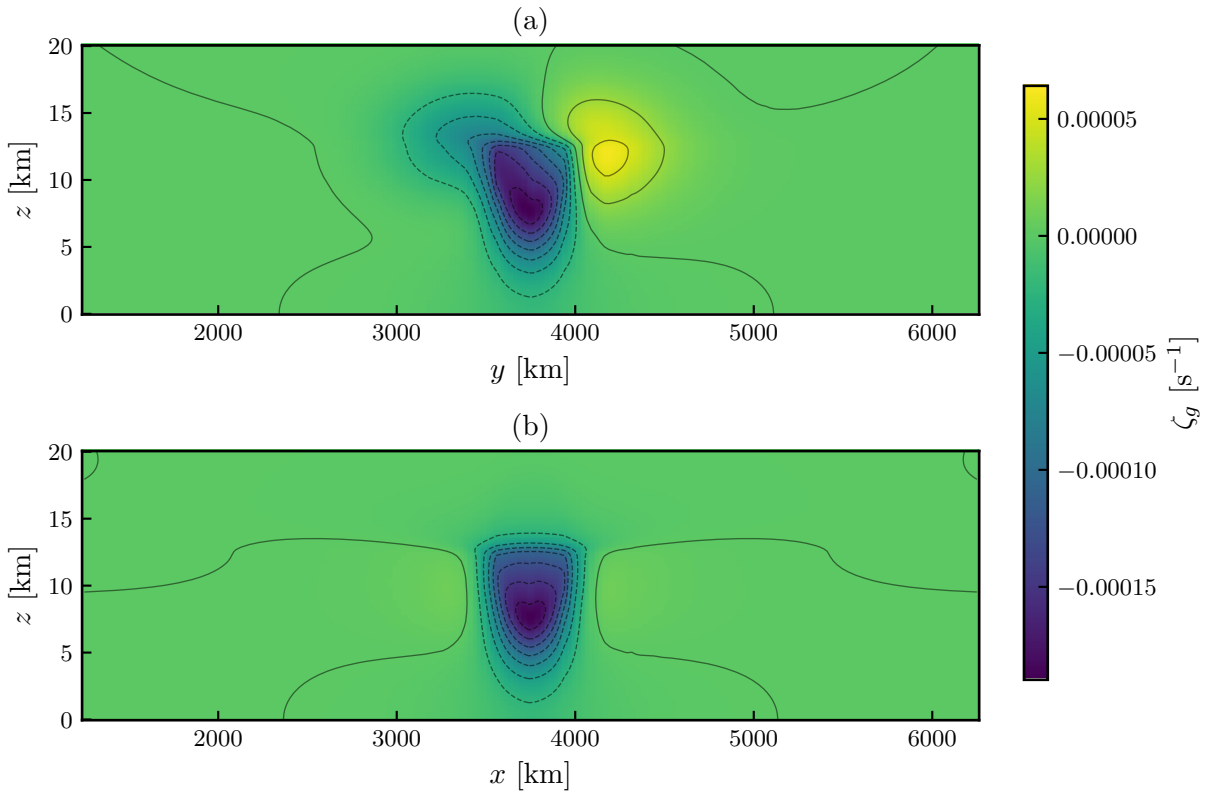


Figure 13: Vertical slices of the geostrophic vorticity of the control state (post-inversion). In this case the vorticity induced by the anomaly has a minor effect on the surface.

where $\psi_0(z)$ is the mean of ψ around the lateral boundary at height z [Spr07]:

$$\psi_0(z) = \frac{1}{2(L_x + L_y)} \left[\int_0^{L_x} [\psi(x, 0, z) + \psi(x, L_y, z)] \, dx + \int_0^{L_y} [\psi(0, y, z) + \psi(L_x, y, z)] \, dy \right] \tag{82}$$

Its implementation is given in section A.7. Subtracting ψ_0 leaves the wind (equation (80)) unchanged, because ψ_0 does not vary horizontally. It also fixes the reference so that p^* averages to zero around the lateral walls at every height, making $\overline{p}(z)$ the far-field pressure. The control state is plotted in figure 14.

- **True potential temperature anomaly** θ^*: Calculated using equation (39). The control state is plotted in figure 15.
- **Temperature anomaly** T^*: Calculated from θ^*, using the definition of potential temperature in equation (1).
- **Pressure** p and **potential temperature** θ: The total pressure and total potential temperature is given by $p = \overline{p} + p^*$ and $\theta = \overline{\theta} + \theta^*$ respectively. The control state is plotted in figures 16 and 17.
- **Minimum dynamical tropopause height** z_{dyn}: The lowest point of the dynamical tropopause, which we define as the $\pm 1.5\,\text{PVU}$ surface that bounds the stratosphere [BNL21]. It is computed by first converting from q to EPV using equation (55). A DoF is marked as stratospheric if $fQ > 0$ and $|Q| \geq 1.5\,\text{PVU}$, so $Q \leq -1.5\,\text{PVU}$ in the Southern Hemisphere. Since the mesh is a uniform rectangle extruded with a constant layer height, the DoFs lie on a regular (x, y, z) lattice, so these marks are gathered from every MPI rank into a single boolean array. Detached pockets of stratospheric air below the tropopause proper, such as those left behind by the cut-off lows of section 2.5, must not be counted as the tropopause [BNL21]. The connected regions of the array are therefore labelled (with `scipy.ndimage.label` [Vir+20], connecting only through shared faces),

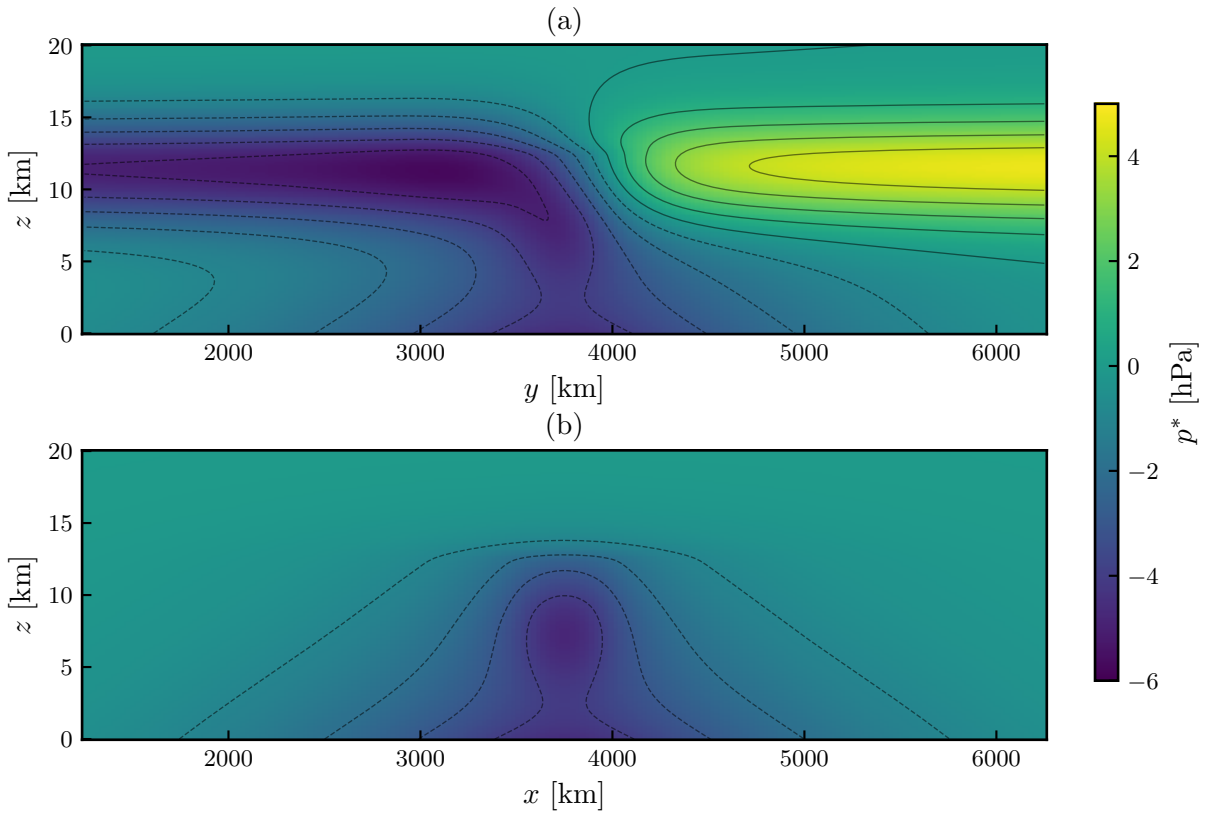


Figure 14: Vertical slices of the pressure anomaly of the control state calculated with equation (81). A contour interval of 1 hPa is used.

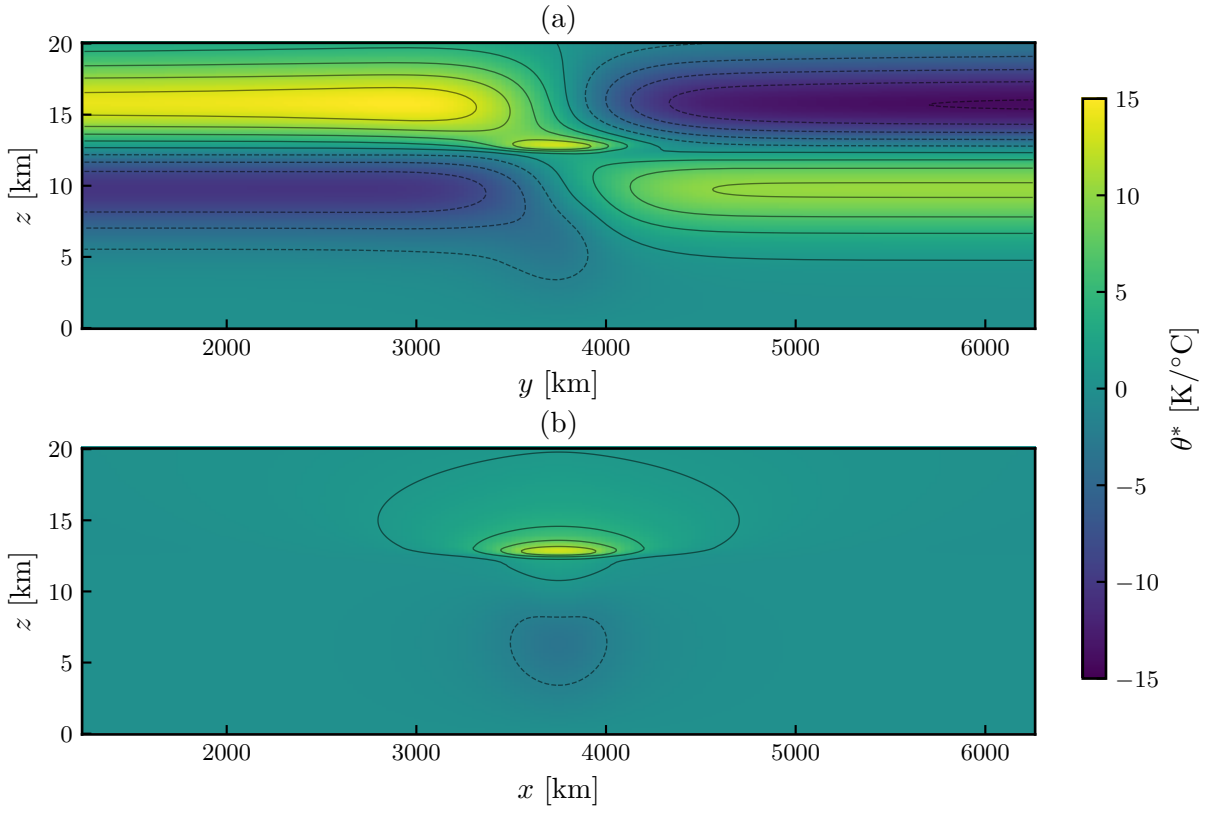


Figure 15: Vertical slices of the potential temperature anomaly of the control state. A contour interval of 3 K is used. In this case most of the potential temperature anomaly is induced by the jet.

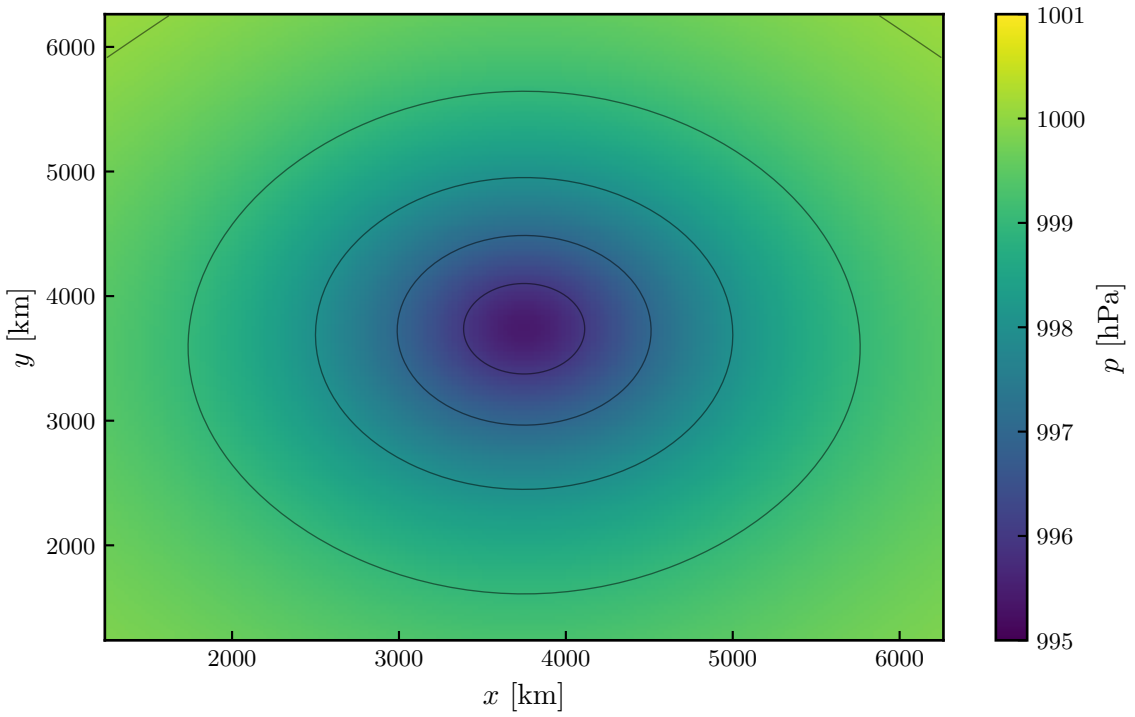


Figure 16: Surface pressure of the control state. A contour interval of 1 hPa is used. The anomaly induces a noticeable pressure gradient on the surface here.

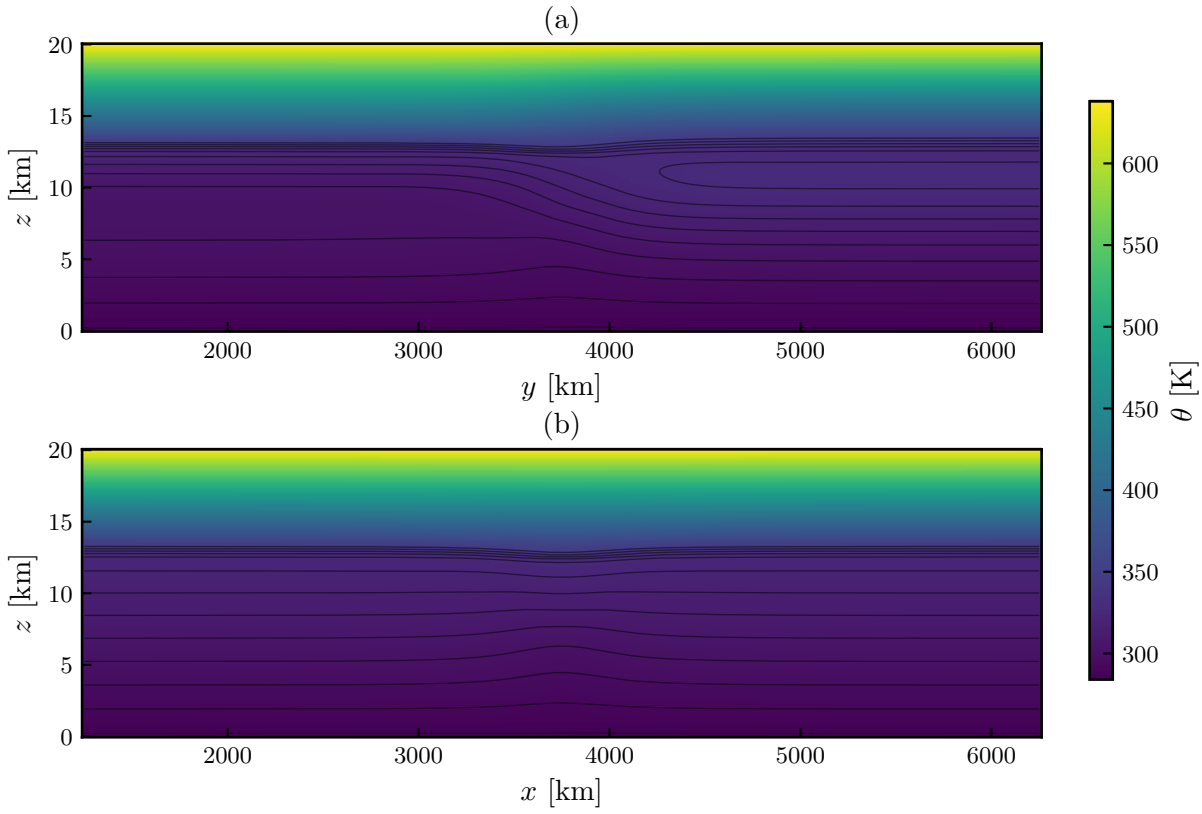


Figure 17: Vertical slices of the potential temperature of the control state. A contour interval of 5 K is used, where the contours are isentropes like those discussed in sections 2.4 and 2.5 [HMR85].

and only the regions that reach the model top at $z = H$ are kept. z_{dyn} is the height of the lowest DoF in these regions, so it is resolved only to the vertical spacing of the DoFs. If no stratospheric air reaches the top of the domain, $z_{\text{dyn}} = H$. The control state has $z_{\text{dyn}} = 6767\,\text{m}$.

- The **maximum Rossby number** is estimated using equation (6):

$$\text{Ro}_{\text{max}} \approx \max_V \left| \frac{\zeta_g}{f} \right| \tag{83}$$

The control state has $\text{Ro}_{\text{max}} \sim 2$, far larger than the $\text{Ro} \ll 1$ required for QGPV to be formally valid [HMR85; Val17].

- The **Froude number** is given by the following [Val17], which we approximate to enable direct calculation:

$$\text{Fr} = \frac{U}{NH} \approx \frac{1}{\overline{N}} \frac{\partial |\boldsymbol{u}|}{\partial z} \tag{84}$$

The control state exhibits $\text{Fr}_{\text{max}} \sim 0.8$.

6.2 Limitations of QGPV

Now that all the numerical tools are in place to set up, compute and plot solutions to equation (37), we are able to examine the effect of various atmospheric and anomaly parameters on surface cyclogenesis.

Before proceeding, we must examine the primary limitations of QGPV and EPV, so that we may restrict our parameter search space not only to physically possible states, but also to states in which QGPV inversion remains a reasonable approximation of the true atmospheric state.

The first constraint is on the sign of the EPV. For full EPV inversion, the requirement $fQ > 0$ makes the inversion problem elliptic and well posed, and is equivalent to the balanced state being statically, inertially and symmetrically stable [HMR85]. To first order, QGPV is related to EPV by linearising about a uniformly stratified reference state at rest, [Feh97], although this relation holds only for small relative anomalies and not in general [HMR85]. Under this relation $fQ > 0$ is equivalent to $q/f > -1$; the QGPV inversion operator itself is linear and elliptic for any q, so this is a condition for the inverted state to be physically realisable rather than for the inversion to be solvable. We set up a direct way to quantify this in equation (59), which brings the implication that for any atmosphere to be physically valid, $\mathcal{V}_- = 0$. As described in section 4.4, $\mathcal{V}_-$ is driven largely by $|\partial\overline{u}/\partial z|$.

A further but looser constraint is placed on the Rossby number. QGPV is only formally valid $\text{Ro} \ll 1$, although it remains qualitatively useful for observed disturbance amplitudes [HH13b]. Davis identified the inaccuracy of QG as the Rossby number becomes $\mathcal{O}(1)$ as its main weakness, while noting, following earlier work, that it may still give useful qualitative information in such situations [Dav92]. Davis did not directly measure the error of QG inversion against full EPV inversion, but did compare a truncated linear piecewise inversion, which nearly reduces to QG for a mean state at rest, with the full linear piecewise method [Dav92]. For an anomaly with maximum vorticity $\zeta_{\text{max}} = 1.9 \times 10^{-4}\,\text{s}^{-1}$ ($\text{Ro} \approx 1.9$ for $f \approx 10^{-4}\,\text{s}^{-1}$), the maximum surface wind attributed to one third of the anomaly differed between these methods by approximately 12%, rising to 24% at $\zeta_{\text{max}} \approx 2.9 \times 10^{-4}\,\text{s}^{-1}$ [Dav92]. Davis further found that the differences among these methods were smaller than their differences from QG inversion [Dav92], so these figures should be read as a lower bound on the error of QG. For positive anomalies, the near-QG method produced stronger winds than the full linear method at the level of the anomaly, but weaker winds at the surface [Dav92]. This is consistent with vertical penetration of the anomaly increasing with its amplitude, whereas QG predicts a fixed penetration depth [Dav92], so QG will tend to underestimate the surface response to strong upper-level anomalies. It must also be noted that Muraki et al. report that, at $\text{Ro} \approx 0.3$, next-order corrections to QG remove qualitative deficiencies and significantly improve quantitative accuracy in the time-evolution of a baroclinic wave on a jet [MSR99]. QGPV therefore already carries noticeable errors well below $\text{Ro} = 1$. Unfortunately further investigating the error of QGPV inversion is beyond the reasonable scope of this Applied Mathematics Honours project.

A restriction is additionally placed on the Froude number (equation (84)), which must be much less than one and comparable to Ro. Finally, EPV is conserved following the flow only for adiabatic, frictionless motion [HMR85]. When condensation or evaporation occurs, heat is added to or removed from the air, and PV is no longer conserved. The inversion itself is instantaneous and does not require adiabatic flow, but moisture still affects it: condensation in rising air reduces the effective static stability, which increases the vertical penetration of an anomaly and enhances surface development [HMR85]. Our inversion uses dry stratification everywhere, so it may underestimate the surface response in moist ascending regions, the same direction of bias as the QG penetration error above.

Sweeps of atmospheric and anomaly parameters must apply these restrictions, so as to limit the chance of non-physical results being derived.

7 Parameter Sweeps

The high efficiency of the diagnostic solver outlined in section 5 enables vast searches of the parameter space. A full atmospheric state can be constructed, inverted and analysed in under 3 seconds on the hardware given in section B.2, meaning these sweeps become highly tractable.

7.1 Parameters to Vary

We establish a list of parameters and ranges in which to vary them that correspond to numerically solvable and reasonably physically plausible atmospheric states.

Table 2: Bounds on the sampled atmospheric and anomaly parameters.

	Bounds			
Quantity	Min	Max	Unit	Justification
Stratification and tropopause structure				
$\overline{N}'_{\text{strat}}$	-0.005	0.003	s^{-1}	Departure from the φ-dependent mean, which is interpolated using $N^2(30°) = 7 \times 10^{-4}\,\text{s}^{-2}$ and $N^2(45°) = 4.5 \times 10^{-4}\,\text{s}^{-2}$ [Bir06]. Upper bound set by $N^2_{\text{max}} \approx 6 \times 10^{-4}\,\text{s}^{-2}$. [11] Lower bound set by $N^2_{\text{min}} \approx 3.5 \times 10^{-4}\,\text{s}^{-2}$ [RWF07, Fig. 4b].
$\overline{N}_{\text{trop}}$	0.008	0.014	s^{-1}	Tropospheric $N^2 \approx 10^{-4}\,\text{s}^{-2}$ with minimal φ variation [Bir06]. Bounds set by [RWF07, Fig. 4b].
w_{trop}	500	2000	m	Observed tropopause inversion layers are $w_{\text{trop}}(33°) \approx 500\,\text{m}$ and $w_{\text{trop}}(48°) \approx 2\,\text{km}$ [Bir06]. The floor is 1 cell for $N_z = 50$, $H = 25\,\text{km}$.
z_{trop}	10	15	km	Same range as [Bar+22]. Zonal-mean tropopause height is 10.0 km at 50°S and 14.5 km at 30°S [GTB10][12].
δ	2	10	–	Purely numerical kink-sharpness parameter for equation (46).
Anomaly				
$\sqrt{x_{\text{size}}\, y_{\text{size}}}$	100	400	km	Same as [Bar+22] but allowing aspect ratio to vary.

Continued on next page

[11] This is the lower bound for the stratospheric global maximum N^2 [Bir06]

[12] Synoptic and seasonal variations are about ±3 km on top of this [SR06]. Not included in bounds due to numerical constraints.

Table 2: Bounds on the sampled parameters (continued).

Quantity	Bounds Min	Bounds Max	Unit	Justification
$\ln(y_{\text{size}}/x_{\text{size}})$	$\ln\frac{1}{4}$	$\ln 4$	–	[Feh97] shows anomalies with aspect ratios of of 2–4 : 1, to at most 4 : 1. [13]
z_{size}	3500	7000	m	Same range as [Bar+22]. Physical feasibility has not been assessed.
Q_{anoclip}	−2	−1	PVU	Same as [Bar+22].
Jet parameters				
L_{jet}	200	1000	km	Jets are about 550 km to 2000 km wide [Stu15]. Converting width to a Gaussian scale[14] gives 160 km to 1200 km. Then capped by the domain and rounded.
z_{jet}	1	6	km	Jets are 5 km to 10 km deep [Stu15], and the S. Hemisphere subtropical jet is confined in 500 hPa to 100 hPa [Bal+01]. Both give 1.45 km to 6.0 km.
U_{jet}	10	80	$\mathrm{m\,s^{-1}}$	Subtropical jets in the S. Hemisphere range from $5\,\mathrm{m\,s^{-1}}$ to $80\,\mathrm{m\,s^{-1}}$ depending on season [Stu15].
y_{jet}	−1500	500	km	Measured from $L_y/2$, where the anomaly sits; negative is poleward. Barnes uses $y_{\text{jet}} = L_y/2$ [Bar+22], but COLs detach equatorward of the jet during their life cycle [BNL21]. Limited to 1500 km by the domain and β=0.
Background state				
φ	−50	−30	°	Brackets the control latitude of 42°S [Bar+22] and covers the latitudes of S. Hemisphere COLs [BNL21].
$\overline{T}'(0)$	−10	14	K	Departure from the annual zonal mean (given in table 3), from daily NCEP/NCAR reanalysis 1991–2020 over southern Africa (15°–35°E) [Kal+96]. The edges cover the 1st and 99th percentiles at almost every latitude.

Table 3: Annual zonal-mean air temperature at 1000 hPa, NCEP/NCAR Reanalysis 1, 1991–2020 monthly long-term means [Kal+96].

Latitude (°)	−50	−45	−40	−35	−30	−25	−20
T (K)	279.20	282.38	285.80	288.85	291.83	294.44	296.47

7.2 One Parameter at a Time

Varying a single parameter at a time, using the control state as a base, allows us to investigate the surface effects of each parameter in isolation. For this experiment we do not place any restrictions on $\mathcal{V}_-$, Ro or Fr. We simply aim to see how each parameter in table 2 affects maximum surface wind speed, minimum dynamical tropopause height, minimum surface pressure anomaly and minimum surface vorticity.

For each parameter g in table 2 that varies in the range $g \in [a, b]$, we define a $g(x) = a + (b - a)x$, with $x \in [0, 1]$. This gives a normalised parameter range so that each parameter may be plotted on the

[13] Filaments beyond 25 : 1 are physically possible [Feh97] but excluded because their short axes give Ro > 1.
[14] under two readings (full extent to 5 % of U, or full width at half maximum)

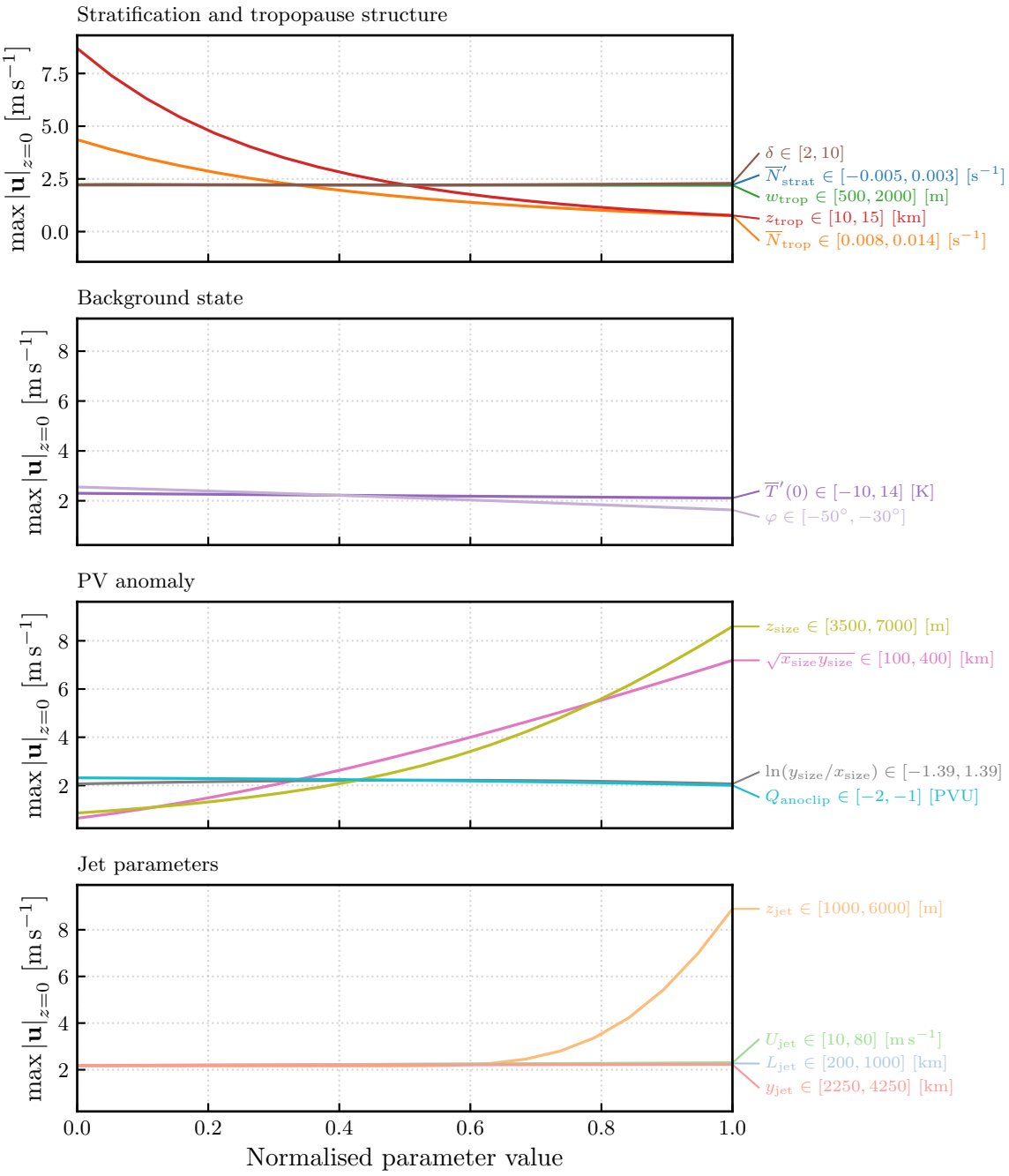


Figure 18: Effect of taking the control state and varying a single parameter at a time on maximum surface wind speed.

same set of axes.

It can be seen in figure 18 that surface wind has a trivial and potentially non-physical relationship with z_{trop}. The Gaussian jet stream (equation (51)) induces its own wind upon the surface once z_{jet} is large enough, and thus the surface wind is not caused by the anomaly. This is supported by the exponential profile of the z_{jet} line in figure 18, following the exponential increase of effectively moving the Gaussian closer. This effect also appears to occur in figure 19, increasing the magnitude of the surface pressure anomaly by inducing baroclinity via non-cyclonic wind.

Comparing figure 20 with figures 18 and 19 reveals that anomaly size affects surface wind and the pressure anomaly much more than minimum surface vorticity. This agrees with the findings of Barnes et al. [Bar+22]. Moreover, figures 18 to 20 show that a weakly stratified troposphere moderately intensifies surface effects, even within physically plausible ranges established in table 2.

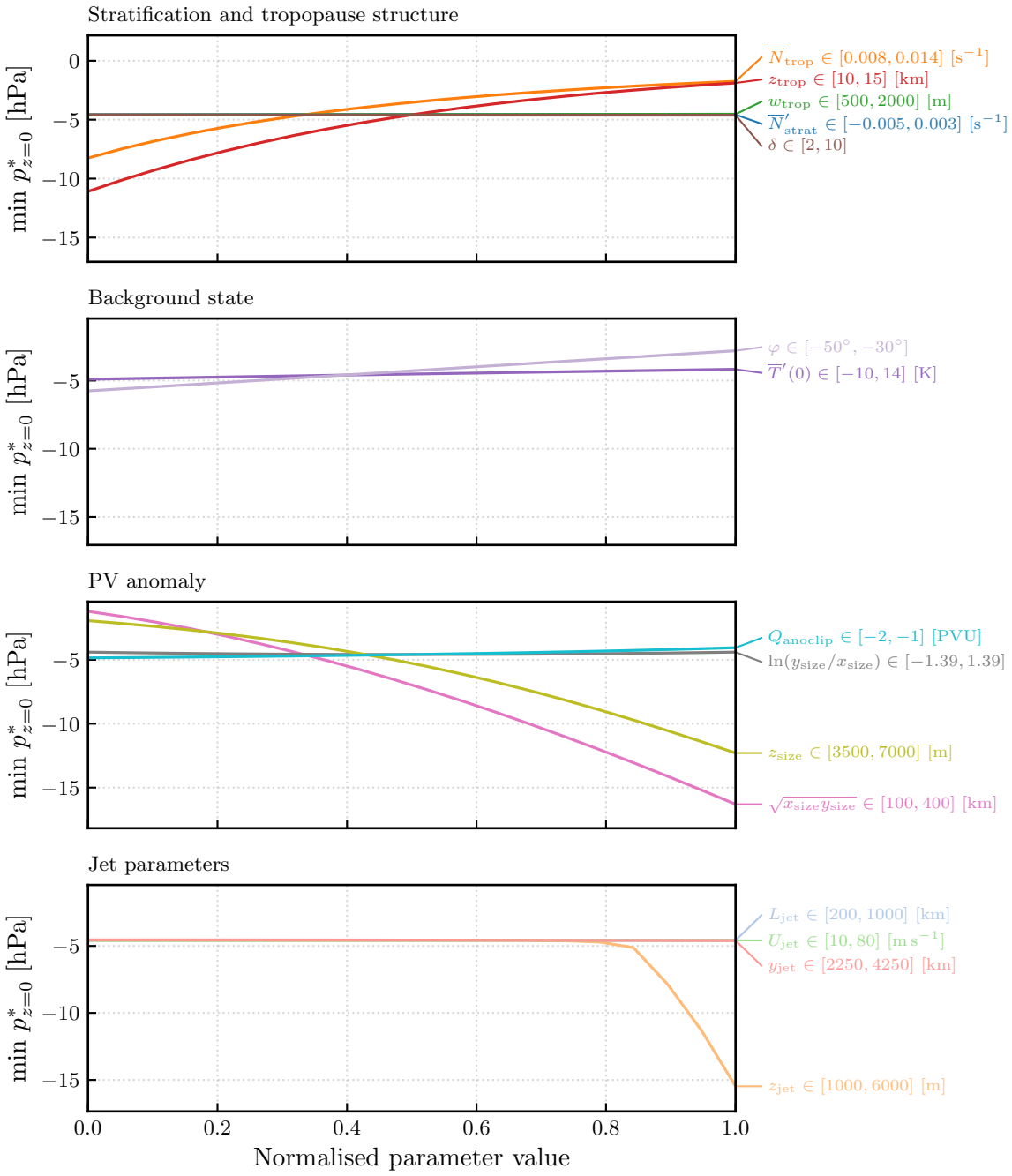


Figure 19: Effect of taking the control state and varying a single parameter at a time on minimum surface pressure anomaly.

Jet stream parameters are what determine $\partial\overline{u}/\partial z$ and hence drive $\mathcal{V}_-$ away from 0, presenting difficulties in justifying the validity of QGPV inversion in certain regions of the parameter space. Interestingly however, U_{jet}, L_{jet} and y_{jet} have almost no effect on the surface, which may justify using weaker jet streams (or potentially none at all) when performing parameter sweeps, further allowable by the linearity of equation (37). Expectedly due to linearity, the surface vorticity (figure 20), the jet has no effect at all.

Increasing latitude consequently increases f, and thereby increases the magnitude of the stretching term in equation (37), likely inducing the slight poleward bias towards the stronger surface effects visible in figures 18 to 20.

A strange and unexplained relationship is the sharp step of the y_{jet} line in figure 21. As the jet is offset further poleward from the anomaly, z_{dyn} exhibits a slight decrease. However at the normalised parameter value of 0.6, z_{dyn} is sharply pulled up to a 10 km altitude. This corresponds to $y_{\text{jet}} \approx L_y/2 - 300$ km. The

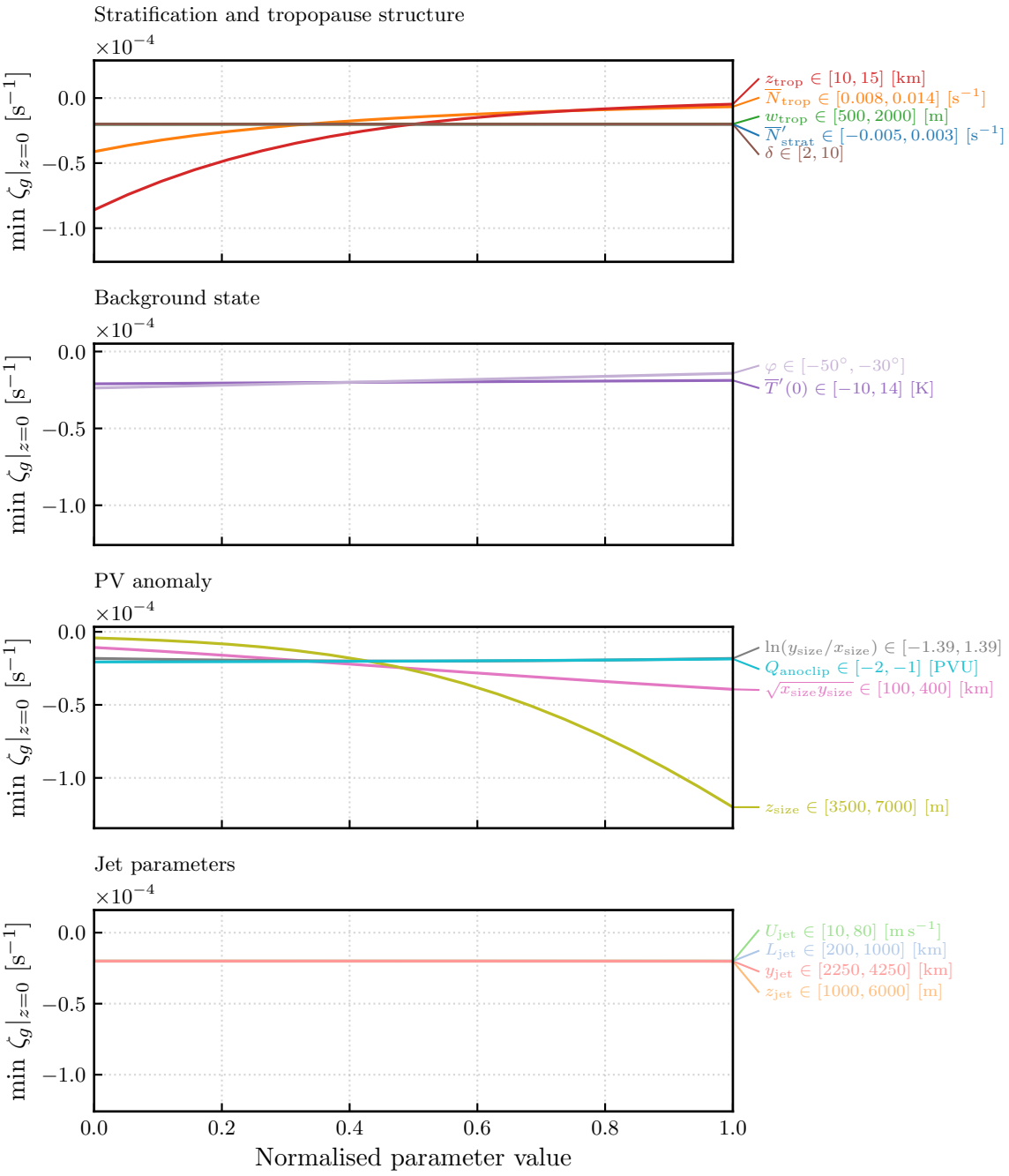


Figure 20: Effect of taking the control state and varying a single parameter at a time on minimum surface vorticity.

phenomenon may be related to $L_{\text{jet}} = 500\,\text{km}$ in the control parameters, but no explanation is offered at present.

Figures 18 to 21 show common behaviour, shedding some light on which atmospheric and anomaly parameters most significantly affect the surface across multiple metrics. The tropopause thickness parameters w_{trop} and δ have almost no measurable effect on any of the plots, most surprisingly figure 21. This is reassuring as it shows that how the tropopause is treated numerically is not important when calculating surface effects. The same is true for $\overline{N}'_{\text{strat}}$, which could be why Barnes et al. use an unrealistic value for $\overline{N}_{\text{strat}}$ [Bar+22]; it simply does not matter on the surface. Furthermore, the aspect ratio of the anomaly $y_{\text{size}}/x_{\text{size}}$ unexpectedly has no effect on the surface wind, vorticity or pressure anomaly. Surface temperature variation $\overline{T}'(0)$ also has very little to no effect on any of the plots, with the exception of causing slight variation in z_{dyn} in figure 21.

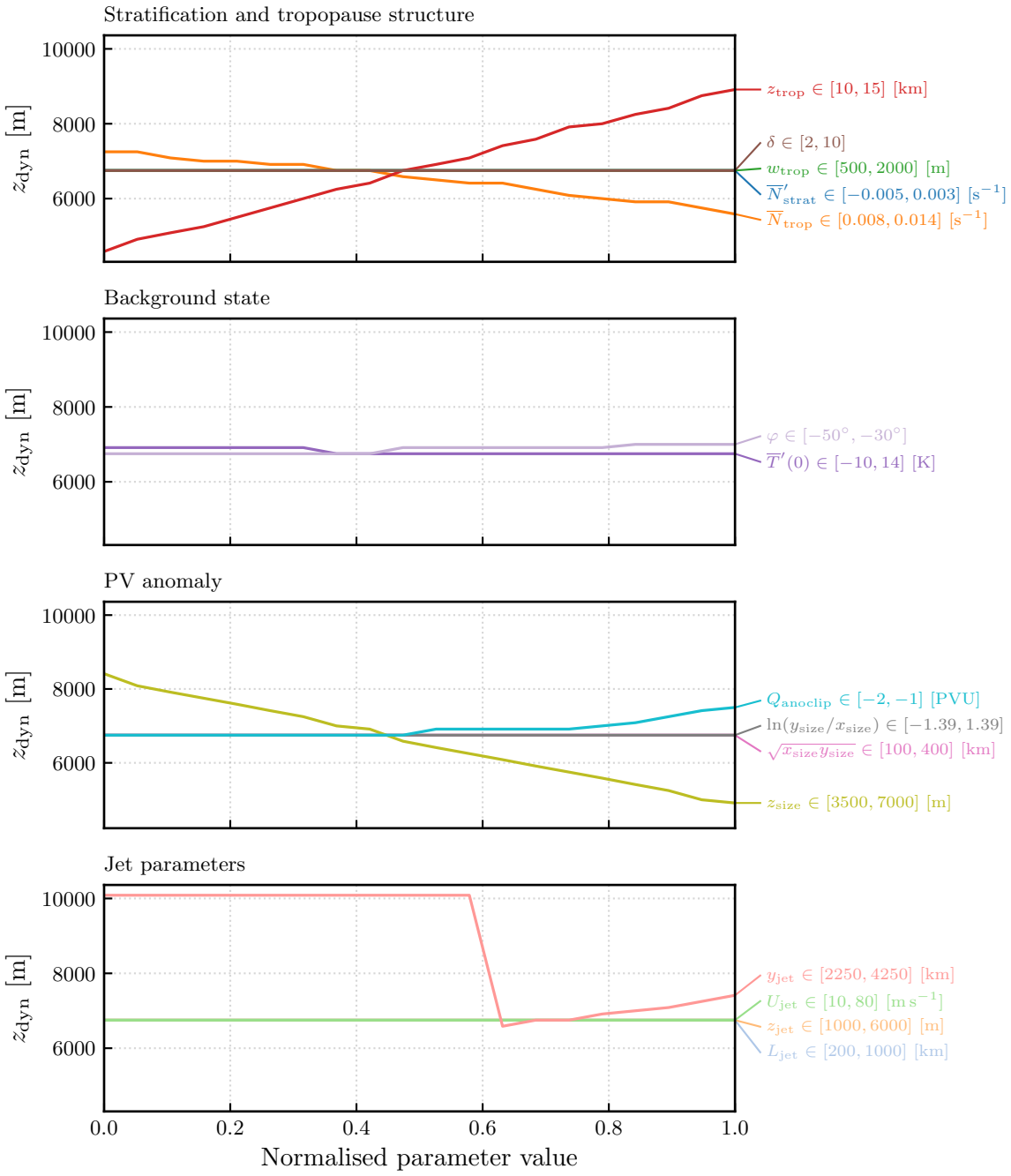


Figure 21: Effect of taking the control state and varying a single parameter at a time on minimum dynamical tropopause height. Step-like behaviour is due to z_{dyn} only being resolved to the nearest degree of freedom, and is hence discrete.

In accordance with the findings of Barnes et al., z_{trop} and z_{size} have a significant effect on all three surface quantities [Bar+22], and have a trivially strong relationship with z_{dyn}.

Of figures 18 to 21, only figure 21 appears to show a non-negligible effect induced by Q_{anoclip}. This is likely caused in part by its range being much smaller than most of the other parameters.

Barnes et al. [Bar+22] found that the minimum dynamical tropopause height is the major factor in surface cyclogenetic forcing, more so than the intrusion's vertical extent. We can view the relationship between the surface variables and minimum dynamical tropopause height by reusing the data from figures 18 to 21.

Most of the observations above arising from figures 18 to 21 are also apparent in figure 22. The parameters that increase surface effects are also correlated with a lowered z_{dyn}. One of the noticeable exceptions is $\overline{N}_{\mathrm{trop}}$. $\overline{N}_{\mathrm{trop}}$ brings *up* z_{dyn} while increasing surface effects. Another exception is anomaly

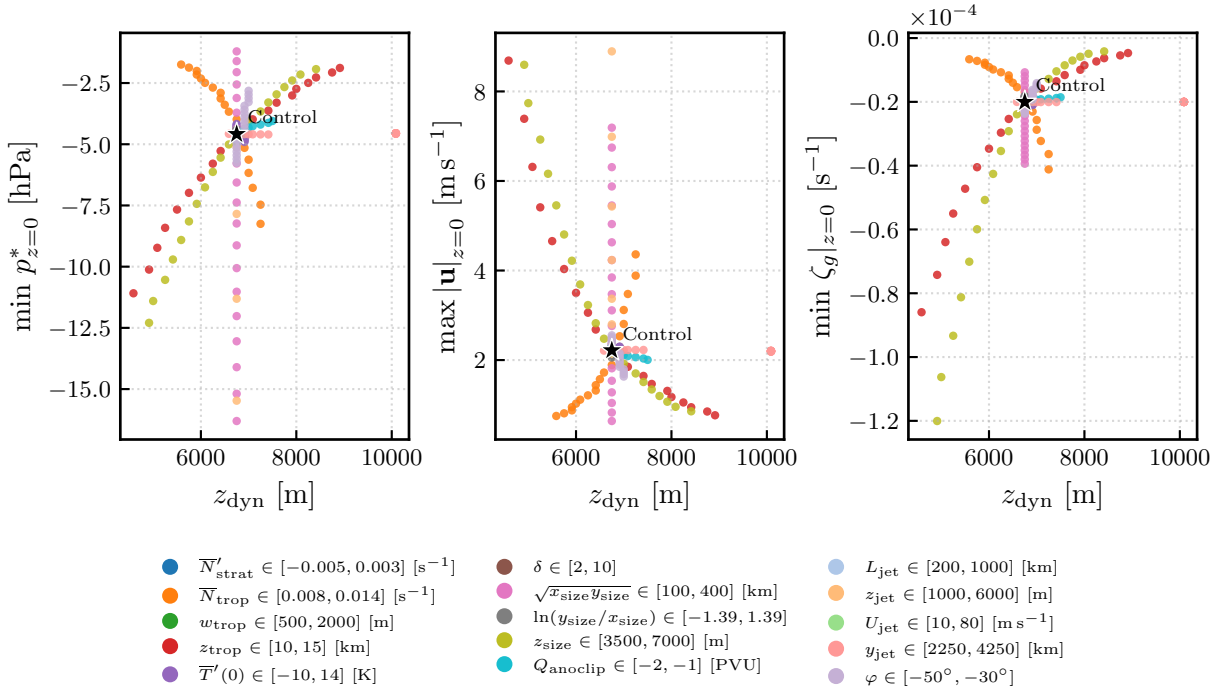


Figure 22: Minimum dynamical tropopause height versus minimum surface pressure anomaly, maximum surface wind speed and minimum surface vorticity. No restrictions are placed on $\mathcal{V}_-$, Ro or Fr. Parameters are varied one at a time from the control parameters.

size. Anomaly size does not affect z_{dyn}, but has a major effect on $p^*_{z=0}$ and $|\boldsymbol{u}|_{z=0}$. A wider anomaly makes the surface low deeper, but it also spreads the low over a larger area. Those two effects largely cancel in the vorticity, while pressure and wind keep growing [Bar+22].

Since no restrictions are placed on whether each parameter evaluation results in a QG-invertible or physically plausible ($\mathcal{V}_- = 0$) atmosphere, these findings may not hold in full EPV inversion. Furthermore, nonlinear relationships between parameters are not possible to investigate by varying a single parameter at a time.

7.3 Random Sampling

In an effort to address the issues stated above, we investigate the relationship between the surface variables and z_{dyn} by randomly and uniformly sampling parameter vectors from the ranges outlined in table 2. The linearity of equation (37) allows us to split the inversion into an anomaly part (inverting q^* gives ψ^*) and a background part (directly calculate $\overline{q}$ from $\overline{\psi}$). The full streamfunction can easily be recovered by computing $\psi = \overline{\psi} + \psi^*$. This allows us to safely remove all jet related parameters from the sweep, and effectively set $U_{\text{jet}} = 0$. We reject any samples with $\mathcal{V}_- > 0$, and colour each sample point by its Rossby number. Accumulating 1000 valid samples gives figure 23.

We employ the Spearman ρ to quantify the strength of the monotonic relationship between each surface variable and z_{dyn}. Figure 23 shows a moderate monotonic relationship between z_{dyn} and the surface variables, with vorticity being on the edge of what could be called a strong relationship [SBS18].

However, most of the atmospheric states in figure 23 are beyond the range of Ro for which QGPV is quantitatively or qualitatively useful (as discussed in section 6.2). To address this we drop any points with Ro ≥ 1 to produce figure 24.

Figure 24 now only consists of points which fall broadly within the range that QGPV has (at minimum) qualitative usefulness. For quantitative usefulness, we require Ro $\ll 1$, but 0.1% of the sampled points have Ro < 0.5, so this was impractical to test. As $|\rho|$ decreases from figure 23 to figure 24, the monotonic relationship weakens in each panel. For the atmospheric states QGPV can model with qualitative accuracy, at most a weak monotonic relationship between minimum surface pressure anomaly and

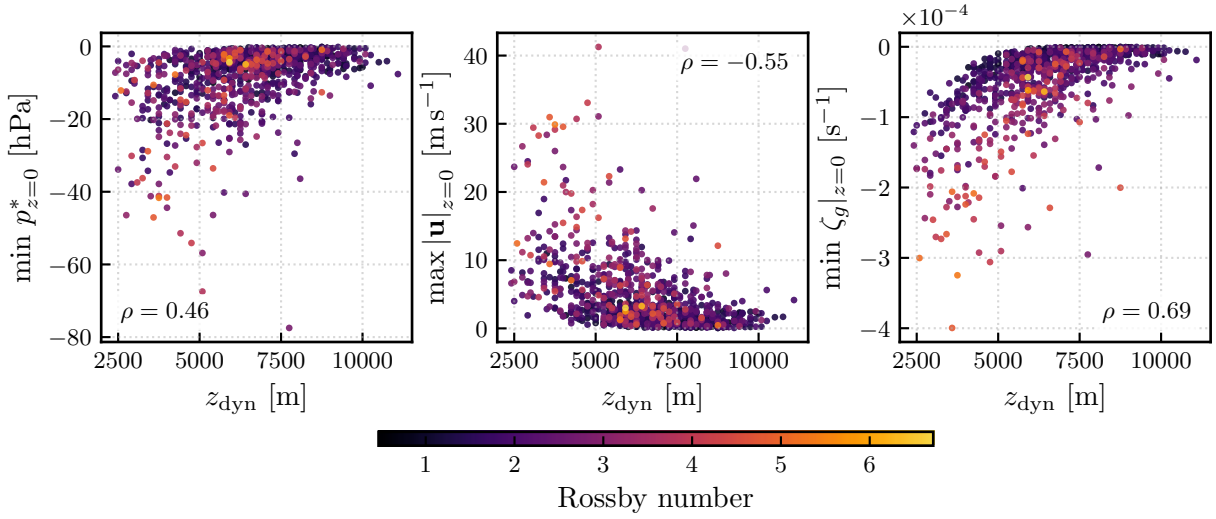


Figure 23: 1000 randomly sampled parameters from table 2 with $\mathcal{V}_- = 0$ and $U_{\text{jet}} = 0$. Each point corresponds to an individual atmospheric state, and is coloured by its Rossby number.

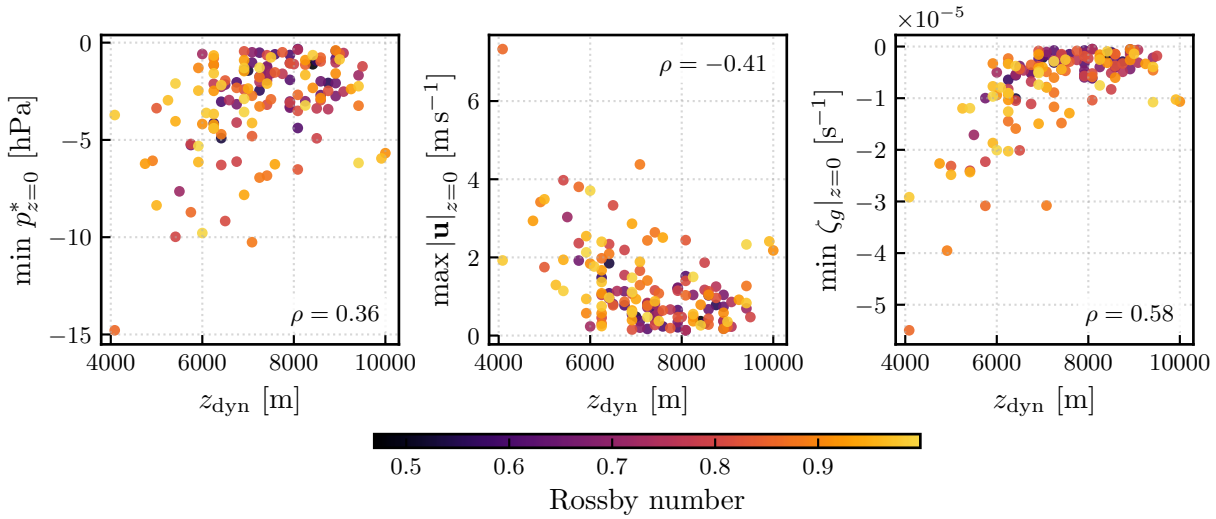


Figure 24: Same as figure 23, but only the 14.5% of points with Ro < 1.

minimum dynamical tropopause height exists. The relationship with maximum surface wind speed is slightly stronger, with minimum surface vorticity being the strongest with a moderate correlation.

The above findings support the conclusion that minimum surface vorticity, minimum surface pressure anomaly and maximum surface wind speed are somewhat driven by the minimum dynamical tropopause height induced by an anomaly, however the combination of other parameters generally appears to be more important than the minimum dynamical tropopause height, even in a highly idealised setting.

7.4 Maximising Surface Response

Automating the search for parameters that maximise surface response requires setting up an optimisation problem. We first reduce the dimensionality of the problem by choosing to not vary w_{trop}, δ and $\overline{N}'_{\text{strat}}$, as section 7.2 shows that they have negligible effect on the surface. We also only look for the surface response induced by the anomaly, by removing the jet parameters as done above. Using the parameter normalisation scheme from section 7.2, we represent any given choice of parameters as a vector $\boldsymbol{x} \in [0,1]^d$, where d denotes the dimensionality or number of parameters we allow to vary. In this case $d = 8$. Computing derivatives of $\min p^*_{z=0}$ and $\min -|\boldsymbol{u}|_{z=0}$ with respect to each of the parameters in table 2 is

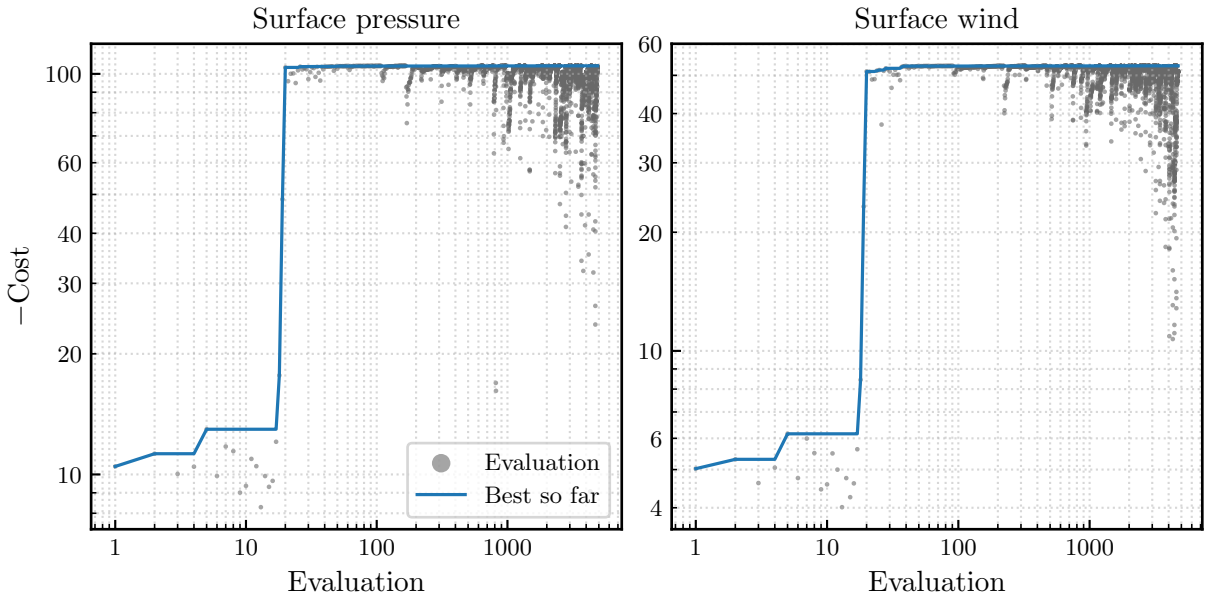


Figure 25: Convergence of Py-BOBYQA towards the global minimum of $C(\boldsymbol{x})$, for both objectives O. 5000 calls were made to $C(\boldsymbol{x})$ in each case, but both reached this minimum in under 100 cost function evaluations.

non-trivial due to the highly nonlinear relationships between some parameters and the surface variables. For example, one of the chains of dependence goes as $\min p^*_{z=0} \to p^* \to \psi \to q \to Q^* \to z_{\text{size}}$, where the arrows each represent complex or nonlinear dependence relations. This rules out any optimsation schemes which require gradients, such as variants of gradient descent.

Instead we opt for a derivative-free bounded optimiser. One such method is BOBYQA [Pow09], implemented in Python as Py-BOBYQA [Car+19]. It can be configured to seek the global minimum, as opposed to a local one [CRS22], which improves the exhaustiveness of our parameter sweep. We begin the implementation by defining a cost function we seek to minimise:

$$C(\boldsymbol{x}) = \frac{O}{\tilde{\text{Ro}} \cdot \tilde{\text{Fr}}} + \mu \mathcal{V}_- \tag{85}$$

Here O is the objective we wish to minimise, either $\min -|\boldsymbol{u}|_{z=0}$ or $\min p^*_{z=0}$. The penalty parameter which drives $\mathcal{V}_- \to 0$ is $\mu = 10^6$, quickly pushing the optimiser away from parameters that cause $\mathcal{V}_- > 0$. The denominator rewards small Rossby and Froude numbers, and punishes large ones, telling the solver to find a balance between surface effect and QG validity. Large $\text{Ro}, \text{Fr} > 1$ push $C(\boldsymbol{x})$ closer to zero, when interesting values of O are ~ -10. We use clipped $\tilde{\text{Ro}} = \max\{\text{Ro}_{\max}, 0.3\}$ and $\tilde{\text{Fr}} = \max\{\text{Fr}_{\max}, 0.3\}$ to prevent the optimiser from finding that sending $\text{Ro}\cdot\text{Fr} \to 0$ is the most effective way to send $C(\boldsymbol{x}) \to -\infty$, thereby polluting the results. We configure Py-BOBYQA to run on a single MPI rank (see section A.5), while allowing the inversion of equation (37) to run on all ranks.

Running the optimiser for both objectives – maximum surface wind speed and minimum surface pressure anomaly – yields figure 25. Py-BOBYQA is configured to call $C(\boldsymbol{x})$ a maximum of 5000 times in each case – an arbitrarily chosen limit dictated by computation time.

The parameters and resulting atmospheric state at the minimum found for each objective are compared to the control state in table 4.

An interesting effect of the penalty scheme in equation (85) is that the pressure-optimised sweep results in stronger surface wind when compared to the wind-optimised sweep. This is caused by the lower Rossby number of the wind sweep pushing $|C(\boldsymbol{x})|$ upwards more than a higher O would.

One would assume that an anomaly with a larger absolute EPV would induce stronger surface effects, however the optimiser finds the optimal Q_{anoclip} to be the smallest $|Q|$ allowed. A possible explanation is that the Ro penalty results in the strategy that keeping $|Q|$ small and anomaly size large is more

Table 4: Control state compared to the minima of $C(\boldsymbol{x})$ found by Py-BOBYQA for each objective O. † marks a value on the edge of its range in table 2. Computed with $N = 50$, $p = 4$.

			Optimised for	
Quantity	Unit	Control	$\max \lvert\boldsymbol{u}\rvert_{z=0}$	$\min p^*_{z=0}$
Background state				
φ	°	−42	$-50^\dagger$	$-50^\dagger$
$\overline{T}(0)$	°C	11.28	$20.05^\dagger$	$-3.95^\dagger$
$\overline{N}_{\text{trop}}$	s^{-1}	0.0100	0.0121	0.0123
$\overline{N}_{\text{strat}}$	s^{-1}	0.0300	0.0289	0.0289
z_{trop}	km	12.5	$10.0^\dagger$	$10.0^\dagger$
U_{jet}	m s^{-1}	35	0	0
Anomaly				
x_{size}	km	200	333	399
$x_{\text{size}}/y_{\text{size}}$	–	1.00	0.693	0.997
z_{size}	km	5.00	$7.00^\dagger$	6.90
Q_{anoclip}	PVU	−1.5	$-1.0^\dagger$	$-1.0^\dagger$
Response				
$\min p^*_{z=0}$	hPa	−4.58	−43.4	−51.7
$\max \lvert\boldsymbol{u}\rvert_{z=0}$	m s^{-1}	2.22	23.8	25.3
$\min \zeta_g\rvert_{z=0}$	$10^{-5}\,\text{s}^{-1}$	−2.00	−16.8	−18.3
z_{dyn}	km	6.75	2.91	3.25
Validity				
$\text{Ro}_{\max}$	–	1.95	1.50	1.65
$\text{Fr}_{\max}$	–	0.800	0.299	0.300
$\mathcal{V}_-$	–	0.0391	0	0

beneficial to minimising $C(\boldsymbol{x})$.

Another effect of the penalty terms is that $\overline{N}_{\text{trop}}$ goes *up* rather than down. It follows that reduced tropospheric stratification would increase the Rossby depth and hence increase the surface response [HH13b]. The $1/\tilde{\text{Fr}}$ penalty term causes the optimiser to find that increasing $\overline{N}_{\text{trop}}$ is more beneficial to the cost function than the increased surface response that would be associated with a smaller $\overline{N}_{\text{trop}}$.

The above findings that maximising z_{size}, x_{size} and $-z_{\text{trop}}$ in turn maximises surface response are reinforced by table 4.

Moving φ as far poleward as the bounds allow is purely beneficial. It increases f, reduces Ro and increases the Rossby depth [HH13b]. All of these decrease $C(\boldsymbol{x})$.

In figures 26 and 27 we see the slightly different anomaly shapes of the parameters from table 4 still have notable effects on $\min p^*_{z=0}$ and $\max \lvert\boldsymbol{u}\rvert_{z=0}$. This implies that anomaly shape is a relevant factor in determining the nature of surface effects caused by an anomaly. A possible explanation is that $\max \lvert\boldsymbol{u}\rvert_{z=0}$ is a gradient quantity at a single point, so an elongated anomaly tightens the gradient along its narrow axis.

Compare the pressure anomaly profile of the pressure-optimised (figure 29) with the control case (figure 14). The pressure-optimised parameters effectively push the region with the greatest magnitude of pressure anomaly to the surface, whereas in figure 14 it sits at an altitude of about 7.5 km. This is part of the reason why the control's $\lvert\min p^*_{z=0}\rvert$ is much smaller than the pressure-optimised case.

Despite providing Py-BOBYQA incentive to bring $\text{Ro} \to 0.3$, while still maximising surface response, the optimal balance found for $C(\boldsymbol{x})$ resulted in $\text{Ro}_{\max} \geq 1.5$ for both objectives. This is far beyond the range for which QGPV has quantitative usefulness. As discussed in section 6.2, QGPV may still have qualitative usefulness at these $\text{Ro} \sim 1$, but to what extent the qualitative findings from above are true is not known. This points towards two possible explanations:

1. Equation (85) improperly balances Ro against O. Perhaps a linear clipped constraint such as the

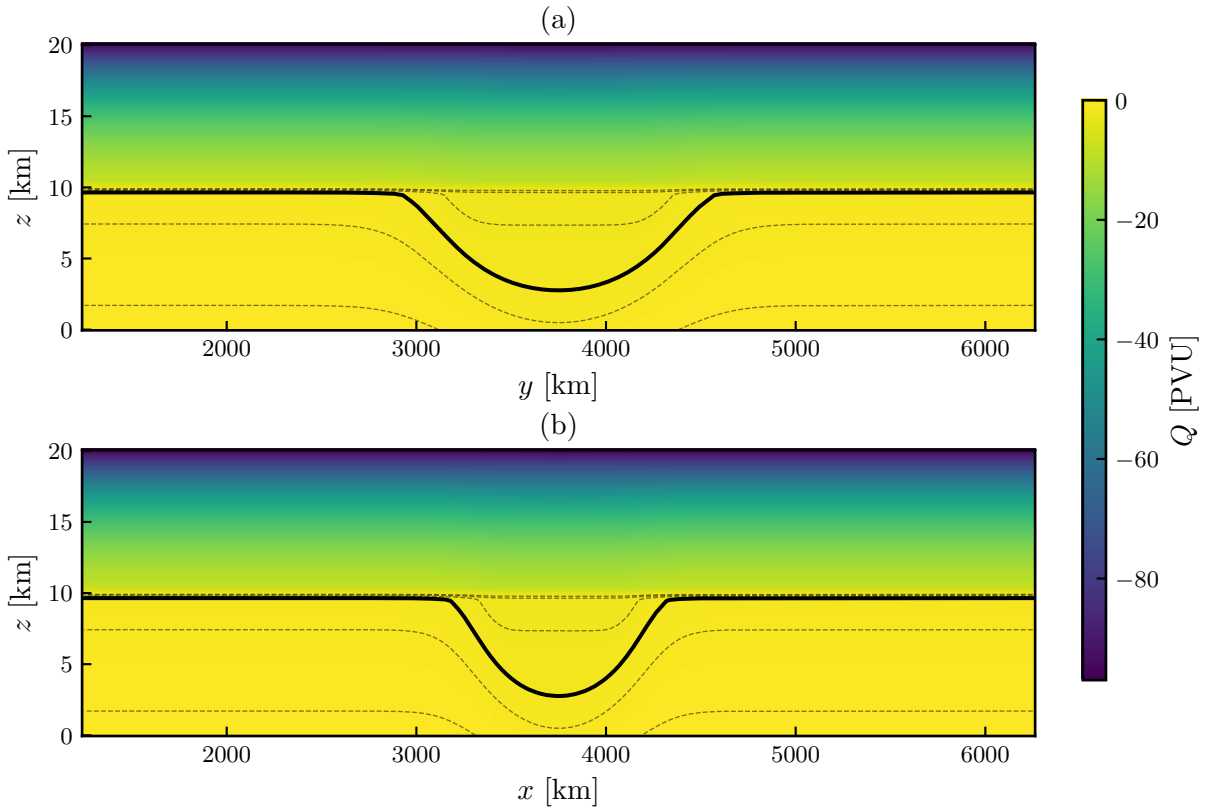


Figure 26: EPV distribution of the wind-optimised parameters in table 4. A contour interval of 0.5 PVU (Potential Vorticity Units) is used. The thick solid line represents the dynamical tropopause – the −1.5 PVU contour.

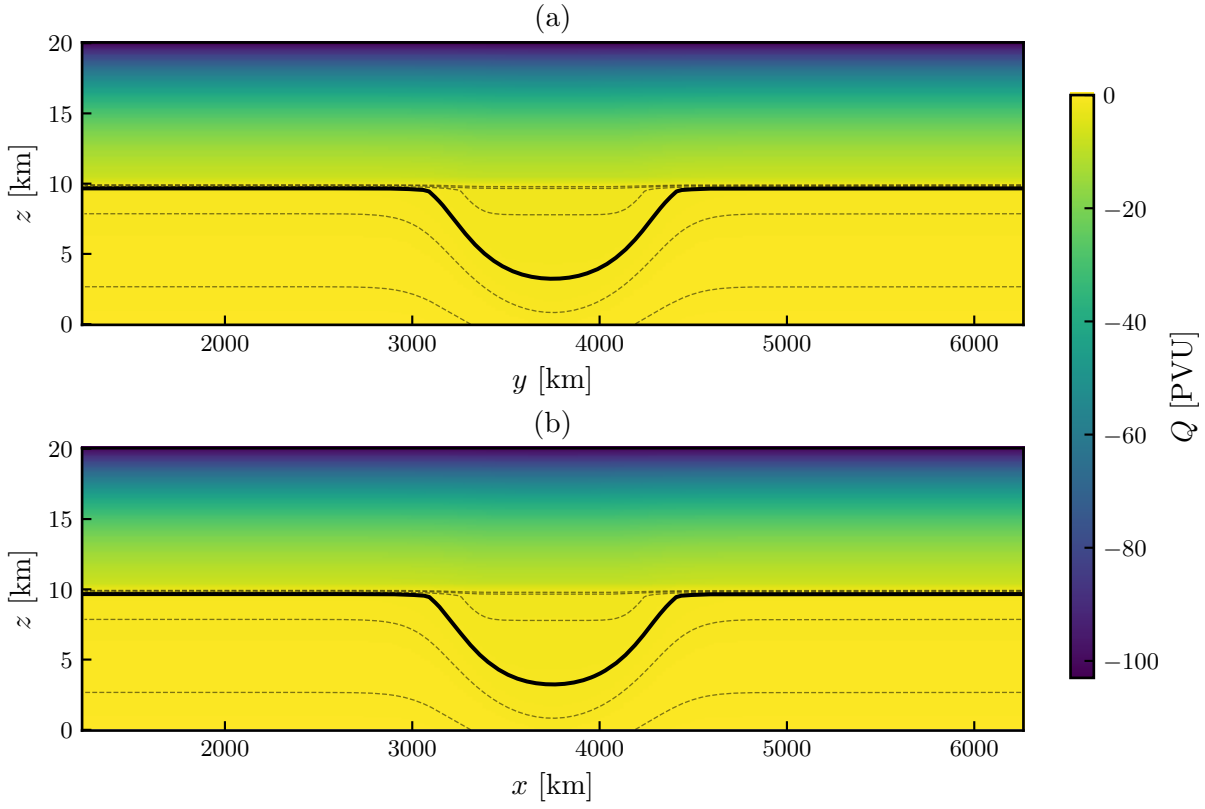


Figure 27: EPV distribution of the pressure-optimised parameters in table 4.

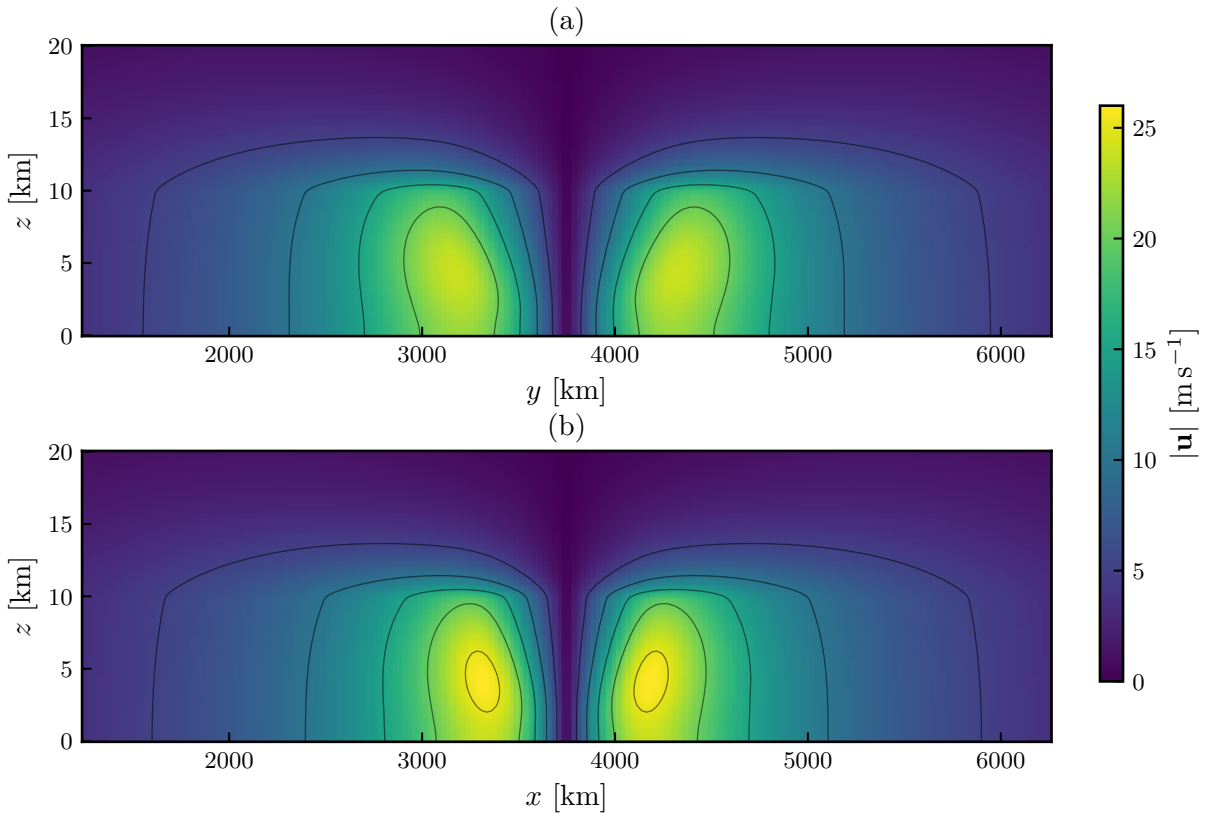


Figure 28: Cross sections of the wind induced by the wind-optimised parameters from table 4.

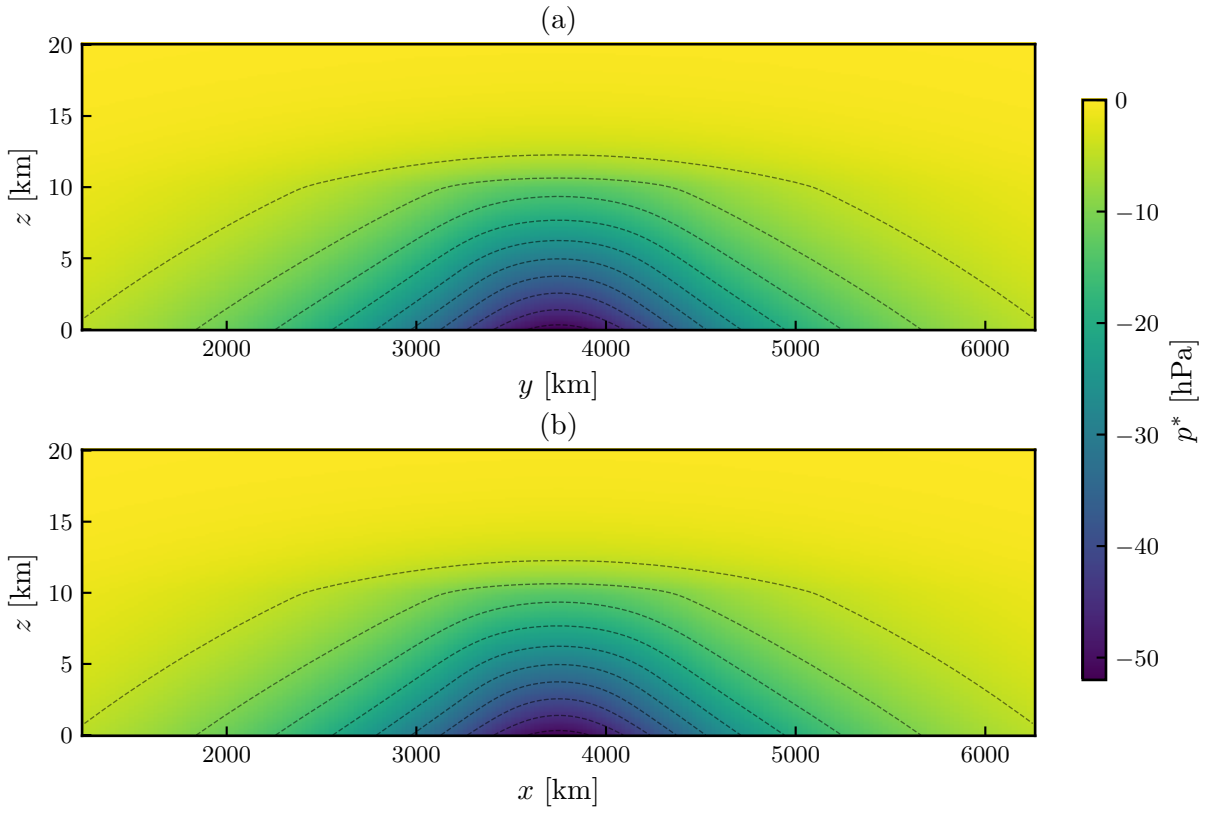


Figure 29: Cross sections of the pressure anomaly induced by the pressure-optimised parameters from table 4.

$\mu\mathcal{V}_-$ term would prove more effective.

2. Alternatively, pure QGPV is ill-suited to a parameter sweep intended to ascertain the relationship between atmospheric parameters and surface response. The Ro $\ll 1$ requirement may be too strict to investigate the effects of a reasonably deep, large or high magnitude stratospheric intrusion.

In extremely rare cases, the -1.5 PVU contour of COLs reaches down to the 700 hPa level [BNL21; Ske+15], which corresponds to similar minimum dynamical tropopause heights (z_{dyn}) as shown in table 4 [Stu15].

8 Numerical Time-Evolution

8.1 Function Space

To assess the longevity and general time-evolution of atmospheric parameters we employ the prognostic QGPV equation (45). This equation consists of purely horizontal advection, so the Continuous-Galerkin elements (the `'Q'` function space) used to compute ψ will certainly cause spurious oscillations if used to advect q[15].

Instead we employ Discontinuous-Galerkin elements (the `'DQ'` function space on tensor-product meshes), which let q jump across cell boundaries. The cells are coupled only through a numerical flux on each facet. Both neighbouring cells share it, so the scheme conserves mass to round-off. We take this flux from the upwind side, which supplies the dissipation that stabilises pure advection [MSP15].

Unlike with finite difference schemes, we can not compute $\partial_t q$ by substituting q directly into equation (45). Instead we must solve the weak form equation (86) for each stage of the timestepping scheme to obtain $\partial_t q$.

8.2 Weak Form

To set up equation (45) to be solvable with the finite element method, we again multiply by a test function ϕ and integrate. Since ϕ and q may now jump across cell boundaries, we integrate over each cell K of the mesh separately, rather than over the whole domain V at once. We use the wind $\boldsymbol{u} = (-\psi_y, \psi_x, 0)$ (equation (80)) to make this process more concise.

$$\int_K \phi \frac{\partial q}{\partial t}\, dV + \underbrace{\int_K \phi\, \boldsymbol{u} \cdot \nabla q\, dV}_{I_K} = 0 \tag{86}$$

Now we apply the divergence theorem to the vector field $\phi q \boldsymbol{u}$ over K. Here ∂K is the surface that encloses K, with outward normal vector $\hat{\boldsymbol{n}}$.

$$\int_K \nabla \cdot (\phi q \boldsymbol{u})\; dV = \int_{\partial K} \phi q\, \boldsymbol{u} \cdot \hat{\boldsymbol{n}}\, dA \tag{87}$$

Expanding the divergence with the product rule, $\nabla \cdot (\phi q \boldsymbol{u}) = \phi\, \boldsymbol{u} \cdot \nabla q + q \nabla \cdot (\phi \boldsymbol{u})$, yields the following for I_K (86):

$$I_K = \underbrace{\int_{\partial K} \phi q\, \boldsymbol{u} \cdot \hat{\boldsymbol{n}}\, dA}_{I_{Kb}} - \int_K q \nabla \cdot (\phi \boldsymbol{u})\; dV \tag{88}$$

Note that $\nabla \cdot \boldsymbol{u} = -\psi_{yx} + \psi_{xy} = 0$, so $\nabla \cdot (\phi \boldsymbol{u}) = \boldsymbol{u} \cdot \nabla \phi$. This holds exactly for the numerical ψ as well, since it is a polynomial within each cell, so its mixed derivatives commute. Choosing $\phi = 1$ on a single cell (and 0 elsewhere) then removes the volume term entirely, so the amount of q in a cell can only change through the flux across its surface.

[15] Discussed in more detail in section 5.3

Since $\boldsymbol{u}$ has no vertical component, $\boldsymbol{u} \cdot \hat{\boldsymbol{n}} = 0$ on every horizontal face, including the top and bottom of the domain, so I_{Kb} only has contributions from vertical faces. Each vertical face either lies on the lateral boundary A_v of the domain, or is an interior face Γ shared by two cells, which we label K^+ and K^- such that $\hat{\boldsymbol{n}}^+ = -\hat{\boldsymbol{n}}^-$. On an interior face $\boldsymbol{u} \cdot \hat{\boldsymbol{n}}^+ = -\psi_y n_x^+ + \psi_x n_y^+$ is the derivative of ψ along the face, which is single-valued because ψ is continuous (`'Q'`). However q takes two values q^+ and q^-, so we replace it with a single numerical flux $\hat{q}$, taken from the upwind side [CS91; Ham+23]:

$$\hat{q} = \begin{cases} q^+ & \boldsymbol{u} \cdot \hat{\boldsymbol{n}}^+ > 0 \\ q^- & \boldsymbol{u} \cdot \hat{\boldsymbol{n}}^+ \leq 0 \end{cases} \tag{89}$$

Summing I_{Kb} over all cells, each interior face receives a contribution from both of its cells, $\phi^+ \hat{q}\, \boldsymbol{u} \cdot \hat{\boldsymbol{n}}^+ + \phi^- \hat{q}\, \boldsymbol{u} \cdot \hat{\boldsymbol{n}}^- = (\phi^+ - \phi^-) \hat{q}\, \boldsymbol{u} \cdot \hat{\boldsymbol{n}}^+$. What leaves one cell therefore enters its neighbour, which is why the scheme conserves q. We write the upwinding in terms of the outflow speed $u_n = \frac{1}{2}(\boldsymbol{u} \cdot \hat{\boldsymbol{n}} + |\boldsymbol{u} \cdot \hat{\boldsymbol{n}}|)$, which is nonzero only on the side of the face that $\boldsymbol{u}$ flows out of:

$$\hat{q}\, \boldsymbol{u} \cdot \hat{\boldsymbol{n}}^+ = u_n^+ q^+ - u_n^- q^- \tag{90}$$

On the lateral boundary A_v there is only one cell, so the interior value of q is used where $\boldsymbol{u}$ flows out of the domain ($\boldsymbol{u} \cdot \hat{\boldsymbol{n}} > 0$). Where $\boldsymbol{u}$ flows in ($\boldsymbol{u} \cdot \hat{\boldsymbol{n}} < 0$), q is carried in from outside the domain, which equation (45) does not determine; we hold it at its initial value, $q_{\text{in}}(x, y, z) = q(x, y, z, 0)$. We then sum equation (86) over all cells, substitute everything and move terms around to get the form $a(\partial_t q, \phi) = L(\phi)$:

$$\begin{aligned} a(\partial_t q, \phi) &= \int_V \phi \frac{\partial q}{\partial t}\, dV \\ L(\phi) &= \int_V q \nabla \cdot (\phi \boldsymbol{u})\, dV - \int_{A_v^{\text{in}}} \phi q_{\text{in}}\, \boldsymbol{u} \cdot \hat{\boldsymbol{n}}\, dA \\ &\quad - \int_{A_v^{\text{out}}} \phi q\, \boldsymbol{u} \cdot \hat{\boldsymbol{n}}\, dA - \int_\Gamma \left(\phi^+ - \phi^-\right) \left(u_n^+ q^+ - u_n^- q^-\right) dA \end{aligned} \tag{91}$$

Here $\int_V$ denotes the sum of the integrals over every cell, and A_v^{in} and A_v^{out} are the parts of A_v where $\boldsymbol{u} \cdot \hat{\boldsymbol{n}} < 0$ and $\boldsymbol{u} \cdot \hat{\boldsymbol{n}} > 0$ respectively. Now equation (45), with the upwind flux and inflow boundary condition embedded, is in a weak form that can easily be expressed in Firedrake's syntax[16].

L depends on q both directly and through $\boldsymbol{u}$, which is calculated from the ψ obtained by inverting equation (73) for the current q. Every evaluation of L therefore requires a diagnostic solve first. Solving $a(\partial_t q, \phi) = L(\phi)$ for $\partial_t q$ then gives the time derivative of q, which the time stepping scheme advances.

The bilinear form a is simply the mass matrix of the `'DQ'` space. Like α, it is symmetric, but evaluating

$$a(\phi, \phi) = \int_V \phi^2\, dV \tag{92}$$

shows that $a(\phi, \phi) > 0$ for all $\phi \neq 0$, so it is positive definite rather than only semi-definite, and no solvability condition (section 5.5) is required. Furthermore, each `'DQ'` basis function is only nonzero within a single cell, so the matrix is block diagonal, with one $(p+1)^3 \times (p+1)^3$ block per cell, and a does not depend on q or t, so the matrix stays the same between time steps.

8.3 Limiter

Since equation (45) describes pure advection by a divergence-free wind, q is only carried around the domain, and is never created or destroyed. With no source term and the inflow held at $q_{\text{in}}(x, y, z) = q(x, y, z, 0)$, the exact solution therefore obeys a maximum principle: it can never leave the range of its initial values [ZS10].

[16] shown in section A.8

$$m \leq q(x,y,z,t) \leq M, \qquad m = \min_V q(x,y,z,0), \quad M = \max_V q(x,y,z,0) \tag{93}$$

The upwind scheme of section 8.2 does not inherit this property. For linear advection, Godunov's theorem states that any linear scheme which does not create new extrema is at most first-order accurate [CS91]. For a given wind, the scheme is linear in q, so at $p = 4$ it must produce overshoots and undershoots wherever q has gradients too steep for the polynomials to resolve, such as the edges of the PV anomaly. These errors do not stay local: every spurious extremum in q is inverted into ψ, and hence into the wind that advects q at the next stage.

Classical slope limiters flatten the polynomial in cells that disagrees with the means of its neighbours [CS91]. However, they can not distinguish an oscillation from a smooth extremum, so they also clip the latter, which reduces the accuracy there unless a problem-dependent parameter is tuned [CS91]. Firedrake's built-in vertex-based limiter is of this type, but implements only the $p = 1$ case of Kuzmin's vertex-based limiter [Kuz10], which clips smooth peaks.

Instead we implement the maximum-principle-satisfying limiter of Zhang and Shu [ZS10], which enforces equation (93) directly rather than searching for oscillations. In each cell K, the polynomial is scaled about its cell mean $\langle q \rangle_K$:

$$\tilde{q} = \langle q \rangle_K + \theta_K \left(q - \langle q \rangle_K \right), \qquad \langle q \rangle_K = \frac{1}{|K|} \int_K q \, dV \tag{94}$$

The scaling factor $\theta_K \in [0, 1]$ is chosen to be as large as possible while keeping $\tilde{q}$ within bounds:

$$\theta_K = \min \left\{ 1, \; \frac{M - \langle q \rangle_K}{M_K - \langle q \rangle_K}, \; \frac{m - \langle q \rangle_K}{m_K - \langle q \rangle_K} \right\} \tag{95}$$

Here M_K and m_K are the maximum and minimum of q over a set of points S_K in K, and each ratio is replaced by 1 unless its bound is violated ($M_K > M$ or $m_K < m$ respectively).

Integrating equation (94) over K shows that $\int_K \tilde{q} \, dV = \int_K q \, dV$, so the limiter leaves every cell mean unchanged, and the scheme still conserves q. Each value of $\tilde{q}$ on S_K lies between $\langle q \rangle_K$ and the corresponding value of q, and θ_K is just small enough to pull the most extreme of them back onto the violated bound. Cells that are already within bounds have $\theta_K = 1$ and are left untouched. Unlike slope limiters, each cell is limited using only its own values and the two global bounds, with no reference to its neighbours.

Since the limiter never changes $\langle q \rangle_K$, it can only succeed if the mean itself lies within $[m, M]$. Zhang and Shu proved that the cell means after a forward Euler step stay within bounds, provided that the values on S_K were within bounds at the start of the step, and that the time step satisfies a CFL condition which, at $p = 4$, is stricter than that required for linear stability [ZS10]. The limiter is therefore applied after every stage of the time stepping scheme, which must in turn be built out of forward Euler steps for the guarantee to carry over (section 8.4) [ZS10]. Should a mean leave the bounds regardless, such as when the time step is too large, the cell is flattened to its mean ($\theta_K = 0$) as a safeguard.

The proof also requires S_K to be a specific set of points: on rectangular cells, the tensor product of Gauss-Legendre points along each face with Gauss-Lobatto points across it, which includes the cell faces where the upwind flux (89) reads q [ZS10]. By default the nodes of the `'DQ'` space are Gauss-Legendre points, which all lie strictly inside the cell [Ham+23], so the values that are actually transported across faces would never be checked. Instead, q is interpolated onto a `'DQ'` space of the same degree with GLL nodes, as used by the `'Q'` space in section 5.3, and M_K and m_K are taken over its $(p+1)^3$ nodes. Both spaces contain exactly the same polynomials, so this interpolation is exact. The GLL nodes include the cell faces, but only approximate Zhang and Shu's point set, so the scheme is close to, rather than strictly, bound-preserving. In the same way, m and M are taken as the extreme values of the initial q at the GLL nodes, computed once before the first step.

To monitor how often the limiter intervenes, the fraction of the domain's volume in which $\theta_K < 1$ is

recorded after each application[17]. A source term breaks equation (93), so the limiter is switched off for verification with the method of manufactured solutions.

8.4 Time Stepping Scheme

Solving $a(\partial_t q, \phi) = L(\phi)$ reduces equation (45) to a system of ordinary differential equations, $\partial_t q = \mathcal{F}(q, t)$, where each evaluation of $\mathcal{F}$ requires one diagnostic inversion followed by one mass matrix solve (section 8.2). We advance this system with the three-stage, third-order strong stability preserving Runge-Kutta scheme (SSPRK3) of Shu and Osher [SO88], applying the limiter Λ (section 8.3) after every stage:

$$\begin{aligned} q^{(1)} &= \Lambda\left[q^n + \Delta t\, \mathcal{F}(q^n, t^n)\right] \\ q^{(2)} &= \Lambda\left[\tfrac{3}{4} q^n + \tfrac{1}{4}\left(q^{(1)} + \Delta t\, \mathcal{F}(q^{(1)}, t^n + \Delta t)\right)\right] \\ q^{n+1} &= \Lambda\left[\tfrac{1}{3} q^n + \tfrac{2}{3}\left(q^{(2)} + \Delta t\, \mathcal{F}(q^{(2)}, t^n + \tfrac{1}{2}\Delta t)\right)\right] \end{aligned} \tag{96}$$

The first stage is a forward Euler step, and each later stage is a convex combination of q^n and a forward Euler step from the previous stage, as the weights are non-negative and sum to one. Any convex property that a forward Euler step preserves, such as the bounds in equation (93), is therefore preserved by the whole scheme under the same time step restriction [GST01]. This is precisely what the limiter requires, since Zhang and Shu's guarantee only covers forward Euler steps [ZS10]. The classical fourth-order Runge-Kutta scheme does not have this property[18], and no explicit four-stage, fourth-order scheme with non-negative coefficients does [GS98].

The stage times t^n, $t^n + \Delta t$ and $t^n + \frac{1}{2}\Delta t$ only matter when q_{in} or a source term depends on time, as in the verification below. Since Δt depends on the current wind, it is chosen after the first evaluation of $\mathcal{F}$, which supplies that wind without an extra inversion.

8.5 MMS Verification

As for the inversion (section 5.7), the time-evolution is verified with the method of manufactured solutions [Roa02]. The inversion has already been verified, so here the wind is prescribed analytically rather than obtained from q, which isolates the transport scheme of section 8.2 and the time stepping of section 8.4. A source term S is added to equation (45), chosen so that a manufactured solution q_a satisfies it exactly:

$$\frac{\partial q}{\partial t} + \boldsymbol{u} \cdot \nabla q = S, \qquad S = \frac{\partial q_a}{\partial t} + \boldsymbol{u} \cdot \nabla q_a \tag{97}$$

The prescribed wind is

$$u = U_0 + U_1 \sin\left(\frac{2\pi y}{L_x}\right), \qquad v = V_0 + V_1 \sin\left(\frac{2\pi x}{L_x}\right) \tag{98}$$

with $U_0 = 20\,\mathrm{m\,s^{-1}}$, $U_1 = 10\,\mathrm{m\,s^{-1}}$, $V_0 = 10\,\mathrm{m\,s^{-1}}$ and $V_1 = 5\,\mathrm{m\,s^{-1}}$, on a domain with $L_x = L_y = 1000\,\mathrm{km}$ and $H = 20\,\mathrm{km}$. Since u depends only on y and v only on x, the wind is divergence-free, like the geostrophic wind it stands in for. Both components are strictly positive, so the wind flows in through the western and southern walls and out through the eastern and northern walls, exercising both the inflow and outflow terms of equation (91). The manufactured solution is

$$q_a(x, y, z, t) = \sin\left(\frac{2\pi x}{L_x}\right) \cos\left(\frac{2\pi y}{L_x}\right)\left(1 + \frac{z}{H}\right) + \sin^2\left(\frac{\pi x}{L_x}\right) \sin^2\left(\frac{\pi y}{L_x}\right) \sin\left(\frac{2\pi t}{\tau}\right) \tag{99}$$

with period $\tau = 2 \times 10^4\,\mathrm{s}$. The time dependence is confined to the second term, which vanishes on the lateral boundary, so the inflow data $q_{\text{in}} = q_a$ is constant in time, just as it is frozen in the real solver. Time-dependent boundary data imposed at the intermediate stage times of a Runge-Kutta scheme can reduce its order of accuracy [Car+95], which would obscure the convergence being measured. Since there

[17] The implementation is shown in section A.9

[18] when $\mathcal{F}$ is nonlinear

is no vertical transport, the mesh has a single layer of cells, in which the linear z dependence of q_a is represented exactly. The source term is derived symbolically by UFL from equations (98) and (99) [Aln+14], and enters the linear form as $\int_V \phi S \, dV$[19]. The limiter is switched off, since S allows q_a to leave the bounds of equation (93), and limiting would clip the exact solution.

The solution is advanced from $q_a(x, y, z, 0)$ to $t = \tau$, and compared with $q_a(x, y, z, \tau)$ using the relative L^2 error. Unlike ψ, q is not defined only up to an additive constant, so its mean is part of what is tested and is not removed as in section A.6:

$$e = \frac{\|q_h - q_a\|_{L^2}}{\|q_a\|_{L^2}} \tag{100}$$

Two convergence tests are run, one for each source of error. For the spatial error, $p = 4$ and the mesh is refined through $N_x = N_y = 4, 8, 16$. The time step allowed by the CFL condition shrinks only in proportion to the cell width h, which would leave the $\mathcal{O}(\Delta t^3)$ error of SSPRK3 dominating the $\mathcal{O}(h^{p+1})$ spatial error. The time step is therefore reduced faster, as $\Delta t \propto h^{(p+1)/3}$, so that both errors fall as h^{p+1}, and the error should converge at order $p + 1 = 5$.

For the temporal error, $p = 2$ and $N_x = N_y = 8$ are fixed, and the number of time steps is doubled three times from a base time step at a Courant number of 1.75. Each solution is compared with the next rather than with q_a, so the spatial error, which is the same for every run on a fixed mesh, cancels. For a third-order scheme, the difference between successive runs should fall by a factor of $2^3 = 8$ with each halving of Δt.

Table 5: Relative L^2 error (equation (100)) of the time-evolved manufactured solution (equation (99)) at $t = \tau$ for $p = 4$, and the observed order of convergence between successive meshes.

$N_x = N_y$	e	Observed order
4	2.01×10^{-4}	–
8	6.28×10^{-6}	5.00
16	1.97×10^{-7}	5.00

The spatial convergence results are given in table 5, where the observed order matches the expected order of $p + 1 = 5$ to within 0.003. In the temporal test, the base time step corresponds to 37 steps to $t = \tau$, and the relative L^2 differences between successive runs fell from 3.50×10^{-5} to 4.35×10^{-6} to 5.43×10^{-7}, giving observed orders of 3.01 and 3.00, consistent with the third-order accuracy of SSPRK3. Together with section 5.7, this verifies the inversion, the transport scheme with its boundary conditions, and the time stepping separately; only the limiter, which is switched off here, is not covered.

8.6 Step size

SSPRK3 is explicit, so the time step Δt must be limited, both for stability and for the guarantee of the limiter (section 8.3) to hold. The wind changes as the flow evolves, so Δt is recomputed at the start of every step from the wind of the current state:

$$\Delta t = \mathrm{C} \frac{s_p \, \Delta x}{\max_V (|u| + |v|)}, \qquad \Delta x = \min\left(\frac{L_x}{N_x}, \frac{L_y}{N_y}\right) \tag{101}$$

Here C is a Courant number, and s_p is the smallest horizontal gap between neighbouring nodes of the `'DQ'` reference cell, including its edges, as a fraction of the cell width, so that Δt is set by the finest spacing the nodes resolve rather than by the cell width. With the default Gauss-Legendre nodes, s_p is the gap between the cell edge and the nearest node, which for $p = 4$ is $s_4 = 0.0469$. On rectangular cells, Zhang and Shu's condition bounds $\Delta t \, (|u|/\Delta x + |v|/\Delta y)$ rather than $\Delta t \sqrt{u^2 + v^2}/\Delta x$ [ZS10], so equation (101) uses $|u| + |v|$; the speed $|\boldsymbol{u}|$ would underestimate it by up to a factor of $\sqrt{2}$ for diagonal flow. The maximum is taken over the nodes of $|u| + |v|$ interpolated into the `'DQ'` space, since $\boldsymbol{u}$, the

[19] shown in section A.8

derivative of the continuous ψ, is discontinuous across cells, and interpolating it into a continuous space would average away its peaks.

To determine how large C can be, the control atmosphere was evolved for 168 h on a mesh with $N_x = N_y = N_z = 40$ and $p = 4$, with the limiter on, once at each $\mathrm{C} \in \{1.3, 1.7, 1.9, 2.1, 2.3, 2.5, 2.7\}$. A run is classed as unstable if $\max|q|$ grows to more than 10^3 times its initial value, or if its roughness grows to more than 1.5 times that of the $\mathrm{C} = 1.3$ run at the same time. The roughness is the largest root-mean-square, over all cells, of the part of q that is of degree p in x or y, found by subtracting the interpolation of q onto polynomials of one degree lower in x and y. Grid-scale noise appears in this highest-degree part, while a smooth q has little of it, however sharp it is in z. The roughness is compared between runs rather than against a fixed threshold, because the flow itself sharpens q: in every run the roughness peaked at about 38.7 times its initial value after 94 h. This sharpening is the same at every Δt, whereas noise caused by the time step is not, so the ratio isolates the latter.

Table 6: Results of evolving the control atmosphere for 168 h at each Courant number C (equation (101)), with $N_x = N_y = N_z = 40$ and $p = 4$. The growth of $\max|q|$ and the L^2 norm of q are relative to their initial values. Δt excludes the final step, which is shortened to land exactly on 168 h. The roughness ratio is the largest, over the run, of its roughness divided by that of the $\mathrm{C} = 1.3$ run at the same time. The limited fraction is the fraction of the domain's volume with $\theta_K < 1$ (equation (95)), averaged over all steps.

C	Steps	Δt [s]	Peak $\max\|q\|$ growth	Final $\|q\|_{L^2}$ ratio	Roughness ratio	Limited fraction
1.3	2937	182–241	1.0842	0.99294	1.000	4.03×10^{-4}
1.7	2246	238–315	1.0836	0.99294	1.006	4.04×10^{-4}
1.9	2010	266–352	1.0837	0.99294	1.004	4.04×10^{-4}
2.1	1818	294–390	1.0848	0.99294	1.006	4.05×10^{-4}
2.3	1660	322–427	1.0836	0.99293	1.007	4.06×10^{-4}
2.5	1527	350–464	1.0850	0.99293	1.009	4.06×10^{-4}
2.7	1414	378–504	1.0842	0.99282	1.057	4.13×10^{-4}

The results are given in table 6, and no run was unstable. The growth of $\max|q|$ and the loss of $\|q\|_{L^2}$ to the dissipation of the upwind flux are the same at every C to within 0.2 %, and the fraction of the domain touched by the limiter to within 3 %, so none of them is caused by the time step. The roughness ratio stays below 1.01 up to $\mathrm{C} = 2.5$. At $\mathrm{C} = 2.7$ it reaches 1.057, but only within the first 0.3 h, and it is back to 1.000 by the end of the run.

These results only establish stability for the flow that was tested. Over the scan, $\max_V(|u| + |v|)$ varied between 47 m s^{-1} and 63 m s^{-1}, but this maximum is only reached near the PV anomaly, whose radius of 200 km is comparable to the cell width $\Delta x = 187.5$ km. Away from the anomaly, $|u| + |v|$ in the jet is only about 37 m s^{-1}, so even the fastest flow there sees only 60 % to 75 % of C. A larger or stronger anomaly, elsewhere in the parameter space, would reach the maximum wind over more of the domain. We therefore take $\mathrm{C} = 1.3$, roughly half of the largest value found to be stable, at the cost of about twice as many steps as $\mathrm{C} = 2.7$ would need. On the mesh above, this gives Δt between 182 s and 241 s, or about 17.5 steps per simulated hour.

8.7 Results

We first test the full numerical prognostic solver by evolving the control parameters for a simulated period of 168 hours, unfortunately introducing further difficulties into the simulation. Firstly, L_x must be significantly increased so that the jet stream does not simply push the anomaly into the domain wall after a few simulated days. This comes with the hindrance that we can not increase Δx to compensate, so we must simply use a far larger mesh and far more degrees of freedom. These domain adjustments resulted in 5×10^7 degrees of freedom, which took approximately 36 hours of processing time with the safe $C = 1.3$ on the high-performance hardware listed in section B.2.

The time evolution of the surface wind is given in figure 30.

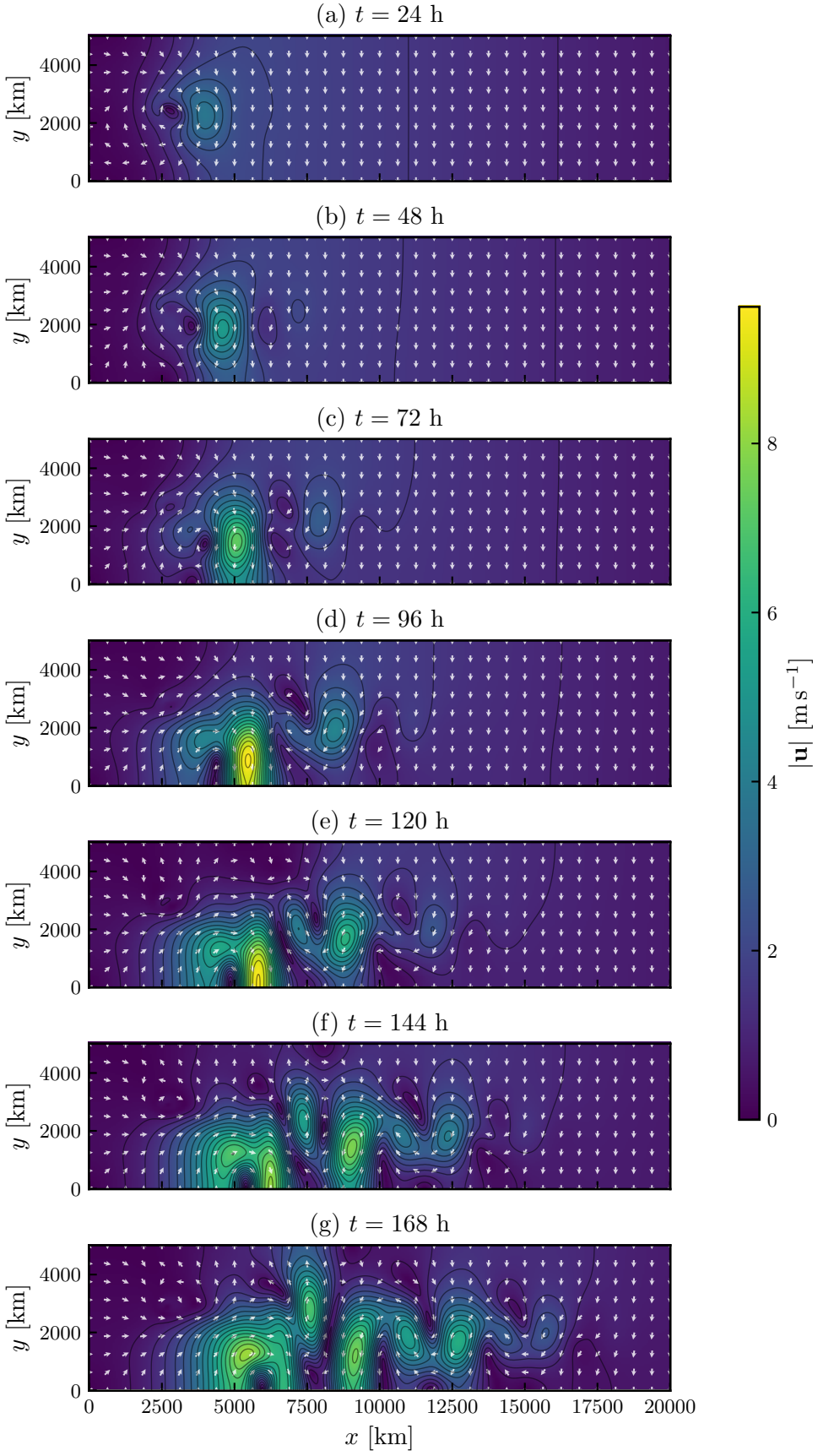


Figure 30: Evolving the control parameters in time on a mesh with increased L_x and N_x. Surface wind is plotted at 24 hour intervals.

Figure 30 makes it immediately clear that the anomaly very quickly moves not only east, but south as well, colliding with the $y = 0$ boundary early on in the simulation. The rest of the plot confirms the stability of $C = 1.3$, but is unusable for physical analysis as no perfectly straight boundary, 20 000 km long, that prescribes tangential wind exists in the Earth's atmosphere. The southward drift is a property of the background state rather than of the solver. Although we set $\beta = 0$, the jet supplies a background PV gradient of its own, which plays the same role as the planetary gradient that supports Rossby waves (section 2.4). Since $\overline{\psi}$ (equation (53)) is independent of x, so is $\overline{q}$ (equation (54)), and the background wind (equations (51) and (52)) is purely zonal, so $\overline{q}$ on its own is a steady solution of equation (45). Splitting $q = \overline{q} + q^*$ and $\psi = \overline{\psi} + \psi^*$ as in section 4.4, with $(u^*, v^*) = (-\psi^*_y, \psi^*_x)$, equation (45) becomes

$$\frac{\partial q^*}{\partial t} + (\overline{u} + u^*)\frac{\partial q^*}{\partial x} + v^*\frac{\partial q^*}{\partial y} + \frac{\partial \overline{q}}{\partial y}v^* = 0 \tag{102}$$

On a β-plane, the $f = f_0 + \beta y$ in $\tilde{q}$ (equation (33)) contributes the same term with β in place of $\partial\overline{q}/\partial y$, so the jet acts on the anomaly as an effective β. Differentiating equation (54) with $\partial_y\overline{\psi} = -\overline{u}$ and evaluating at the jet core, where $\partial_z\overline{u} = 0$, gives

$$\frac{\partial \overline{q}}{\partial y} = -\frac{\partial^2\overline{u}}{\partial y^2} - \frac{f_0^2}{\overline{N}^2}\frac{\partial^2\overline{u}}{\partial z^2} = \frac{2U_{\text{jet}}}{L_{\text{jet}}^2} + \frac{2U_{\text{jet}}f_0^2}{\overline{N}^2 z_{\text{jet}}^2} \tag{103}$$

Both terms are positive, like the planetary $\beta = \partial f/\partial y \approx 1.7 \times 10^{-11}\,\text{m}^{-1}\,\text{s}^{-1}$ at 42°S, but much larger: for the control jet the first term alone is $2.8 \times 10^{-10}\,\text{m}^{-1}\,\text{s}^{-1}$. Even for the weakest sampled jet ($U_{\text{jet}} = 10\,\text{m}\,\text{s}^{-1}$, table 2) the first term exceeds β for every sampled L_{jet}. A cyclone on a β-plane drifts poleward and westward relative to the surrounding flow [Hol83]. The control anomaly sits on the jet core ($y_{\text{ano}} = y_{\text{jet}}$), so on top of being carried east by the jet it drifts south, as seen in figure 30. Away from the core the curvature of $\overline{u}$ changes sign, beyond $|y - y_{\text{jet}}| = L_{\text{jet}}/\sqrt{2}$ horizontally, so $\partial\overline{q}/\partial y$ weakens and eventually reverses, and an anomaly placed far enough from the jet would be expected to drift equatorward instead.

The nonlinearity of equation (45) prevents simply removing the jet as was done in section 7.3, which would otherwise stop the meridional drift. Another technique to handle this drift – without making the mesh intractably large – would be to make the mesh periodic in x and y, though this would require redoing a significant portion of the derivations above, making it out of scope for this Honours Thesis.

9 Conclusions

The aim of this project (section 3) was to quantify how the structure of a stratospheric intrusion, and the background atmosphere in which it is embedded, determine its surface response. We first review each objective in turn, and then summarise what the results say about this aim.

9.1 Objectives

1. **Recreating the control state of Barnes et al.** The background state and anomaly were recreated from the published values (section 4.4), with the quantities Barnes et al. did not publish (w_{trop}, δ and L_{jet}) taken from Fehlmann or matched to their figures [Bar+22; Feh97]. The recreation is therefore only approximate. It also exposed two problems with the control state itself: $\overline{N}_{\text{strat}} = 0.03\,\text{s}^{-1}$ gives $\overline{\theta} > 600\,\text{K}$ at 20 km (figure 2), and $\mathcal{V}_- \approx 0.04$ (equation (59)), so about 4 % of the domain has $fQ \leq 0$, where an EPV inversion would fail. This objective was achieved, subject to these caveats.

2. **Developing and validating the inversion solver.** Equation (37) was solved with $p = 4$ continuous Galerkin elements on an extruded hexahedral mesh (section 5). The solvability condition (equation (77)) was verified to round-off, and the method of manufactured solutions gave the expected $\mathcal{O}(\Delta z^{p+1})$ convergence for every p tested (figure 10). The scheme is fifth-order accurate, compared with the second-order scheme of Barnes et al. [Bar+22]. This objective was achieved.

3. **Optimising the solver.** Out of more than 40 configurations, a matrix-free conjugate gradient solver with a p-multigrid pre-conditioner was the fastest on both the consumer hardware and the HPC cluster (section 5.6). Only the $p = 1$ coarse operator is assembled. Its solve time scales as $\mathcal{O}(\mathrm{DoF})$ (figure 9), and a full atmospheric state can be constructed, inverted and analysed in under 3 seconds on the HPC cluster. Since only the coarse matrix is stored, the solver also needs a small fraction of the memory of an assembled one, which permits large meshes on consumer hardware. This objective was achieved.

4. **Assessing the limitations of QGPV.** The constraints on $\mathcal{V}_-$, the Rossby number, the Froude number and the assumption of adiabatic flow were set out in section 6.2. The Rossby number is by far the most restrictive. The control state has $\mathrm{Ro}_{\max} \approx 2$, and of the randomly sampled states with $\mathcal{V}_- = 0$, only 14.5 % have $\mathrm{Ro} < 1$, and 0.1 % have $\mathrm{Ro} < 0.5$ (section 7.3). Most of the parameter space in table 2 therefore lies where QGPV is at best qualitatively useful. Both of the known biases, the fixed penetration depth of QG [Dav92] and the use of dry stratification [HMR85], point towards an underestimated surface response for strong anomalies. This objective was achieved qualitatively, but quantifying the error of QG inversion against full EPV inversion was beyond the scope of this project.

5. **Isolating the effect of each parameter.** Varying one parameter at a time (section 7.2) showed that the tropopause height z_{trop}, the anomaly depth z_{size} and its horizontal size dominate the surface response, in agreement with Barnes et al. [Bar+22]. A weaker tropospheric stratification and a more poleward latitude intensify it moderately. The tropopause smoothing parameters w_{trop} and δ, $\overline{N}_{\mathrm{strat}}$, the anomaly aspect ratio and $\overline{T}'(0)$ have almost no effect on the surface, so neither the numerical treatment of the tropopause nor the unrealistic $\overline{N}_{\mathrm{strat}}$ of the control state affects the surface results. The jet parameters also have almost no effect on the surface, which, together with the linearity of equation (37), justified removing the jet from the later sweeps. Random sampling (section 7.3) showed a moderate monotonic relationship between the minimum dynamical tropopause height z_{dyn} and the surface response, but once states with $\mathrm{Ro} \geq 1$ are removed it weakens to at most a weak relationship for the surface pressure anomaly, and a moderate one for the surface vorticity (figure 24). So, unlike Barnes et al., who found z_{dyn} to be the major factor in surface cyclogenesis [Bar+22], we find that within the range where QGPV is useful, the combination of the other parameters matters more than z_{dyn} alone. This objective was achieved, although one feature, the sharp step in z_{dyn} as y_{jet} varies (figure 21), remains unexplained.

6. **Searching for the most severe surface response.** Py-BOBYQA minimised the cost function equation (85) in under 100 evaluations for both objectives (section 7.4). The optimised states have a surface pressure anomaly of $-51.7\,\mathrm{hPa}$ and surface wind of $25.3\,\mathrm{m\,s^{-1}}$, about ten times those of the control state, with $z_{\mathrm{dyn}} \approx 3\,\mathrm{km}$, comparable to the rare COLs whose $-1.5\,\mathrm{PVU}$ contour reaches 700 hPa [BNL21] (table 4). They place the tropopause as low, the latitude as far poleward and the anomaly as deep and wide as the bounds allow, and the difference between their anomaly shapes shows that shape also matters for the surface wind. However, despite the penalty in equation (85), both have $\mathrm{Ro}_{\max} \geq 1.5$, so these results are qualitative only. The optimiser itself was implemented as intended, but it did not find states within the range where QGPV is quantitatively valid.

7. **Integrating forward in time.** The prognostic solver uses an upwind discontinuous Galerkin transport scheme (section 8.2), the Zhang-Shu limiter (section 8.3) and SSPRK3 (section 8.4). The scheme converges at fifth order in space and third order in time (section 8.5), was stable for Courant numbers up to $\mathrm{C} = 2.7$, and is run at $\mathrm{C} = 1.3$ (section 8.6). The control state was evolved for 168 h on a mesh of 5×10^7 degrees of freedom (section 8.7). However, the anomaly drifted south and reached the $y = 0$ wall early in the run, because the jet's background PV gradient acts as an effective β (equations (102) and (103)). The longevity of the optimised configurations was therefore not assessed. This objective was only partly achieved: the time-evolution solver exists and is verified, but in the current domain its results cannot be used for physical analysis.

Taken together, within the quasigeostrophic framework the surface response of an intrusion is set mainly by how low and how deep it reaches and by its horizontal size, with the background stratification and latitude playing a secondary role. The jet has almost no effect on the instantaneous surface response, yet it dominates the evolution of the anomaly through its PV gradient. The strongest responses, however, lie in the part of the parameter space where QGPV is beyond its formal validity, so these findings should be read as qualitative.

9.2 Future Work

- **Inversion beyond QG.** The Rossby number constraint limits most of the results above. Next-order corrections to QG improve its accuracy substantially at Ro ≈ 0.3 [MSR99], and a full EPV inversion would avoid the problem entirely, although it would also require background states with $\mathcal{V}_- = 0$. Comparing such an inversion with equation (37) for the states in table 4 would also quantify the error of QG directly, which Davis did not do [Dav92].

- **A realistic lower boundary.** At present, θ^* on the lower boundary (equation (40)) is a single horizontally uniform value of about $-0.1\,\mathrm{K}$, chosen only to satisfy the solvability condition (section 5.5). However, whether a COL reaches the surface depends strongly on a surface θ anomaly in phase with the upper anomaly: 66% of deep COLs have one when they form, compared with 12% of shallow COLs [BNL21] (section 2.5). Prescribing a horizontally varying surface θ^*, such as a warm anomaly just downstream of the intrusion, with equation (77) adapted to fix only its mean, would allow the interaction between the two anomalies to be studied.

- **The optimiser's cost function.** Replacing the division by $\tilde{\mathrm{Ro}} \cdot \tilde{\mathrm{Fr}}$ in equation (85) with a clipped penalty, like the $\mu\mathcal{V}_-$ term, or a hard constraint on $\mathrm{Ro}_{\max}$, may steer the optimiser towards states where QGPV remains quantitatively valid.

- **The time-evolution domain.** The lateral walls and their prescribed tangential wind are the main obstacle to physical analysis of the time-evolution. Since the jet is uniform in x, a zonally periodic channel would remove the need for a very large L_x and for the inflow condition on the eastern and western walls. A larger L_y, or an anomaly offset from the jet core where the effective β weakens, would delay the collision with the southern wall. With these changes, the optimised configurations of table 4 could be evolved to complete the final objective. In addition, the limiter was switched off in the verification of section 8.5; a test without a source term, whose exact solution obeys equation (93), would verify it as well.

- **Moisture.** The inversion uses dry stratification throughout, while condensation in rising air reduces the effective static stability and increases the vertical penetration of an anomaly [HMR85]. A reduced effective $\overline{N}$ in regions of ascent would give a first estimate of this effect.

- **A more realistic background state.** The control state was kept for parity with Barnes et al., but a realistic $\overline{N}_{\mathrm{strat}} \approx 0.022\,\mathrm{s}^{-1}$ [GTB10] and a jet with $\mathcal{V}_- = 0$ would give a physically valid reference state for the sweeps.

- **Solver performance.** A transformation that absorbs the spatially varying coefficients of equation (37) into ψ or into rescaled coordinates would remove the remaining memory traffic of the matrix-free solver (section 5.6), and speed up both the sweeps and the diagnostic solve at every stage of the time-evolution. It is possible that this transformation would allow for a closed form Green's function solution to equation (37), further enabling an implicit solution to equation (45) using the method of characteristics, bypassing the need for a numerical solver entirely.

10 References

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Appendices

Appendix A Firedrake

Firedrake has a more involved installation process than most Python libraries. The instructions for which can be found here.

A.1 Mesh Implementation

Before setting up the mesh, we specify a `SolverParams` and `PhysicalParams` class to store parameters for the simulation. These objects will be passed through each part of the solver.

The Firedrake implementation for generating the extruded mesh according to section 5.2 is as follows:

```
temp_mesh = RectangleMesh(
    solver_params.nx, solver_params.ny, # Specify the number of elements in the x and
    ↪ y direction
    phys_params.Lx, phys_params.Ly, # Specify the size of the domain in the x and y
    ↪ directions
    quadrilateral=True # Enables use of the 'Q' function space (crucial)
)

# Extrude the flat 2D mesh upwards in 3D
mesh = ExtrudedMesh(
    temp_mesh,
    layers=solver_params.nz, # Specify the number of elements in the z direction
    layer_height=(phys_params.H / solver_params.nz) # Calculate the correct vertical
    ↪ distance between layers
)
```

A.2 Numerical Vertical Integrator

Firedrake does not provide a simple way to evaluate integrals of the form:

$$I(z) = \int_0^z f(\xi)\, d\xi \tag{104}$$

$I(z)$ could be solved with Firedrake by rewriting it as an initial value problem (IVP):

$$\frac{\partial I}{\partial z} = f(z), \quad I(0) = 0 \tag{105}$$

On our 3D mesh defined in section 5.2, we would be wasting an enormous amount of CPU time and memory by repeatedly solving an identical IVP in every vertical column, when the integrand depends exclusively on z. Additionally, the IVP consists of only advection, which is unsuitable for the `'Q'` function space we chose in section 5.3.

To circumvent this the `'DQ'` function space could be implemented, but this would require expensive projections back and forth between both function spaces [Ham+23]. A more efficient approach would be to make use of the pre-existing high-order numerical integration system in the package `scipy` [Vir+20], and then write the result back into a Firedrake `Function`. The annotated code that performs the integration up to 4th order accuracy is given:

```
def compute_vertical_integral(integrand, cg_space):
    """
    Computes the vertical integral I(z) = \int_{z_{bottom}}^z f(\xi) d\xi.
```

```
    """
    Collapses the mesh into a single 1D vertical profile. The integration is handled
    ↪ by SciPy's CubicSpline, which maintains high-order accuracy compatible with
    ↪ higher-degree Firedrake function spaces, before mapping the result back to a
    ↪ 3D Firedrake Function.
    """
    mesh = cg_space.mesh()

    # Extract Z-coordinates and integrand values at all mesh Degrees of Freedom
    ↪ (DoFs)
    z_expr = SpatialCoordinate(mesh)[2]

    z_fd = Function(cg_space).interpolate(z_expr)
    f_fd = Function(cg_space).interpolate(integrand)

    z_data = z_fd.dat.data_ro
    f_data = f_fd.dat.data_ro

    # Sort DoFs vertically. SciPy's CubicSpline requires strictly increasing
    ↪ coordinates.
    sort_idx = np.argsort(z_data)
    z_sorted = z_data[sort_idx]
    f_sorted = f_data[sort_idx]

    # Collapse 3D mesh into a 1D vertical profile. This prevents redundant
    ↪ integration across identical horizontal columns and removes duplicate
    ↪ Z-coordinates that would crash the spline.
    z_rounded = np.round(z_sorted, decimals=8)
    _, unique_idx = np.unique(z_rounded, return_index=True)

    z_1d = z_sorted[unique_idx]
    f_1d = f_sorted[unique_idx]

    # Fit a continuous piecewise cubic polynomial to the 1D profile
    spline = CubicSpline(z_1d, f_1d)

    # Analytically integrate the spline (producing a 4th-degree polynomial)
    integral_spline = spline.antiderivative()

    # Evaluate the integral back at the original 3D mesh DoF coordinates.
    # Subtracting the bottom boundary value enforces the condition I(0) = 0.
    integral_dofs = integral_spline(z_data) - integral_spline(z_1d[0])

    # Write the 1D evaluated data back into a native Firedrake Function
    result = Function(cg_space, name="Vertical_Integral")
    result.dat.data[:] = integral_dofs

    return result
```

When running with MPI, this function does not collate the data between ranks onto a single rank. This means that each rank evaluates the integral only on the data in its subset of the mesh. If the mesh

were to be subdivided vertically, this would completely break the function and incorrect results would be given. However, Firedrake only subdivides extruded meshes horizontally [Ham+23; Lan+16; CP08], so each MPI rank is operating on its own independent vertical column.

For this reason, the `compute_vertical_integral` function will **only** produce accurate results for **extruded meshes**. This results in a slight increase in efficiency as communication between ranks is not necessary.

A.3 Diagnostic Weak Form

Equation (73) is expressed in Firedrake's syntax, which is based on Unified Form Language (UFL) [Aln+14]. This code is included as certain parts of the syntax can be quite unintuitive, especially with extruded meshes.

```
# Define the trial and test functions
psi = TrialFunction(cg_space)
phi = TestFunction(cg_space)
n = FacetNormal(mesh) # The normal vector n=[n_x, n_y, n_z]

# The value of the below expression will be evaluated at both vertical boundaries
vertical_boundary = phys_params.g * theta_star / (phys_params.f * theta_bar)

# Calculate the quotient in advance
# Prevents huge quadrature degrees from being estimated by the form compiler if a
↪ quadrature degree is not explicitly set
rho_N2 = Function(cg_space).interpolate(rho_bar / (N_bar**2))

x_base = rho_bar * phi * v * n[0]
y_base = rho_bar * phi * -u * n[1]
z_base = n[2] * (f**2) * phi * rho_N2

L_x = x_base * ds_v((1, 2)) # Tuple that refers to both x-boundaries on the extruded
↪ mesh
L_y = y_base * ds_v((3, 4)) # Same but for y-boundaries
L_z = z_base * self._vertical_boundary * ds_tb # ds_bt represents the vertical
↪ boundaries in an extruded mesh
L_vol = -q * rho_bar * phi * dx

# Linear form L
L = L_x + L_y + L_z + L_vol

# Bilinear form alpha
a = (rho_bar * psi.dx(0) * phi.dx(0) +
     rho_bar * psi.dx(1) * phi.dx(1) +
     (f**2) * rho_N2 * psi.dx(2) * phi.dx(2)) * dx
```

An important detail to note is that constants that may change depending on atmospheric parameters (such as `phys_params.f`) should be wrapped with `Constant(...)` once, and subsequently updated with its `assign()` method. If a plain Python float is used instead, its value is embedded in the UFL form as a literal, so each new parameter value produces a different form, for which Firedrake must generate and compile new C kernels [Ham+23]. A `Constant` is instead a symbolic coefficient whose value is passed to the already compiled kernel at run time, so the form, and hence the generated code, is reused across parameter changes.

A.4 Solver Parameters

A direct Python equivalent for the solver set-up and parameters given in section 5.6 is provided.

```python
psi_soln = Function(cg_space) # The solution will be stored here

# Stop the form compiler from estimating massive quadrature degrees
form_compiler_params = {
    "quadrature_degree": 3*polynomial_order
}

problem = LinearVariationalProblem(
    a, L, psi_soln,
    constant_jacobian=True, # Saves a lot of time with the assembled solver parameters and repeated solves
    form_compiler_parameters=form_compiler_params
)

params = {
    "mat_type": "matfree",
    "ksp_type": "cg",
    "ksp_rtol": ksp_rtol,
    "ksp_atol": ksp_atol,
    # p-multigrid for outer preconditioner
    "pc_type": "python",
    "pc_python_type": "firedrake.PMGPC",
    # For p=2,3,4
    "pmg_mg_levels_ksp_type": "chebyshev",
    "pmg_mg_levels_pc_type": "jacobi",
    "pmg_mg_coarse_ksp_type": "preonly",  # Don't iterate, just apply the direct solver once
    "pmg_mg_coarse_pc_type": "python",
    "pmg_mg_coarse_pc_python_type": "firedrake.AssembledPC",  # Force assembly of ONLY the p=1 matrix
    "pmg_mg_coarse_assembled_pc_type": "cholesky",
    "pmg_mg_coarse_assembled_pc_factor_mat_solver_type": "mumps",
    "pmg_mg_coarse_assembled_mat_mumps_icntl_24": 1  # detect & null out the known null pivot
}

# Define a constant null space to prevent the matrix from being singular
nullspace = VectorSpaceBasis(constant=True, comm=mesh.comm)

# Build the reusable solver object
solver = LinearVariationalSolver(
    problem,
    solver_parameters=params,
    nullspace=nullspace
)

# Finally solve the equation!
solver.solve()
```

A value of `ksp_rtol = 1e-9` was selected for the convergence conditions, as minimal benefit arises from setting it much smaller [Ham+23]. An absolute tolerance of `ksp_atol = 1e-3` was used.

A.4.1 Unused Assembled Matrix Parameters

The parameters used to produce figure 9 are given. These are inferior to the matrix-free set of parameters given above and in section 5.6, but only by an additive constant. Both parameter sets achieve $\mathcal{O}(\text{DoF})$ time complexity. This parameter set requires substantially more memory than the matrix-free alternative.

```
params = {
    "mat_type": "aij",
    "ksp_type": "cg",
    "pc_type": "python",
    "ksp_rtol": ksp_rtol,
    "ksp_atol": ksp_atol,
    "pc_python_type": "firedrake.PMGPC",
    "pmg_mg_levels_pc_type": "jacobi",
    "pmg_mg_coarse_ksp_type": "preonly",
    "pmg_mg_coarse_pc_type": "hypre",
    "pmg_mg_coarse_pc_hypre_type": "boomeramg"
}
```

A.5 Running in Parallel with MPI

To run Firedrake in parallel, it is necessary to use a Message Passing Interface (MPI) Library [Ham+23; Dal+11]. For example:

```
mpiexec -n 16 python simulation.py
```

It is worth noting that standard python functions will run redundantly on every single MPI rank, including `print()` statements. Firedrake's standard functions know how to handle multiple ranks seamlessly, however normal python libraries will not necessarily handle this correctly. Instead Firedrake's functions should be used where possible, such as `PETSc.Sys.Print()` instead of `print()` [Ham+23; Dal+11].

A.6 Relative Error Norm

Because equation (37) is a pure Neumann problem, the numerical solution ψ_h can only be compared with the manufactured solution ψ_a up to an additive constant. The constant chosen is the one that minimises $\|\psi_h - \psi_a - c\|_{L^2}$, which is the mean of the difference, $c = \langle \psi_h - \psi_a \rangle$, where

$$\langle g \rangle \equiv \frac{1}{|V|} \int_V g \, dV, \quad \|g\|_{L^2} \equiv \left(\int_V g^2 \, dV \right)^{1/2} \tag{106}$$

The relative error is then:

$$e = \frac{\|(\psi_h - \psi_a) - \langle \psi_h - \psi_a \rangle\|_{L^2}}{\|\psi_a - \langle \psi_a \rangle\|_{L^2}} \tag{107}$$

The mean is also removed from ψ_a in the denominator, since its additive constant is just as arbitrary and would otherwise inflate $\|\psi_a\|_{L^2}$ and shrink e.

Since ψ_a is not a polynomial, the integrals in equation (107) cannot be computed exactly. The quadrature degree is therefore set well above that of the finite element space, so that the quadrature error is not what limits the measured error. The annotated code, simplified to compute only the L^2 norm, is given:

```python
def relative_error(exact, numerical : Function):
    """
    Relative L2 error between an exact UFL expression and a numerical Function, after the mean of each is removed.
    """
    # Quadrature degree well above that of the finite element space, so that integrating against the non-polynomial exact solution does not set the error floor
    degree = 3 * numerical.function_space().ufl_element().embedded_superdegree + 6
    dx_q = dx(domain=numerical.function_space().mesh(), degree=degree)

    volume = assemble(Constant(1) * dx_q)

    # The solution is only defined up to an additive constant, so remove the mean
    def mean_removed(expr):
        return expr - assemble(expr * dx_q) / volume

    def l2_norm(expr):
        return np.sqrt(assemble(expr**2 * dx_q))

    return float(l2_norm(mean_removed(numerical - exact)) / l2_norm(mean_removed(exact)))
```

Each `assemble` call sums its integral over every MPI rank, so unlike `compute_vertical_integral`, this function gives the same result however the mesh is partitioned.

A.7 Lateral Boundary Mean

Firedrake has no built-in way to integrate over a curve at each height, as equation (82) requires. Instead, the test function is integrated over the lateral walls of the domain, which gives one weight per DoF that is zero away from the walls. The basis functions of the `'Q'` space are products of a horizontal and a vertical part, so every DoF on a given z level shares the same vertical factor in its weight. That factor cancels when the weighted sum of ψ on a level is divided by the sum of the weights, which leaves exactly the perimeter mean of ψ_h at that level. The annotated code is given:

```python
def compute_psi_0(psi : Function):
    """
    Computes psi_0(z), the mean of psi around the lateral boundary at each height, as a Function that varies with z alone.
    """
    V = psi.function_space()
    mesh = V.mesh()

    # Height of every DoF on this rank, including the halo
    z = Function(V).interpolate(SpatialCoordinate(mesh)[2]).dat.data_ro_with_halos

    # Collapse the heights into the distinct z levels of the extruded mesh
    z_sorted = np.sort(z)
    tol = 1e-8 * (z_sorted[-1] - z_sorted[0])
    levels = z_sorted[np.concatenate(([True], np.diff(z_sorted) > tol))]

```

```
    # Index of the level each DoF sits on
    above = np.clip(np.searchsorted(levels, z), 1, levels.size - 1)
    level = np.where(z - levels[above - 1] <= levels[above] - z, above - 1, above)

    # Integrate each basis function over the lateral walls (ds_v); zero for DoFs off
    ↪ the walls
    weights = assemble(TestFunction(V) * ds_v).dat.data_ro
    psi_data = psi.dat.data_ro
    owned_level = level[:weights.size] # data_ro holds only owned DoFs, so none are
    ↪ counted twice

    # Sum the weighted psi, and the weights, on each level
    local = np.concatenate((
        np.bincount(owned_level, weights=weights * psi_data, minlength=levels.size),
        np.bincount(owned_level, weights=weights, minlength=levels.size)
    ))

    # The lateral walls are split between ranks, so add up every rank's contribution
    totals = np.empty_like(local)
    mesh.comm.Allreduce(local, totals, op=MPI.SUM)

    # Weighted mean on each level
    profile = totals[:levels.size] / totals[levels.size:]

    # Write the same profile into every column, so psi_0 is exactly horizontally
    ↪ uniform
    psi_0 = Function(V, name="psi_0")
    psi_0.dat.data_with_halos[:] = profile[level]

    return psi_0
```

As with `compute_vertical_integral`, this relies on Firedrake only subdividing extruded meshes horizontally, so that every rank holds whole columns, sees every z level, and numbers the levels identically. Unlike it, the lateral walls are shared between many ranks, so the per-level sums must be combined with an explicit `Allreduce` before the mean is taken. Writing the profile directly into the DoFs, rather than interpolating an expression, guarantees that ψ_0 has no horizontal variation, so none can leak into p^* through equation (81).

A.8 Prognostic Weak Form

Equation (91) is expressed in Firedrake's syntax below. Here `q` is a `Function` on the DG space whose values are overwritten in place before each solve, so, like a `Constant`, the form is only compiled once.

```
# Define the trial and test functions on the DG space
dq_dt = TrialFunction(dg_space)
phi = TestFunction(dg_space)
n = FacetNormal(mesh)

# The wind, from the latest streamfunction in the CG space
u = as_vector([-psi.dx(1), psi.dx(0), 0])

# The outflow speed u_n, nonzero only on the downwind side of a facet
```

```
un = 0.5 * (dot(u, n) + abs(dot(u, n)))

# Bilinear form a, the mass matrix
a = phi * dq_dt * dx

# Linear form L
L = (q * div(phi * u) * dx
     - conditional(dot(u, n) < 0, phi * dot(u, n) * q_in, 0.0) * ds_v # Inflow
     - conditional(dot(u, n) > 0, phi * dot(u, n) * q, 0.0) * ds_v # Outflow
     - (phi('+') - phi('-')) * (un('+') * q('+') - un('-') * q('-')) * dS_v) #
     ↪ Interior facets
```

The measure `ds_v` integrates over the lateral boundary A_v, and `dS_v` over the interior vertical facets Γ, with `('+')` and `('-')` restricting an expression to either side of the facet [Ham+23]. Firedrake picks which side is `'+'`, but equation (91) is unchanged by swapping the labels, so the choice does not matter. The horizontal facets (`ds_tb` and `dS_h`) are left out, as $\boldsymbol{u} \cdot \hat{\boldsymbol{n}} = 0$ on them. For verification with the method of manufactured solutions, a source term `phi * s * dx` is added to `L`, which is otherwise zero. Since `a` never changes, the problem is set up with `constant_jacobian=True` so that Firedrake does not rebuild it between solves.

A.9 Zhang-Shu Limiter

The limiter from section 8.3 is given below, simplified to omit the diagnostic. Firedrake has no built-in operation for the minimum and maximum of a `Function` within each cell, so these are computed with a `par_loop`, which runs a short user-written kernel over every cell of the mesh [Ham+23]. Every other step is expressed with standard Firedrake operations on the `'DQ'` space with $p = 0$, which holds one value per cell.

```
class ZhangShuLimiter:
    def __init__(self, q, p):
        mesh = q.function_space().mesh()

        # q on the GLL nodes, which include the cell faces read by the upwind flux
        self._nodes = Function(FunctionSpace(mesh, "DQ", p, variant="gll"))
        self._limited = Function(q.function_space())

        # One value per cell: its volume, and the mean, min, max and theta of q
        W0 = FunctionSpace(mesh, "DQ", 0)
        self._test = TestFunction(W0)
        self._volume = assemble(self._test * dx)
        self._cell_sum = Cofunction(W0.dual())
        self._mean, self._min, self._max, self._theta = (Function(W0) for _ in
        ↪ range(4))

        # The global bounds m and M, from the initial q
        self._nodes.interpolate(q)
        with self._nodes.dat.vec_ro as v:
            self.lower, self.upper = v.min()[1], v.max()[1]

    def apply(self, q):
        # Exact, since both spaces contain the same polynomials
        self._nodes.interpolate(q)
```

```
        # m_K and M_K: loop over the nodes of each cell
        self._min.assign(float('inf'))
        self._max.assign(float('-inf'))
        instructions = """
        for i
            lo[0] = fmin(lo[0], f[i])
            hi[0] = fmax(hi[0], f[i])
        end
        """
        par_loop(("{[i]: 0 <= i < f.dofs}", instructions), dx,
                 {"lo": (self._min, RW), "hi": (self._max, RW), "f": (self._nodes,
                  ↪  READ)})

        # Cell means: the integral of q over each cell, divided by the cell's volume
        assemble(q * self._test * dx, tensor=self._cell_sum)
        self._mean.dat.data_wo[:] = self._cell_sum.dat.data_ro /
        ↪  self._volume.dat.data_ro

        lo, hi, mean = self._min, self._max, self._mean
        L, U = self.lower, self.upper
        # Each ratio is only used where its bound is violated
        # A mean already out of bounds can't be fixed by scaling about it, so that
        ↪  cell goes flat (theta = 0)
        upper = conditional(hi > U, conditional(mean < U, (U - mean) / (hi - mean),
        ↪  0.0), 1.0)
        lower = conditional(lo < L, conditional(mean > L, (L - mean) / (lo - mean),
        ↪  0.0), 1.0)
        self._theta.interpolate(max_value(0.0, min_value(1.0, min_value(upper,
        ↪  lower))))

        # Scale q about each cell's mean, which leaves the mean unchanged
        self._limited.interpolate(mean + self._theta * (q - mean))
        q.assign(self._limited)
        return q
```

Assembling the test function of the $p = 0$ space gives the volume of each cell, and assembling it against q gives the integral of q over each cell, so dividing the two arrays entry by entry yields $\langle q \rangle_K$. The cell volumes are computed once, since the mesh does not change. The limited q is written to a separate `Function` before being copied back, since q itself appears in the expression being interpolated. The global bounds are found with PETSc's `min()` and `max()`, which reduce over every MPI rank [Bal+25; Dal+11]. Every other operation acts on one cell at a time, so, as with `relative_error`, the result does not depend on how the mesh is partitioned.

Appendix B Hardware Used

B.1 Consumer Grade Hardware Used for Primary Testing

The majority of benchmarking and solver optimisation was conducted on an 8 core-16 thread AMD Ryzen 7 7730U with a base clock speed of 2 GHz and a boost clock speed of 4.5 GHz. This processor has a total of 512 kB of L1 cache, 4 MB of L2 cache and 16 MB of L3 cache. Firedrake scripts were executed on

Windows Subsystem for Linux (WSL) with 12 GB of DDR4 memory (3200 MT/s) available.

B.2 High Performance Cluster

Heavy computations were performed using facilities provided by the University of Cape Town's ICTS High Performance Computing team: hpc.uct.ac.za [UCL23] I was given access to a maximum of 40 cores and 320GB of memory per job. Each node has 2 Intel Xeon Gold 6442Y processors, each with 24 cores, 24 threads, 1.92 MB of L1 cache, 48 MiB of L2 cache and 60 MiB of L3 cache.